

5G RAN

MIMO (TDD) Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
 - Revised the descriptions of the cell requirements for inter-cell cable sequence detection. For details, see [Cells](#).
 - Change the name of a function from "32 CSI-RS ports" to "CSI-RS port number configuration". For details, see [4.1.3.2 Downlink Optimization](#).
 - Added MML command examples for the **NRDUCellCsirs.FR1MaxCellCsirsPortNum** parameter. For details, see [4.4.1.2 Using MML Commands](#).
 - Revised the descriptions of the principles of inter-cell cable sequence detection. For details, see [10.1 Principles](#).
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 08 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added impact relationships of time-domain power saving BWP with PDSCH MU-MIMO and PUSCH MU-MIMO. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added impact relationships of time-domain power saving BWP with uplink SU-MIMO and downlink SU-MIMO. For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added impact relationships of NR DU cell resource management between network slices with mmWave PUSCH MU-MIMO. For details, see 8.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added an impact relationship of multi-beam reception enhancement with PUSCH MU-MIMO. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship of real-time-information-based precise channel estimation with PUSCH beam domain enhancement. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship of uplink 4-layer transmission for a single FWA user with PUSCH MU-MIMO. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship of enhanced uplink L4S congestion control based on rate adjustment with PUSCH MU-MIMO. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship of enhanced downlink L4S congestion control based on rate adjustment with PDSCH MU-MIMO. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added impact relationships of hyper cells in FSD transmission mode with the following functions: PUSCH MU-MIMO, PDSCH MU-MIMO, and spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added mmWave PUSCH MU-MIMO. For details, see 8 mmWave MU-MIMO (High-Frequency TDD) .	Added parameters: NRDUCellUIMimo.UIMuSrsPreSinrThld NRDUCellUIMimo.UIMuIsolationMeasThld NRDUCellUIMimo.UIMuSpacelsoThld NRDUCellUIMimo.UIMuBackToSuSeGainThld	High-frequency TDD	5900 series base stations (macro base stations)
Added the requirement of 2:3 slot configuration in high frequency bands on Fixed Weight Type . For details, see 4.3.2.2 Mutually Exclusive Functions .	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between "uplink channel densified equalization for high-speed UEs" and PUSCH MU-MIMO. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
Added descriptions of the mutually exclusive relationship between super uplink and CCE resource optimization. For details, see [7.1.3.2.2 Mutually Exclusive Functions](#).
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010003	MIMO Basic Package	4 Basic Functions of MIMO
FOFD-010020	SU-MIMO Multiple Layers	5 SU-MIMO
FOFD-010010	MU-MIMO Basic Pairing	6 MU-MIMO (Low-Frequency TDD) 8 mmWave MU-MIMO (High-Frequency TDD)
FBFD-010025	Basic O&M Package	10 Inter-Cell Cable Sequence Detection

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic functions of MIMO	None	4 Basic Functions of MIMO
SU-MIMO	None	5 SU-MIMO
MU-MIMO	None	6 MU-MIMO (Low-Frequency TDD) 8 mmWave MU-MIMO (High-Frequency TDD)
Massive MIMO multi-layer enhancement	None	7 Massive MIMO Multi-Layer Enhancement (Low-Frequency TDD)
Multi-beam FDM (high-frequency TDD)	None	9 Multi-Beam FDM (High-Frequency TDD)
Inter-cell cable sequence detection	None	10 Inter-Cell Cable Sequence Detection

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic functions of MIMO	- PMI-based weights but not SRS-based weights are supported in high frequency bands. Both PMI-based weights and SRS-based weights are supported in low frequency bands. - PUSCH beam domain enhancement and PUSCH time domain enhancement are supported in low frequency bands, but not in high frequency bands.	4 Basic Functions of MIMO
SU-MIMO	Uplink UE SINR optimization is supported only in high frequency bands.	5 SU-MIMO
MU-MIMO	None	6 MU-MIMO (Low-Frequency TDD) 8 mmWave MU-MIMO (High-Frequency TDD)
Massive MIMO multi-layer enhancement	Supported only in low frequency bands	7 Massive MIMO Multi-Layer Enhancement (Low-Frequency TDD)
Multi-beam FDM (high-frequency TDD)	Supported only in high frequency bands	9 Multi-Beam FDM (High-Frequency TDD)
Inter-cell cable sequence detection	Supported only in low frequency bands	10 Inter-Cell Cable Sequence Detection

3 Overview

With the rapid development of wireless communications, increasingly high requirements are imposed on system capacity and spectral efficiency. To meet these requirements, various technologies have emerged. Common technologies include extending system bandwidth and increasing signal modulation order. However, extending system bandwidth can only increase system capacity but cannot effectively improve spectral efficiency, while increasing signal modulation order can only improve spectral efficiency to a limited extent on live networks, as modulation order is strongly correlated with signal quality.

Multiple-input multiple-output (MIMO) is a technology that uses multiple antennas at the transmitter and receiver to multiply spectral efficiency. MIMO is supported in 2T2R/4T4R/8T8R/32T32R/64T64R in NR TDD. **The capability of MIMO to improve spectral efficiency is closely related to the number of antennas. Therefore, MIMO in 32T32R/64T64R mode (that is, massive MIMO) is recommended for NR TDD.** (32T32R or 64T64R cells are referred to as massive MIMO cells.) To make full use of MIMO performance, the following signal processing technologies are required:

- **Receive diversity**
It is used to improve uplink reception performance. For details, see [4 Basic Functions of MIMO](#).
- **Beamforming**
It is used to improve downlink transmission performance. For details, see [4 Basic Functions of MIMO](#).
- **Spatial multiplexing**
For details about spatial multiplexing of a single UE's multiple data streams, see [5 SU-MIMO](#).
For details about spatial multiplexing of multiple UEs' data streams in a cell, see [6 MU-MIMO \(Low-Frequency TDD\)](#) and [8 mmWave MU-MIMO \(High-Frequency TDD\)](#).
- **Multi-beam FDM**
For high-frequency TDD cells, the gNodeB can also use multi-beam frequency division multiplexing (FDM) to improve spectral efficiency. For details, see [9 Multi-Beam FDM \(High-Frequency TDD\)](#).

In massive MIMO scenarios, more and more layers are paired for MU-MIMO, with the development of technologies. This imposes higher requirements on inter-UE

interference suppression, multi-UE scheduling, and weight accuracy. To satisfy these requirements, massive MIMO multi-layer enhancement is introduced. It optimizes interference suppression, scheduling, and weights for a large number of layers, increasing the downlink UE throughput. For details, see [7 Massive MIMO Multi-Layer Enhancement \(Low-Frequency TDD\)](#).

Moreover, in a MIMO system, the polarization modes of physical and logical antenna ports corresponding to the channels of RF modules (also referred to as RRUs in this document) must be matched to ensure system performance. For details, see [4.3.4 Networking](#). Inter-cell antenna connections are prone to be incorrect and faults are difficult to locate. The cable sequence detection function can be used to detect faults. For details, see [10 Inter-Cell Cable Sequence Detection](#).

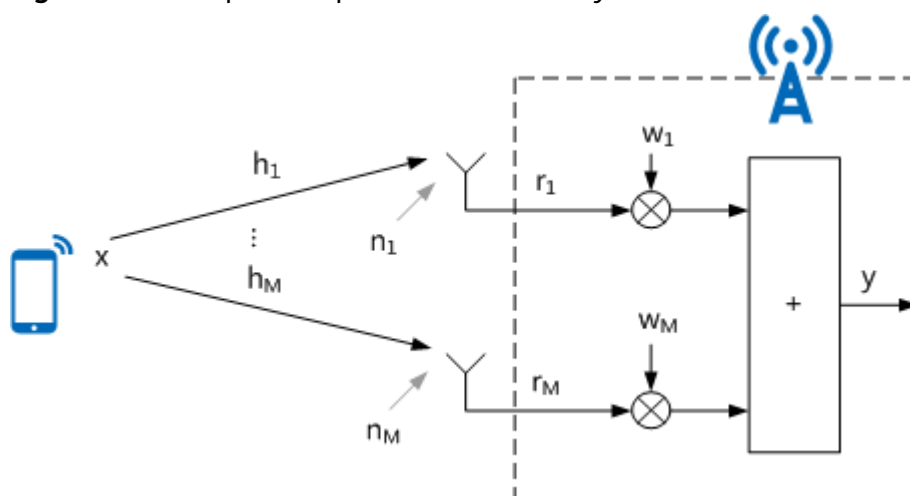
4 Basic Functions of MIMO

4.1 Principles

4.1.1 Uplink Receive Diversity

Figure 4-1 shows the principles of uplink receive diversity.

Figure 4-1 Principles of uplink receive diversity



The UE sends signal x , which passes through different channels to the gNodeB's M antennas r_1 to r_M . The gNodeB applies a weight w_i to each received signal, and then combines the weighted signals into signal y . The combined signal can be expressed as follows:

$$y = W (Hx + N)$$

where

- $W = (w_1, \dots, w_M)$: $1 \times M$ vector composed of RX antenna weights.
- $H = (h_1, \dots, h_M)^T$: $M \times 1$ vector composed of spatial channel coefficients. h_i indicates the coefficient of channel i , and T indicates a transpose operator.

The channel coefficients are used to obtain the signals that change in amplitude and phase after passing through channels.

- $N = (n_1, \dots, n_M)^T$: $M \times 1$ vector composed of received noises. n_i indicates the coefficient of channel i , and T indicates a transpose operator.
- x : TX signal.

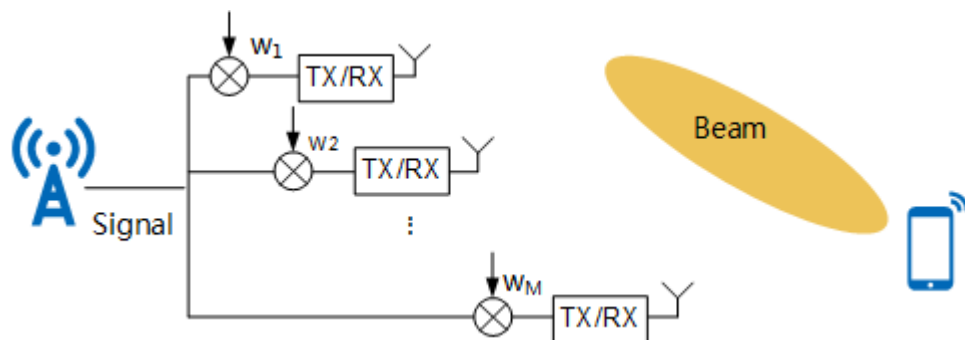
Radio channels between a transmitter and a receiver may experience time-varying deep fading, which leads to SINR fluctuation at the receiver. If the receiver uses multiple antennas for reception, there is a relatively low probability that deep fading occurs simultaneously on different antennas. As a result, the combined signals experience a much lower probability of deep fading than the signals received by a single antenna, which brings diversity gains. Moreover, white noises on different antennas are uncorrelated, and therefore the power of the combined noise remains unchanged while the energy of the combined signal increases several-fold, which brings array gains.

4.1.2 Downlink Beamforming

Downlink beamforming mentioned in this document refers to a technique in which a gNodeB weights a to-be-transmitted signal and generates a directional narrow beam. The more accurately the beam points to the UE, the better the UE performance. The beam direction is determined by weights.

Figure 4-2 shows the weighting principle. The signal to be transmitted is weighted using different weights ($w_1, \dots, w_M$) on different logical antenna ports to change the signal amplitudes and phases. After the signal energy radiated by the antennas is superimposed, a beam towards the UE is generated. The more the antennas, the narrower the beam and the more flexible the beam direction. For each data stream, a directional beam is formed based on independent weights.

Figure 4-2 Weighting principle



The weights ($w_1, \dots, w_M$) are used to change the beamwidth and direction. They are calculated based on downlink channel status.

There are two methods of weight calculation:

- Weight calculation based on sounding reference signal (SRS): The calculation procedure is illustrated in **Figure 4-3**.
- Weight calculation based on precoding matrix indication (PMI): The calculation procedure is illustrated in **Figure 4-4**.

Figure 4-3 SRS-based weight calculation

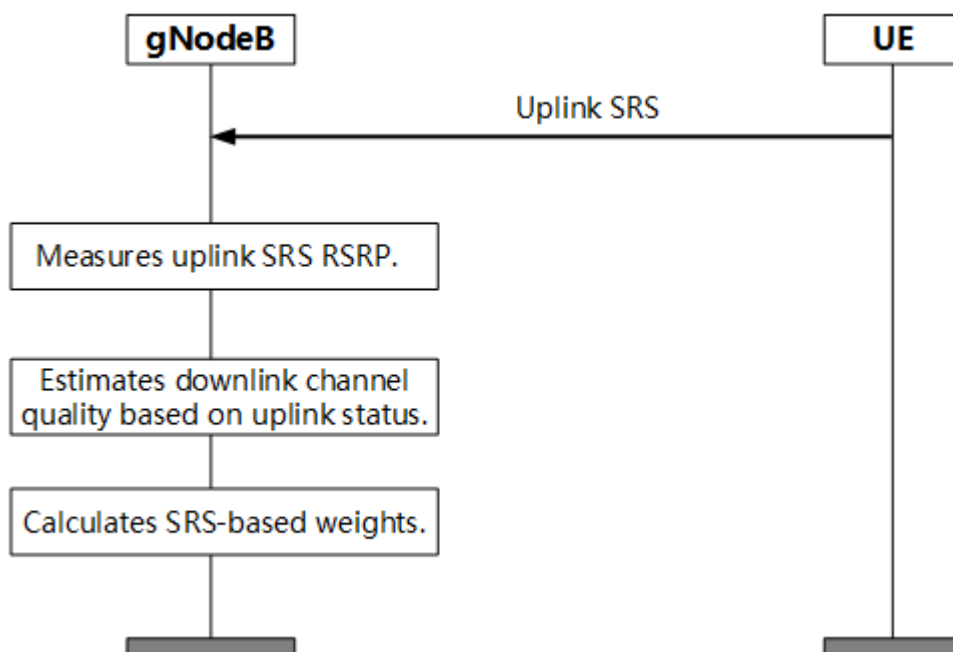
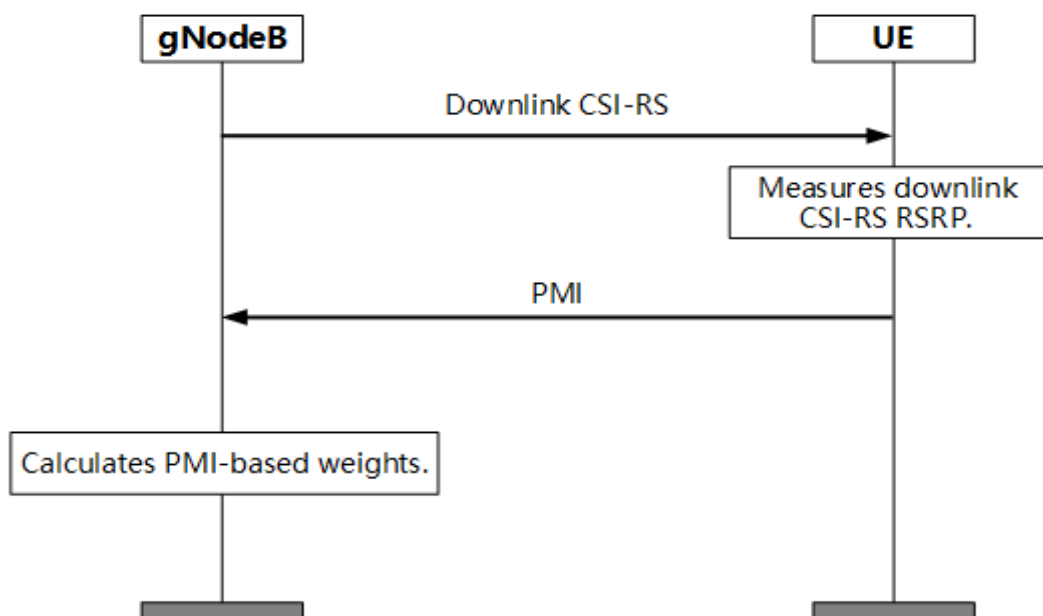


Figure 4-4 PMI-based weight calculation



PMI-based weights are always used for cells with less than 4T4R. Both SRS-based weights and PMI-based weights can be used for cells with at least 4T4R.

SRS-based weights or PMI-based weights can be adaptively selected when both are supported, so that the weights can more accurately reflect the quality of traffic channels and increase the downlink throughput of cell edge users (CEUs). SRS-based weights are preferentially used when SRS resources are available and the signal quality of SRS is good. Otherwise, PMI-based weights are preferentially used. SRS- and PMI-based weight adaptation is controlled by the **DL_PMI_SRS_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter.

Table 4-1 details how SRS- or PMI-based weights are selected when SRS- and PMI-based weight adaptation is enabled and when it is disabled.

Table 4-1 Selection of SRS- or PMI-based weights

Scenario	Selection of SRS- or PMI-based Weights	Parameter
SRS- and PMI-based weight adaptation enabled	During initial access, SRS-based weights are used by default. After the access, SRS- or PMI-based weights are selected as follows: - When the SRS SINR is greater than the SRS PreSINR threshold^a plus 3 dB, SRS-based weights are used. (SRS PreSINR is short for SRS SINR before equalization.) - When the SRS SINR is less than the SRS PreSINR threshold^a minus 3 dB, PMI-based weights are used. - When the SRS SINR is in the range of [SRS PreSINR threshold^a – 3 dB, This threshold + 3 dB], the weight type remains unchanged.	a: The SRS PreSINR threshold is specified by the NRDUCellPdsch.SrsPreSinrJudgeThld parameter.
SRS- and PMI-based weight adaptation disabled	- For antenna selection UEs, SRS- or PMI-based weights are always selected based on the configured fixed weight type^b. - For non-antenna-selection UEs, SRS- or PMI-based weights are always selected based on the fixed precoding type configured for such UEs^c.	b: The fixed weight type is specified by the NRDUCellPdsch.FixedWeightType parameter. c: The fixed precoding type for non-antenna-selection UEs is specified by the NRDUCellPdschPre-code.NonAsFixedPrecodingType parameter.

 NOTE

For a high-frequency TDD cell, it is recommended that downlink SRS- and PMI-based weight adaptation be enabled when more than 80% of PUSCH MCS indexes in the cell are greater than 13.

An antenna selection UE refers to a UE that transmits SRSs on multiple antennas in turn, and a non-antenna-selection UE refers to a UE that transmits SRSs on a fixed antenna. For details about antenna selection, see section 6.2.1.2 "UE sounding procedure for DL CSI acquisition" in 3GPP TS 38.214 V17.2.0.

FWA users are generally stationary, and channel quality changes are mainly caused by external noises. The downlink beamforming weight filtering

optimization function is introduced to improve the accuracy of channel quality estimation by the gNodeB for stationary UEs through SRS filtering. This function is controlled by the **DL_BF_WEIGHTFILTER_OPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter when the **DL_PMI_SRS_ADAPT_SW** option of the same parameter is selected. If this function is enabled, the gNodeB performs SRS filtering and then calculates SRS-based weights for identified FWA users. This improves the accuracy of downlink beamforming. For details about FWA users, see *FWA*.

SRS-based weights are valid in a certain period, which is specified by the **NRDUCellPdschPrecode.SrsWeightValidityPeriod** parameter. When the SRS reporting interval exceeds the value of this parameter, SRS-based weights become invalid and PMI-based weights are used instead for the downlink.

For a fast track through the principles of beamforming, see [\(Video\) Beamforming in a Multi-Antenna System](#).



4.1.3 MIMO Performance Enhancements

MIMO can improve the uplink and downlink throughputs of UEs through weight optimization, measurement optimization, scheduling optimization, and beam selection optimization. In addition, as MetaAAUs have much better performance than common AAUs, MIMO provides dedicated optimization measures for cells served by MetaAAUs. As RRUs do not have the antenna polarization capability of AAUs, the polarization modes of physical and logical antenna ports corresponding to RRU channels must be consistent to ensure MIMO performance when RRUs are used. If the polarization modes are inconsistent during antenna installation, beam coverage will deteriorate. Therefore, antenna port mapping is required for cells served by RRUs to ensure that physical and logical antenna ports have consistent polarization modes. For details, see [4.3.4 Networking](#).

4.1.3.1 Uplink Optimization

[Table 4-2](#) describes the uplink optimization functions.

Table 4-2 Uplink optimization functions

Optimization Method	Function Name	Function Switch	Function Description
Weight optimization	PUSCH time domain enhancement	PUSCH_TIME_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	This function takes effect only for SU-MIMO CEUs in low-frequency TDD massive MIMO cells. After this function is enabled, the gNodeB retains useful signals and suppresses white noise based on the power delay profile (PDP) of PUSCH DMRSs to improve the channel estimation accuracy, uplink demodulation performance, and uplink CEU throughput.
Measurement optimization	SRS SINR measurement optimization	SRS_SINR_MEAS_OPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	For low-frequency TDD massive MIMO cells as well as high-frequency TDD cells, this function corrects the SRS measurement results reported by UEs based on interference strength, and improves the accuracy of SRS SINR measurement results and uplink MCS indexes.
Scheduling optimization	Fast uplink rank reduction based on SRS SINR	UL_RANK_FAST_DECREASE_SW option of the NRDUCellUIRank.UlRankAlgoSwitch parameter	This function identifies whether UEs are in power saving or dual-SIM dual-pass mode based on the SRS SINR difference between streams. If so, it quickly reduces ranks to increase the uplink UE throughput.
	Enhanced PUSCH SNR level for channel estimation	PUSCH_CE_SINR_LEVEL_ENH_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	This function takes effect only in low-frequency TDD massive MIMO cells. This function improves the PUSCH demodulation performance in interference scenarios and increases the uplink cell throughput and uplink UE throughput.

Optimization Method	Function Name	Function Switch	Function Description
Beam selection optimization	PUSCH beam domain enhancement	PUSCH_BEAM_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCoverageEnhSwitch parameter	This function takes effect only in low-frequency TDD massive MIMO cells. After this function is enabled, the gNodeB can adaptively adjust the number of PUSCH receive beams. In multipath scenarios, it increases the number of receive beams to improve the base station demodulation performance and uplink UE throughput. In multipath scenarios in dense urban areas, the gains are higher. In open areas, the gains are lower but there are no negative impacts.
	PUSCH beam selection optimization	PUSCH_BEAM_SELECTION_OPT_SW option of the NRDUCellAlgoSwitch.FullChannelCoverageEnhSwitch parameter	This function takes effect only in low-frequency TDD massive MIMO cells. After this function takes effect, different numbers of receive beams can be adaptively selected for different UEs in a cell. This improves the interference suppression effect, uplink demodulation performance of the base station, and uplink UE throughput.

4.1.3.2 Downlink Optimization

Table 4-3 describes the downlink optimization functions.

Table 4-3 Downlink optimization functions

Optimization Method	Function Name	Function Switch	Function Description
Weight optimization	CSI-RS port number configuration	NRDUCellCsirs.FR1MaxCellCsirsPortNum	The number of CSI-RS ports can be configured for low-frequency TDD massive MIMO cells, and the maximum number is 32. When 32 CSI-RS ports are configured, the following options must be selected for the configuration to take effect: • For cells in non-CA scenarios or PCells in CA scenarios: SGL_OR_PCC_CSIRS_32PORT_SW option of the NRDUCellCsirs.CsiSwitch parameter • For SCells in CA scenarios: CA_SCC_CSIRS_32PORT_SW option of the NRDUCellCsirs.CsiSwitch parameter

Optimization Method	Function Name	Function Switch	Function Description
Scheduling optimization	Packet length-based scheduling optimization	NRDUCellPdsch.DLSchOptTimeThld and NRDUCellDLSch.DLAdaptSchTimeThld	This function is applicable when the traffic models of UEs differ greatly. It enables small-packet UEs with a small amount of buffered data and good signal quality to be preferentially scheduled and therefore increases the downlink UE throughput. Specifically: - Assume that the duration threshold for optimized downlink scheduling (threshold 1^a) is not zero and the duration threshold for adaptive downlink scheduling (threshold 2^b) is zero. - If the estimated downlink data transmission duration of a UE is less than threshold 1^a, the downlink scheduling priority of the UE will be increased. - If the estimated downlink data transmission duration of a UE is greater than or equal to threshold 1^a, the downlink scheduling priority of the UE will remain unchanged. - Assume that neither threshold 1^a nor threshold 2^b is zero. - If the estimated downlink data transmission duration of a UE is less than min(Threshold 1^a, Threshold 2^b), the downlink scheduling priority of the UE will be increased. - If the estimated downlink data transmission duration of a UE is greater than or equal to either threshold 1^a or threshold 2^b, the downlink scheduling priority of the UE will remain unchanged.

Optimiz ation Method	Function Name	Function Switch	Function Description
			• If threshold 1^a is set to zero, this function does not take effect. The involved parameters are as follows: a: The duration threshold for optimized downlink scheduling is specified by the NRDUCellPdsch.DISchOptTimeThld parameter. b: The duration threshold for adaptive downlink scheduling is specified by the NRDUCellDISch.DIAdaptSchTimeThld parameter.

Optimization Method	Function Name	Function Switch	Function Description
MetaAAU optimization	MetaAAU beam sensing	NRDUCellBeamAlgo.BeamPerceiveMode parameter set to DISTRIBUTED_MODE	This function takes effect only in cells served by low-frequency TDD MetaAAUs. This function adaptively enhances the beam accuracy in a cell based on the information about UE data transmission and service distribution, increasing the uplink and downlink UE throughputs in the cell. When this function is enabled: • If MetaAAU beam adaptation^a is also enabled, the beam accuracy for large-packet UEs in the cell can be improved. • If SRS-based weight estimation for downlink large-packet UEs^b is enabled in addition to MetaAAU beam adaptation, SRS-based weights can be estimated for downlink large-packet UEs, and the downlink throughput of these UEs can increase. The involved parameters are as follows: a: MetaAAU beam adaptation is controlled by the NRDUCellBeamAlgo.BeamAdaptSwitch parameter. b: SRS-based weight estimation for downlink large-packet UEs is controlled by the SRS_WEIGHT_ESTIMATE_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter.
	PMI-based weight optimization	PMI_WEIGHT_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter	This function takes effect only in cells served by low-frequency TDD MetaAAUs. This function improves the PMI-based weight accuracy and downlink UE throughput and may increase the number of scheduled UEs and RB utilization in both uplink and downlink.

Optimiz ation Method	Function Name	Function Switch	Function Description
	Open-loop weight optimization	OPEN_LOOP_WEIGHT_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgorithm parameter	This function takes effect only in cells served by low-frequency TDD MetaAAUs. This function improves the open-loop weight performance and downlink UE throughput.

4.2 Network Analysis

4.2.1 Benefits

Array, diversity, and beamforming gains are offered by multiple TX and RX antennas, as shown in [Figure 4-5](#), [Figure 4-6](#), and [Figure 4-7](#). Cell coverage improves; specifically, coverage direction is more flexible, coverage scope is wider, and three-dimensional coverage can be provided in massive MIMO mode. As a result, the average UE throughput increases in both uplink and downlink.

Figure 4-5 Array gain

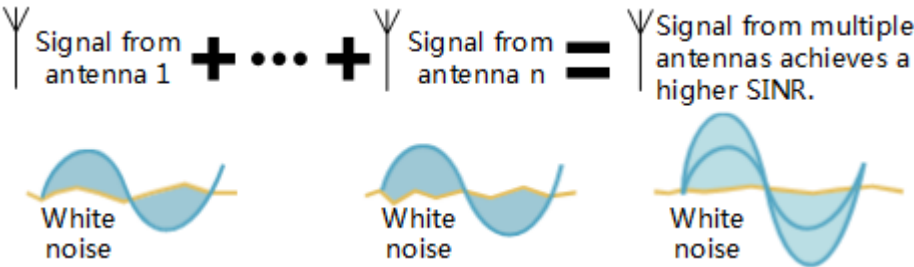


Figure 4-6 Diversity gain

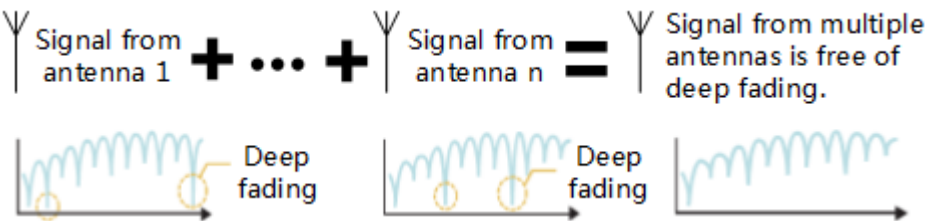
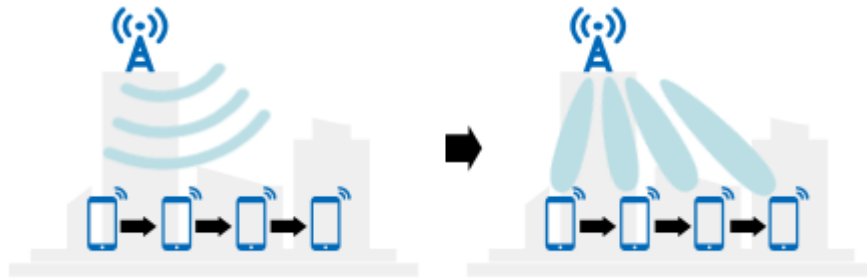


Figure 4-7 Beamforming gain



4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

MIMO is mainly used in heavy-load scenarios where the PRB usage of a cell is greater than or equal to 70% and less than 95%. However, the BBP CPU usage is high in heavy-load scenarios. To reduce the BBP CPU usage and improve gNodeB MIMO performance by optimizing the MIMO measurement algorithm, the **CHN_MEASURE_CPU_DEC_SW** option of the **gNodeBParam.NrBoardPerformanceSw** parameter can be selected to enable the function of CPU usage reduction for channel measurement. CPU usage reduction for channel measurement is not supported in high-frequency TDD.

When 32 CSI-RS ports are configured for a cell, CSI-RS resource overhead increases. If the proportion of scheduling with PMI-based weights is small or the proportion of UEs supporting 32 CSI-RS ports is small, the average downlink UE throughput and average downlink cell throughput may decrease, and the downlink PRB usage, proportion of HARQ retransmissions, average downlink MCS index, average downlink rank, and downlink initial block error rate (IBLER) of the cell may change. The impact on the average downlink UE throughput depends on the measurement performance of UEs.

In addition, certain functions may generate the impacts described in [Table 4-4](#).

Table 4-4 Network impacts of certain functions

Function	Network Impact
MetaAAU beam sensing and MetaAAU beam adaptation	• In both uplink and downlink, the UE throughput increases, the cell PRB usage decreases, and the IBLER, average MCS index, and HARQ retransmission proportion fluctuate. The downlink average CQI also fluctuates. In addition, indicators related to UE access and service drops fluctuate. If the CSI-RS measurement resources supported by UEs are insufficient, there are negative impacts. • The values of the following indicators fluctuate: N.UL.NI.Avg, N.UL.RSSI.Avg, N.UL.First.SymbolX.Pwr ($X = 0, 2, 4, 6$), N.UL.Last.SymbolX.Pwr ($X = 6, 11, 13$), N.GAP.LastSymbol.Pwr, N.CCE.UL.AggLvlXNum ($X = 1, 2, 4, 8, 16$), and N.CCE.DL.AggLvlXNum ($X = 1, 2, 4, 8, 16$).
PMI-based weight optimization	• The number of RRC reconfiguration messages increases, and the average downlink MCS index, average downlink rank, and downlink IBLER of a cell change. • The average uplink UE throughput decreases, and the average uplink MCS index, average uplink rank, and uplink IBLER of a cell decrease.
Open-loop weight optimization	• The average downlink MCS index, average downlink rank, and downlink IBLER of a cell change. • The number of scheduled downlink UEs and downlink RB usage in a cell increase.
SRS-based weight estimation for downlink large-packet UEs	The average throughput, average MCS, IBLER, HARQ retransmission rate, and PRB usage fluctuate in the downlink for downlink large-packet UEs.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MetaAAU beam sensing	NRDUCellBeamAlgo.BeamPerceiveMode set to DISTRIBUTED_MODE	<i>MIMO (TDD)</i>	PMI-based weight optimization takes effect only after MetaAAU beam sensing is enabled. Open-loop weight optimization takes effect only after MetaAAU beam sensing is enabled.
Low-frequency TDD	SRS- and PMI-based weight adaptation	DL_PMI_SRS_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>MIMO (TDD)</i>	The setting of NRDUCellPdschPre-code.SrsNonASFxedWeightType takes effect only if SRS- and PMI-based weight adaptation is enabled.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	PUSCH beam domain enhancement cannot be enabled for 32T32R or 64T64R cells.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	If the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL , the NRDUCellBeamAlgo.BeamPerceiveMode parameter must be set to CENTRALIZED_MODE .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive CSI-RS multi-beam adjustment	NRDUCellCsirs.CsirsCellResourceNum set to FD_RESOURCE	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	When both adaptive CSI-RS multi-beam adjustment and PMI-based weight optimization are enabled for a cell, the following conditions must be met: - If the cell works in 32T32R, the NRDUCellCsirs.FdCsiPeriod parameter cannot be set to SLOT40 or SLOT80. - If the cell works in 64T64R, the NRDUCellCsirs.FdCsiPeriod parameter cannot be set to SLOT40, SLOT80, or SLOT160.
Low-frequency TDD	CSI-RS for CM reuse 3	CSIRS_REUSE3_SW option of the NRDUCellCsirs.CsirsSwitch parameter	<i>Channel Management</i>	CSI-RS for CM reuse 3 must be disabled if PMI-based weight optimization is enabled and the NRDUCellCsirs.CsirsPeriod parameter is set to SLOT10 , SLOT20 , or SLOT40 .
Low-frequency TDD	Optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration	CSIRS_INCOMPATIBLE_UE_SW option of the NRDUCellCsirs.CsirsSwitch parameter	<i>Channel Management</i>	PMI-based weight optimization cannot be enabled together with "optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration".
Low-frequency TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSwitch.MultiCarrierSwitch parameter	None	PMI-based weight optimization cannot be enabled together with Bi-Carrier.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE capability adaptation based on configurable CSI-RS for CM resources	CSIRS_CONFIG_CAPP_ADAPTIVE_SW option of the NRDUCellCsirs.CsiSwitch parameter	<i>Channel Management</i>	PMI-based weight optimization cannot be enabled together with "UE capability adaptation based on configurable CSI-RS for CM resources".
Low-frequency TDD	Weight optimization in scenarios of aggregation into 2T	AGG_2T_WEIGHT_OPT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	SRS- and PMI-based weight adaptation cannot be enabled together with "weight optimization in scenarios of aggregation into 2T".
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDualNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	For an NR TDD cell with Joint Transmission Cell Combination enabled and the DL_PMI_SRS_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter deselected, the NRDUCellPdsch.FixedWeightType parameter must be set to PMI_WEIGHT .

4.3.2.3 Function Impacts

- Uplink receive diversity
None
- Downlink beamforming

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Downlink beamforming weight filtering optimization does not take effect when RF channel intelligent shutdown takes effect.

- Uplink optimization

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink power reduction optimization	UL_256QAM_MPR_OPT_SW option of the NRDUCellULAmc.ULAmcAlgoSw parameter	<i>Uplink Scheduling</i>	It is recommended that SRS SINR measurement optimization be enabled together with uplink power reduction optimization to reduce the uplink BLER during modulation order switching.
High-frequency TDD	Uplink UE SINR optimization	UL_USER_SINR_OPT_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	None	It is recommended that SRS SINR measurement optimization be enabled together with uplink UE SINR optimization to improve the accuracy of SINR calculation in single-UE multi-stream scenarios.
High-frequency TDD	PUSCH beam allocation optimization	PUSCH_BEAM_ALLOC_OPT_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter	None	It is recommended that SRS SINR measurement optimization be enabled together with PUSCH beam allocation optimization.
High-frequency TDD	Rank Decrease Spct Eff Hys Threshold	NRDUCellUIRank.RankDecSpctEffHysThld	None	The value 15 is recommended if SRS SINR measurement optimization is enabled.
Low-frequency TDD	PUCCH grouping adaptation enhancement for inter-FR CA	PUCCH_GROUP_ADAPT_ENH_SW option of the NRDUCellCarrMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	SRS SINR measurement optimization does not take effect for UEs for which PUCCH grouping adaptation enhancement for inter-FR CA is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Real-time-information-based precise channel estimation	PREC_CHANNEL_EST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Real-time-information-based precise channel estimation does not take effect when it is enabled together with PUSCH beam domain enhancement.

- Downlink optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	It is recommended that MetaAAU beam sensing be enabled together with UE-characteristic-based SRS period adaptation.
Low-frequency TDD	TRS rate matching	TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	It is recommended that MetaAAU beam sensing be enabled together with TRS rate matching.
Low-frequency TDD	Enhanced aperiodic uplink CSI scheduling	UL_APERIODIC_CSI_SCH_ENH_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	None	It is recommended that MetaAAU beam sensing be enabled together with enhanced aperiodic uplink CSI scheduling.
Low-frequency TDD	UE capability adaptation based on configurable CSI-RS for CM resources	CSIRS_CONFIG_CAP_ADAPTIVE_SW option of the NRDUCellCsirs.CsiSwitch parameter	<i>Channel Management</i>	It is recommended that UE capability adaptation based on configurable CSI-RS for CM resources be disabled when MetaAAU beam sensing is enabled.
Low-frequency TDD	U-TDOA- and AoA-based high-precision positioning	AOA_UTDOA_HYBRID_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	U-TDOA- and AoA-based high-precision positioning preferentially takes effect when it is enabled together with MetaAAU beam sensing (enabled by setting NRDUCellBeamAlgo.BeamPerceiveMode to DISTRIBUTED_MODE).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low power high accuracy positioning	LPHAP_REDCAP_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	Low power high accuracy positioning preferentially takes effect when it is enabled together with MetaAAU beam sensing (enabled by setting NRDUCellBeamAlgo.BeamPerceiveMode to DISTRIBUTED_MODE).
Low-frequency TDD	Spectral efficiency threshold optimization	SPCT_EFF_THLD_OPT_SW option of the NRDUCellDlRank.AiRankParaOptSwitch parameter	None	Spectral efficiency threshold optimization preferentially takes effect when it is enabled together with MetaAAU beam sensing (enabled by setting NRDUCellBeamAlgo.BeamPerceiveMode to DISTRIBUTED_MODE).
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For UEs in their SCells of downlink intra-band CA, the activation of packet length-based scheduling optimization may be affected by the mechanism of data distribution to the SCells.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For UEs in their SCells of downlink intra-FR inter-band CA, the activation of packet length-based scheduling optimization may be affected by the mechanism of data distribution to the SCells.
Low-frequency TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	For UEs in their SCells of inter-gNodeB CA, the activation of packet length-based scheduling optimization may be affected by the mechanism of data distribution to the SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlogoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When MetaAAU beam adaptation takes effect, the benefits of CCE partial symbol adaptation may decrease.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For functions other than MetaAAU beam sensing:

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

For MetaAAU beam sensing:

All NR-capable main control boards and all NR TDD-capable baseband processing units except the UBBPfw support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Only MetaAAUs support the following functions: MetaAAU beam sensing, PMI-based weight optimization, open-loop weight optimization, and SRS-based weight estimation for downlink large-packet UEs.

Any other function:

All NR TDD-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Networking

When physical and logical antenna ports have the same polarization mode, the downlink multi-antenna performance improves. For UEs accessing the network,

the access success rate increases. For UEs performing data transmission, the downlink throughput increases. The antenna port mapping function is controlled by the **NRDUCellCoverage.AntennaPortMapping** parameter for low-frequency TDD 4T4R cells or controlled by the **NRDUCellCoverage.AntennaPortMapping** or **NRDUCellCoverage.AntennaPortMappingExt** parameter for low-frequency TDD 8T8R cells.

For low-frequency TDD 4T4R cells, see [Low-Frequency TDD 4T4R Cell Antenna Port Mapping](#) for details.

For low-frequency TDD 8T8R cells, see [Low-Frequency TDD 8T8R Cell Antenna Port Mapping](#) for details.

For details about connections between antennas and RRU channels, see the RF cable connection description in the RRU hardware description in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

The **NRDUCellCoverage.AntennaPortMappingExt** parameter can be set to any antenna port mapping and takes effect only when the **NRDUCellCoverage.AntennaPortMapping** parameter is set to **DEFAULT**. If the **NRDUCellCoverage.AntennaPortMappingExt** parameter has been set to a value, this value can be cleared in the following two ways: (1) set the parameter to a null string by running the command **MOD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, AntennaPortMapping=DEFAULT, AntennaPortMappingExt=""**; (2) enter a space on the GUI, which is equivalent to running the command **MOD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, AntennaPortMapping=DEFAULT, AntennaPortMappingExt=" "**;

Low-Frequency TDD 4T4R Cell Antenna Port Mapping

For TDD 4T4R cells, the gNodeB applies one polarization weight to logical antenna ports 0 and 2 and another polarization weight to logical antenna ports 1 and 3. Therefore, antenna port mapping has the following requirements:

- Port 0 is always mapped to RRU channel A. The RRU channels corresponding to ports 0 and 2 must be connected to a pair of physical antenna ports with the same polarization.
- The RRU channels corresponding to ports 1 and 3 must be connected to another pair of physical antenna ports with the same polarization.

Table 4-5 lists the recommended configurations for different connections, taking a TDD 4T4R RRU as an example.

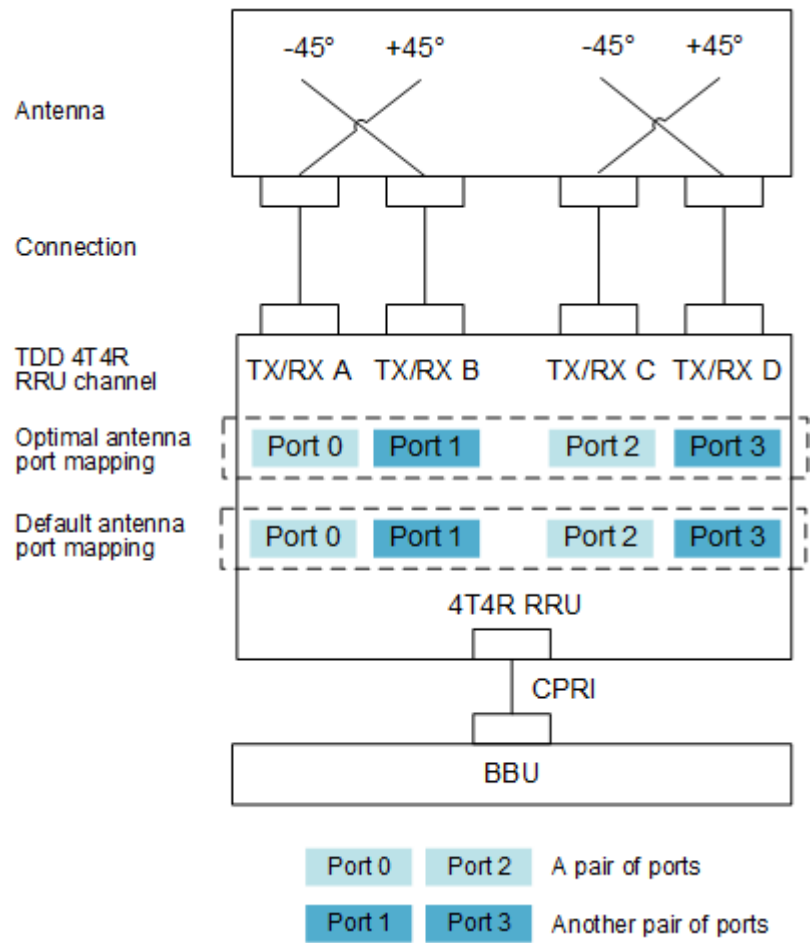
Table 4-5 Recommended antenna port mapping configurations for different TDD 4T4R RRU connections

RRU Channels Connected to Physical Antenna Ports (+45°, -45°, +45°, -45°)	Recommended NRDUCellCoverage.AntennaPortMapping Parameter Setting	Logical Antenna Ports Mapped to RRU Channels (A, B, C, D)
(A, B, C, D)	4T_0123 or DEFAULT	(0, 1, 2, 3)
(A, C, D, B)	4T_0312	(0, 3, 1, 2)
(A, B, D, C)	4T_0132	(0, 1, 3, 2)

RRU Channels Connected to Physical Antenna Ports (+45°, -45°, +45°, -45°)	Recommended NRDUCellCoverage.AntennaPortMapping Parameter Setting	Logical Antenna Ports Mapped to RRU Channels (A, B, C, D)
(A, C, B, D)	4T_0213	(0, 2, 1, 3)
(A, D, B, C)	4T_0231	(0, 2, 3, 1)
(A, D, C, B)	4T_0321	(0, 3, 2, 1)
Note: Physical antenna ports (+45°, -45°, +45°, -45°) indicate that the polarization directions of physical antenna ports are +45°, -45°, +45°, and -45°, respectively.		

Figure 4-8 shows the mapping. The optimal mapping is that channels A, B, C, and D are mapped to ports 0, 1, 2, and 3, respectively. In this case, channel A (port 0) and channel C (port 2) are connected to a pair of +45° physical antenna ports, and channel B (port 1) and channel D (port 3) are connected to another pair of -45° physical antenna ports. Therefore, the optimal setting of the NRDUCellCoverage.AntennaPortMapping parameter is 4T_0123.

Figure 4-8 Antenna port mapping for a TDD 4T4R RRU

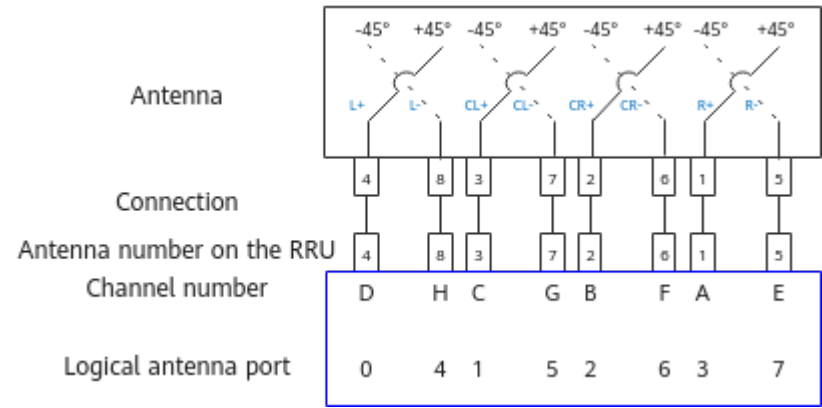


Low-Frequency TDD 8T8R Cell Antenna Port Mapping

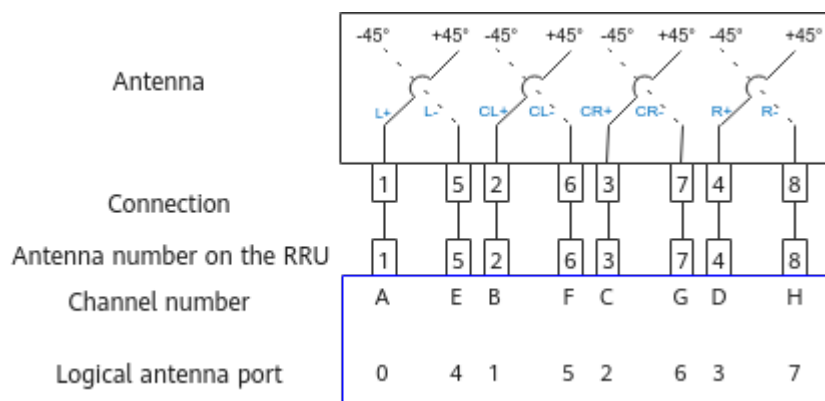
For low-frequency TDD 8T8R cells, the mapping between logical antenna ports and RRU channels can be adjusted based on the polarized antenna sequence using a parameter so that the mapping between antennas and logical antenna ports is as follows: (L+, port 0), (L-, port 4), (CL+, port 1), (CL-, port 5), (CR+, port 2), (CR-, port 6), (R+, port 3), and (R-, port 7). Specifically,

- When a low-frequency TDD 8T8R RRU and a Huawei TDD 8T8R passive antenna are connected through feeders based on the antenna numbers shown in [Figure 4-9](#):
 - It is recommended that the **NRDUCellCoverage.AntennaPortMapping** parameter be set to **DEFAULT**, corresponding to the actual value 8T_32107654.
 - It is recommended that the **NRDUCellCoverage.AntennaPortMappingExt** parameter be set to **32107654**.

Figure 4-9 Antenna port mapping for a low-frequency TDD 8T8R RRU and a Huawei TDD 8T8R passive antenna



- When a low-frequency TDD 8T8R RRU and a Huawei AAU's passive antenna are connected through feeders based on the antenna numbers shown in [Figure 4-10](#):
 - It is recommended that the **NRDUCellCoverage.AntennaPortMapping** parameter be set to **8T_01234567**.
 - It is recommended that the **NRDUCellCoverage.AntennaPortMappingExt** parameter be set to **01234567**.

Figure 4-10 Antenna port mapping for a low-frequency TDD 8T8R RRU and a Huawei AAU's passive antenna

It is recommended that Huawei 8T8R antennas or NGMN-compliant 8T8R antennas be used. Using other antennas may result in mismatched polarized antenna sequences, affecting cell coverage. (NGMN stands for Next Generation Mobile Network.)

4.3.5 Cells

PUSCH beam domain enhancement: The **NRDUCellTrp.TrpType** parameter cannot be set to **MASTER_DM_MIMO_LIGHT**.

PMI-based weight optimization:

- For a massive MIMO cell, if **NRDUCellCsirs.CsiPeriod** is set to **SLOT20**, **NRDUCellCsirs.CsirsCellResourceNum** cannot be set to **4_RESOURCE** or **2_RESOURCE**; if **NRDUCellCsirs.CsiPeriod** is set to **SLOT40**, **NRDUCellCsirs.CsirsCellResourceNum** cannot be set to **4_RESOURCE**.
- For a massive MIMO cell, **NRDUCellCsirs.CsiPeriod** cannot be set to **SLOT10**.

Fixed weights: 2:3 slot configuration in high frequency bands requires the **NRDUCellPdsch.FixedWeightType** parameter to be set to **PMI_WEIGHT**.

4.3.6 Others

SRS-based Weights

SRS-based weights require that the uplink and downlink channels of a cell have reciprocity. In a TDD system, uplink and downlink channels over the air interface are inherently reciprocal. Therefore, the reciprocity of TX and RX channels within an RF module must be ensured. This reciprocity can be ensured through channel calibration.

- 3900 and 5900 series base stations (macro base stations) always perform channel calibration.
- DBS3900 LampSite and DBS5900 LampSite support channel calibration only in distributed massive MIMO cells for which the **INTER_RRU_CHN_CALIB_SW** option of the **NRDUCellAlgoSwitch.ChnCalibSwitch** parameter is selected. For details about distributed massive MIMO cells, see *Distributed Antenna Solutions (Low Frequency TDD)*.

- PMI-based weights are always used for modules that do not support channel calibration.

NOTE

If channel calibration is not supported in a cell, the message "The NR DU cell does not support channel calibration" will be displayed after the **STR NRDUCELLCHNCALIB** command is executed. Channel calibration is not supported in the following scenarios:

- RF modules do not have the required capability.
- Channel shutdown takes effect.
- The self-contained slot of a LampSite cell contains only one GP symbol.
- The number of pRRUs that meet calibration conditions in a LampSite cell is less than 2. The possible causes are as follows:
 - The number of available pRRUs is less than 2.
 - pRRUs are far away from each other.
 - The number of pRRUs whose transmit power is the same as the configured TRP power is less than 2 in a cell. In a distributed massive MIMO cell, the TRP refers to the master TRP.

32 CSI-RS Ports

If the **NRDUCellCsirs.FR1MaxCellCsirsPortNum** parameter is set to **32PORT**, there are the following requirements:

- For 32T, the **NRDUCellCsirs.CsiPeriod** parameter must be set to at least **SLOT20**.
- For 64T, the **NRDUCellCsirs.CsiPeriod** parameter must be set to at least **SLOT40**, and the **CSIRS_REUSE3_SW** option of the **NRDUCellCsirs.CsiSwitch** parameter must be deselected.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This function does not need to be activated. [Table 4-6](#) describes the required parameters.

Table 4-6 Required parameters

Parameter Name	Parameter ID	Option	Setting Notes
Transmit and Receive Mode	NRDUCellTrp. TxRxMode	None	Set this parameter based on the numbers of TX and RX antennas of an RF module.

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	DL_PMI_SRS_ADAPT_SW	Select this option if SRS- and PMI-based weight adaptation is required.
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	PUSCH_BEAM_DOMAIN_ENH_SW	Select this option if PUSCH beam domain enhancement is required.
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	PUSCH_TIME_DOMAIN_ENH_SW	Select this option if PUSCH time domain enhancement is required.
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	PUSCH_BEAM_SEL_OPT_SW	Select this option if PUSCH beam selection optimization is required.
SRS Pre-SINR Judge Threshold	NRDUCellPdsch.SrsPreSinrJudgeThld	None	Retain the default value.
Fixed Weight Type	NRDUCellPdsch.FixedWeightType	None	Set this parameter based on cell capability.
Channel Calibration Switch	NRDUCellAlgoSwitch.ChnCalibSwitch	INTER_RRU_CHN_CALIB_SW	For a distributed massive MIMO cell served by a LampSite base station, select this option to enable channel calibration so as to ensure the reciprocity between uplink and downlink channels if the DL_PMI_SRS_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter is selected or the NRDUCellPdsch.FixedWeightType parameter is set to SRS_WEIGHT .
NR Board Performance Switch	gNodeBParam.NrBoardPerformanceSw	CHN_MEASURE_CPU_DEC_SW	It is recommended that this option be selected in heavy-load scenarios to reduce the BBP CPU usage.

Parameter Name	Parameter ID	Option	Setting Notes
SRS Algorithm Switch	NRDUCellSrs. SrsAlgoSwitch	SRS_SINR_MEAS_OPT_SW	This option is selected by default for new high-frequency TDD cells. Select this option when SRS SINR measurement optimization is required in other scenarios.
Uplink Rank Algorithm Switch	NRDUCellUplink. UlRankAlgorithmSw	UL_RANK_FAST_DECREASE_SW	Select this option if fast uplink rank reduction based on SRS SINR is required.
SRS-based Weight Validity Period	NRDUCellPdschPrecode. SrsWeightValidityPeriod	None	Use the recommended value.
Downlink Schedule Opt Time Threshold	NRDUCellPdsch. DlSchOptTimeThld	None	It is recommended that this parameter be set to 300 in medium- and heavy-load scenarios (where the PRB usage is greater than 60%). Retain the default value in other scenarios.
PUSCH Performance Switch	NRDUCellPusch. PuschPerformanceSwitch	PUSCH_CE_SINR_LEVEL_ENH_SW	Select this option if enhanced PUSCH SNR level for channel estimation is required.
FR1 Max Cell CSI-RS Port Num	NRDUCellCsirs. FR1MaxCellCsirsPortNum	None	Use the recommended value.
SRS Non-AS Fixed Weight Type	NRDUCellPdschPrecode. SrsNonASFixedWeightType	None	Set it based on the network plan.
Beam Perceive Mode	NRDUCellBeamAlgo. BeamPerceiveMode	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
Weight Algorithm Switch	NRDUCellBeamAlgo.WeightAlgoSwitch	SRS_WEIGHT_ESTIMATE_SW	Select or deselect this option according to the network plan for a low-frequency TDD cell served by a MetaAAU if both MetaAAU beam sensing and MetaAAU beam adaptation are enabled for the cell. In other scenarios, you are advised to deselect this option.
Weight Algorithm Switch	NRDUCellBeamAlgo.WeightAlgoSwitch	PMI_WEIGHT_OPT_SW	It is recommended that this option be selected when the cell is the only one configured on a low-frequency TDD MetaAAU and the cell is enabled with MetaAAU beam sensing. It is recommended that this option be deselected in other scenarios.
Weight Algorithm Switch	NRDUCellBeamAlgo.WeightAlgoSwitch	OPEN_LOOP_WEIGHT_OPT_SW	It is recommended that this option be selected when the cell is configured on a low-frequency TDD MetaAAU and enabled with MetaAAU beam sensing. It is recommended that this option be deselected in other scenarios.
DL Adaptive Scheduling Time Threshold	NRDUCellDisch.DLAdaptSchTimeThld	None	The value 30 is recommended when packet length-based scheduling optimization is required in NSA and SA hybrid networking scenarios. Use the default value in other scenarios.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3.2 Functions](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

NOTE

Before setting the maximum number of CSI-RS ports for a cell, you are advised to run the **LST NRDUCELLCSIRS** command to query the value of **NRDUCellCsirs.CsirsCellResourceNum** and then set the maximum number to this value.

```
//Setting the TX/RX mode of a cell
MOD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=64T64R;
//Configuring downlink SRS- and PMI-based weight adaptation (only for low-frequency TDD)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_PMI_SRS_ADAPT_SW-1;
//Configuring the SRS-based weight validity period
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, SrsWeightValidityPeriod=MS400;
//Configuring other coverage enhancement functions
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
FullChannelCovEnhSwitch=PUSCH_BEAM_DOMAIN_ENH_SW-1&PUSCH_TIME_DOMAIN_ENH_SW-1&PUSCH_BEAM_SEL_OPT_SW-1;
//Setting the SRS Pre-SINR Judge Threshold parameter
MOD NRDUCELLPDSCH: NrDuCellId=0, SrsPreSinrJudgeThld=-100;
//Optional) Configuring packet length-based scheduling optimization in medium- or heavy-load scenarios (PRB usage > 60%)
MOD NRDUCELLPDSCH: NrDuCellId=0, DLSchOptTimeThld=300;
//Optional) Setting parameters when packet length-based scheduling optimization is used in NSA and SA hybrid networking scenarios
MOD NRDUCELLDLSCH: NrDuCellId=0, DLSchAdaptSchTimeThld=30;

//Enabling CPU usage reduction for channel measurement
MOD GNODEBPARAM: NrBoardPerformanceSw=CHN_MEASURE_CPU_DEC_SW-1;
//Configuring enhanced functions
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_SINR_MEAS_OPT_SW-1;
MOD NRDUCELLULRANK: NrDuCellId=0, UIRankAlgoSw=UL_RANK_FAST_DECREASE_SW-1;
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=PUSCH_CE_SINR_LEVEL_ENH_SW-1;
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, SrsNonASFixedWeightType=PMI_WEIGHT;
//Configuring the maximum number of CSI-RS ports for the cell
MOD NRDUCELLCSIRS: NrDuCellId=0, FR1MaxCellCsirsPortNum=8PORT;
//Configuring the MetaAAU beam sensing function
MOD NRDUCELLBEAMALGO: NrDuCellId=0, BeamPerceiveMode=DISTRIBUTED_MODE;
//Configuring the SRS-based weight estimation for downlink large-packet UEs function
MOD NRDUCELLBEAMALGO: NrDuCellId=0, WeightAlgoSwitch=SRS_WEIGHT_ESTIMATE_SW-1;
//Configuring the PMI-based weight optimization function
MOD NRDUCELLBEAMALGO: NrDuCellId=0, WeightAlgoSwitch=PMI_WEIGHT_OPT_SW-1;
//Configuring the open-loop weight optimization function
MOD NRDUCELLBEAMALGO: NrDuCellId=0, WeightAlgoSwitch=OPEN_LOOP_WEIGHT_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink SRS- and PMI-based weight adaptation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_PMI_SRS_ADAPT_SW-0;
//Disabling other coverage enhancement functions
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
FullChannelCovEnhSwitch=PUSCH_BEAM_DOMAIN_ENH_SW-0&PUSCH_TIME_DOMAIN_ENH_SW-0&PUSCH_BEAM_SEL_OPT_SW-0;
//Disabling packet length-based scheduling optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DLSchOptTimeThld=0;
//Disabling CPU usage reduction for channel measurement
MOD GNODEBPARAM: NrBoardPerformanceSw=CHN_MEASURE_CPU_DEC_SW-0;
//Disabling enhanced functions
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoSwitch=SRS_SINR_MEAS_OPT_SW-0;
MOD NRDUCELLULRANK: NrDuCellId=0, UIRankAlgoSw=UL_RANK_FAST_DECREASE_SW-0;
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=PUSCH_CE_SINR_LEVEL_ENH_SW-0;
//Disabling the SRS-based weight estimation for downlink large-packet UEs function
MOD NRDUCELLBEAMALGO: NrDuCellId=0, WeightAlgoSwitch=SRS_WEIGHT_ESTIMATE_SW-0;
//Disabling the PMI-based weight optimization function
```



```
MOD NRDUCELLBEAMALGO: NrDuCellId=0, WeightAlgoSwitch=PMI_WEIGHT_OPT_SW-0;  
//Disabling the open-loop weight optimization function  
MOD NRDUCELLBEAMALGO: NrDuCellId=0, WeightAlgoSwitch=OPEN_LOOP_WEIGHT_OPT_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

NR cells are multi-antenna cells by default. The basic functions of MIMO do not require additional activation verification.

4.4.3 Network Monitoring

NR cells are multi-antenna cells by default. The basic functions of MIMO do not require additional network monitoring.

You can observe the service performance of a cell by observing the following KPIs related to average UE throughput:

- **User Uplink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**

5 SU-MIMO

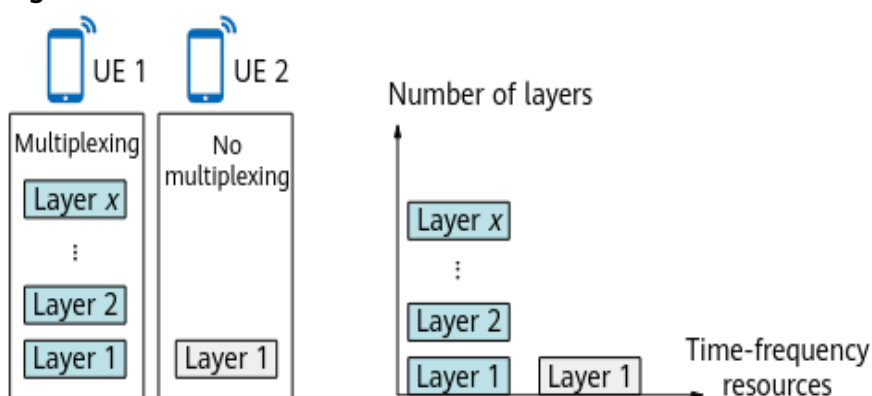
5.1 Principles

5.1.1 SU-MIMO Basics

Overview

Single-user MIMO (SU-MIMO) is a multi-antenna technique that implements spatial multiplexing of time-frequency resources for a single UE on the PUSCH and PDSCH. It enables multi-layer data transmission, as shown in [Figure 5-1](#), and increases the single UE peak rate in both uplink and downlink.

Figure 5-1 SU-MIMO



Uplink SU-MIMO

Uplink SU-MIMO allows multi-layer transmission in the uplink. The maximum number of PUSCH multiplexing layers for a single UE varies as follows:

- For low-frequency TDD cells with 32 or more RX antennas on the gNodeB side and high-frequency TDD cells, the maximum number of PUSCH multiplexing layers for a single UE is equal to the minimum among the number of gNodeB RX antennas, the number of UE TX antennas, and the value of **NRDUCellPusch.MaxMimoLayerCnt**.

- For low-frequency TDD cells with eight or fewer RX antennas on the gNodeB side, the maximum number of PUSCH multiplexing layers for a single UE is equal to the smaller one between the number of gNodeB RX antennas and the number of UE TX antennas.

Table 5-1 lists the maximum number of PUSCH layers supported for a single UE, using a 2T UE and the value **LAYER_4** for **NRDUCellPusch.MaxMimoLayerCnt** as an example.

Table 5-1 Maximum number of PUSCH layers for a single UE

Number of gNodeB RX Antennas	Number of UE TX Antennas	Value of NRDUCellPusch.MaxMimoLayerCnt	Maximum Number of PUSCH SU-MIMO Layers
64R	2T	LAYER_4	2
32R	2T	LAYER_4	2
8R	2T	-	2
4R	2T	-	2
2R	2T	-	2

This function is controlled by the **UL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch.SuMimoMultipleLayerSw** parameter. The number of gNodeB RX antennas is specified by the **NRDUCellTrp.TxRxMode** parameter. The RF module hardware capability needs to be considered during the configuration.

If uplink scheduling enhancement (controlled by the **UL_SCHEDULE_ENH_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter) is enabled, the gNodeB enhances uplink scheduling based on whether SU-MIMO or MU-MIMO is enabled in the cell. For details, see [6.1.4 PUSCH MU-MIMO Enhancements](#).

Downlink SU-MIMO

Downlink SU-MIMO allows multi-layer transmission in the downlink. The maximum number of PDSCH multiplexing layers for a single UE varies as follows:

- For low-frequency TDD cells with 32 or more TX antennas on the gNodeB side and high-frequency TDD cells, the maximum number of PDSCH multiplexing layers for a single UE is equal to the minimum among the number of gNodeB TX antennas, the number of UE RX antennas, and the downlink maximum number of MIMO layers. The downlink maximum number of MIMO layers for this channel is specified by the **NRDUCellPdsch.MaxMimoLayerNum** parameter.
- For cells with eight or fewer TX antennas on the gNodeB side, the maximum number of PDSCH multiplexing layers for a single UE is equal to the smaller one between the number of gNodeB TX antennas and the number of UE RX antennas.

Table 5-2 lists the maximum number of PDSCH layers supported for a single UE, using a 4R UE with the downlink maximum number of MIMO layers being eight as an example.

Table 5-2 Maximum number of PDSCH layers for a single UE

Number of gNodeB TX Antennas	Number of UE RX Antennas	Value of <i>NRDUCellPdsch.MaxMimoLayerNum</i>	Maximum Number of PDSCH SU-MIMO Layers
64T	4R	LAYER_8	4
32T	4R	LAYER_8	4
8T	4R	-	4
4T	4R	-	4
2T	4R	-	2

This function is controlled by the **DL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch.SuMimoMultipleLayerSw** parameter. The number of gNodeB TX antennas is specified by the **NRDUCellTrp.TxRxMode** parameter. The RF module hardware capability needs to be considered during the configuration.

5.1.2 Downlink SU-MIMO Enhancements

Downlink SU-MIMO can be optimized from multiple dimensions such as weight, scheduling, and power to increase the average downlink UE throughput.

5.1.2.1 Weight Optimization

Table 5-3 describes the weight optimization functions.

Table 5-3 Weight optimization functions

Function Name	Function Switch	Function Description
Downlink channel prediction for mobility enhancement	DL_SU_PREDICT_MOBILITY_ENH_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayerSw parameter	For low-frequency TDD massive MIMO cells, assume that 1T4R and 2T4R antenna selection UEs in movement use SRS-based weights and the wideband SRS period is a short period. Then, channel information in the next SRS period can be predicted based on historical and current channel information. This increases the accuracy of SRS-based weights and the average downlink throughput of moving UEs.

Function Name	Function Switch	Function Description
Downlink single-UE mobility enhancement in intra-base-station CoMP	DL_SU_INTRA_CO MP_MOB_ENH_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayerSw parameter	For low-frequency TDD massive MIMO cells, assume that 1T4R or 2T4R antenna selection UEs in movement use SRS-based weights, the wideband SRS period is a short period, and DL CoMP is used for the UEs. Then, channel information in the next SRS period can be predicted in the cooperating cells of the UEs based on the historical and current channel information. This increases the accuracy of SRS-based weights and the average downlink throughput of moving UEs in their cooperating cells. This function takes effect for 1T4R antenna selection UEs in NSA networking and 2T4R antenna selection UEs in SA networking.
SRS-based precoding information update delay shortening	NRDUCellPdschPrecode.SrsPrecodeUpdateDelayReduPol	For 1T4R and 2T4R antenna selection UEs moving in low-frequency TDD cells with at least 8T8R, the SRS-based weight update delay can be shortened to increase the weight accuracy and average downlink UE throughput. SRS-based precoding information update delay shortening can be enabled only when either of the following conditions is met: - For a cell serving mobile 1T4R antenna selection UEs, the slot assignment (specified by the NRDUCell.SlotAssignment parameter) is set to 8_2_DDDDDDDSUU, 8_2_DDDSUUDDDD, or 4_1_DDDSU. - For a cell serving mobile 2T4R antenna selection UEs, the slot assignment is set to 4_1_DDDSU.
SRS-based precoding optimization	SRS_PRECODE_OPT_SW option of the NRDUCellPdschPrecode.DlPrecodeOptOnSrsDtxSw parameter	For low-frequency TDD cells, this function improves the downlink SRS-based weight accuracy for UEs in the DTX state to increase the average downlink UE throughput.

5.1.2.2 Scheduling Optimization

Table 5-4 describes the scheduling optimization functions.

Table 5-4 Scheduling optimization functions

Function Name	Function Switch	Function Description
Downlink rank adaptation	DL_RANK_ADAPT_SW option of the NRDUCellPdsch.DLinkAdaptEnhancementSw parameter	For low-frequency TDD cells, the gNodeB can adjust the number of downlink layers to better match the actual UE channel quality and increase the average downlink UE throughput when SRS-based weights are used.
Adaptive DMRS overhead deduction for SU-MIMO	SU_DMRS_OH_ADAPT_DEDUCT_SW option of the NRDUCellDmrs.DlDmrsSwitch parameter	For low-frequency TDD cells, the RB allocation policy can be optimized to increase the average downlink UE throughput.

5.1.2.3 Power Optimization

Table 5-5 describes the power optimization functions.

Table 5-5 Power optimization functions

Function Name	Function Switch	Function Description
Power adjustment for antenna selection UEs	Step 1: DL_SU_MULTI_LAYER_PWR_CTRL_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayerSw parameter Step 2: DL_SU_PWR_CTRL_ENH_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayerSw parameter Step 3: NRDUCellDLAmc.SuMimoPwrCtrlProtectThld	With this function, the power of each layer can be adjusted for antenna selection UEs by performing the following steps: If the option in step 1 is selected and single-DCI scheduling is used for UEs, the gNodeB can adjust the power of each layer based on the channel quality of respective layers to increase the average downlink UE throughput. (DCI is short for downlink control information.) On condition that the option in step 1 is selected, if the option in step 2 is selected and some UEs are in movement, the gNodeB can adjust the power of each layer based on the MCS to further increase the average downlink throughput of moving UEs. In addition, the parameter in step 3 can be set so that an inter-layer power control protection procedure can be triggered when the inter-layer power allocation difference of a UE is greater than the threshold indicated by this parameter. This increases the average downlink throughput of UEs with a large inter-layer power allocation difference.

5.1.2.4 Other Optimizations

For low-frequency TDD massive MIMO cells, precoding enhancement can be used for 1T4R or 2T4R non-antenna-selection UEs to increase the average downlink throughput of these UEs. This "precoding enhancement for non-antenna-selection UEs" function is controlled by the **DL_NAS_PRECODE_ENH_SW** option of the **NRDUCellPdschPrecode.SuNonAsPrecodeSw** parameter. The function involves the following parameters:

- **NRDUCellPdschPrecode.NonAsPrecodeEnhMcsThld**, which specifies the MCS index threshold for this function to take effect
- **NRDUCellPdschPrecode.Srs1TNonAsPrecodeEnhCoeff** and **NRDUCellPdschPrecode.Srs2TNonAsPrecodeEnhCoeff**, which specify the correction coefficients of precoding enhancement for 1T4R and 2T4R UEs, respectively

5.2 Network Analysis

5.2.1 Benefits

SU-MIMO provides the following benefits:

- Uplink SU-MIMO
If a UE supports uplink data transmission over N layers, its uplink peak rate is theoretically N times that of data transmission over a single layer.
- Downlink SU-MIMO
If a UE supports downlink data transmission over N layers, its downlink peak rate is theoretically N times that of data transmission over a single layer.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

5.3 Requirements

5.3.1 Licenses

The license for SU-MIMO Multiple Layers has been purchased.

RAT	Feature ID	Feature Name	Model	Sales Unit
Low-frequency TDD High-frequency TDD	FOFD-010020	SU-MIMO Multiple Layers	NR0S0PRE UM00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

For low-frequency gNodeBs with at least 32T32R, the license for the number of spatial multiplexing layers is also required. The maximum number cannot exceed the licensed number. One license unit corresponds to two layers. For details about licensing rules, see *License Control Item Lists*.

RAT	Model	Description	Sales Unit
Low-frequency TDD	NR0S0DLEPU00	Massive MIMO DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell
Low-frequency TDD	NR0S0ULEPU00	Massive MIMO UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell

For high-frequency TDD, the license for the number of spatial multiplexing layers is also required. The maximum number cannot exceed the licensed number. One license unit corresponds to two layers. For details about licensing rules, see *License Control Item Lists*.

RAT	Model	Description	Sales Unit
High-frequency TDD	NR0SMMWDLE00	mmWave DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per 100MHz per Cell
High-frequency TDD	NR0SMMWULE00	mmWave UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per 100MHz per Cell

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Downlink channel prediction for mobility enhancement	DL_SU_PREDICT_MOBILITY_ENH_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayerSw parameter	<i>MIMO (TDD)</i>	Downlink single-UE mobility enhancement in intra-base-station CoMP requires that downlink channel prediction for mobility enhancement be enabled.
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	Downlink single-UE mobility enhancement in intra-base-station CoMP requires that intra-base-station DL CoMP be enabled.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO cell	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Downlink channel prediction for mobility enhancement is not supported in distributed massive MIMO cells.

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Reducing the number of MIMO layers	UE_MIMO_LAYER_REDUCTION_SW option of the NRCCellAlgoSwitch.UeAssistanceInfoSwitch parameter and OVERHEATING_ASSISTANCE_INFO_SW option of the NRCCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	When both SU-MIMO and the function of reducing the number of MIMO layers are enabled, the value of the N.ChMeas.MIMO.DL.Transmission.Layer counter may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	UEs performing low latency and high reliability services support only single-stream rank 1.
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Enabling the power saving BWP function after enabling MIMO may decrease the number of layers and BLER, increase the number of scheduling times, and decrease the average cell throughput and UE throughput.
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Uplink SU-MIMO and downlink SU-MIMO are not supported for UEs working on BWP2 for power saving. Enabling time-domain power saving BWP after enabling uplink SU-MIMO or downlink SU-MIMO may cause the following impacts: • The average number of downlink layers, maximum number of downlink layers, and downlink BLER change. • The average number of uplink layers, maximum number of uplink layers, and uplink BLER change. • The numbers of uplink and downlink scheduling times decrease. • The average cell throughput and average UE throughput change. For the definitions of the preceding indicators, see <i>UE Power Saving</i>.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-domain high-resolution power control	FREQ_HIGH_RESOLUT_PWR_CTRL_SW option of the NRDUCellPdsch.DIPdsc hAlgoSwitch parameter	<i>MIMO (TDD)</i>	After power reduction for CCUs takes effect, power is no longer allocated to CCUs for which frequency-domain high-resolution power control also takes effect.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DIRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, the function of precoding enhancement for non-antenna-selection UEs does not take effect for the user.
Low-frequency TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Coverage performance of low power consumption mode may deteriorate when both low power consumption mode and precoding enhancement for non-antenna-selection UEs are enabled and a customized backup power saving policy is used in low power consumption mode.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Precoding enhancement for non-antenna-selection UEs: All NR TDD-capable 32T32R and 64T64R RF modules support this function.

All NR TDD-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

- Uplink SU-MIMO is not supported in SUL cells (with **NRDUCell.DuplexMode** set to **CELL_SUL**).
- Uplink SU-MIMO is not supported in SDL cells (with **NRDUCell.DuplexMode** set to **CELL_SDL**).
- SRS-based precoding information update delay shortening requires that **NRDUCell.NrDuCellNetworkingMode** be set to **NORMAL_CELL** or **FUSION_CELL**.
- Precoding enhancement for non-antenna-selection UEs requires that **NRDUCell.NrDuCellNetworkingMode** be set to **NORMAL_CELL**.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-6](#) describes the parameters used for SU-MIMO activation.

Table 5-6 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
SU-MIMO Multiple Layers Switch	NRDUCellAlgoSwitch.SuMimoMultipleLayerSw	• DL_SU_MU_LTI_LAYER_SW • UL_SU_MU_LTI_LAYER_SW	You are advised to select both options if both downlink SU-MIMO and uplink SU-MIMO are required.
Downlink Maximum MIMO Layer Number Quota	NRDUCellPdsch.MaxMimoLayerNum	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Maximum MIMO Layer Count Quota	NRDUCellPusch.MaxMimoLayerCnt	None	Retain the default value.
DL Link Adaptation Enhancement Switch	NRDUCellPdsch.DLinkAdaptEnhancementSw	DL_RANK_ADAPT_SW	You are advised to select this option if downlink rank adaptation is required.
SU-MIMO Multiple Layers Switch	NRDUCellAlgoSwitch.SuMimoMultipleLayerSw	- DL_SU_MULTI_LAYER_PWR_CTRL_SW - DL_SU_PWR_CTRL_ENH_SW - DL_SU_PREDICT_MOBILITY_ENH_SW - DL_SU_INTRA_COMP_MOB_ENH_SW	- Select the DL_SU_MULTI_LAYER_PWR_CTRL_SW option if step 1 of power adjustment for antenna selection UEs (detailed in Table 5-5) is to be performed. - Select the DL_SU_PWR_CTRL_ENH_SW option if downlink single-UE power control enhancement is required. - Select the DL_SU_PREDICT_MOBILITY_ENH_SW option if downlink channel prediction for mobility enhancement is required. - Select the DL_SU_INTRA_COMP_MOB_ENH_SW option if downlink single-UE mobility enhancement in intra-base-station CoMP is required.
SU-MIMO Power Control Protection Threshold	NRDUCellIDAmc.SuMimoPwrCtrlProtectThld	None	Retain the default value.
DL DMRS Switch	NRDUCellDmrs.DlDmrsSwitch	SU_DMRS_OH_ADAPT_DEDUCT_SW	Select this option if adaptive DMRS overhead deduction for SU-MIMO is required.
DL SRS DTX-based Precode Optimization Switch	NRDUCellPdschPrecode.DlPrecodeOptOnSrsDtxSw	SRS_PRECODE_OPT_SW	Select this option if SRS-based precoding optimization is required.

Parameter Name	Parameter ID	Option	Setting Notes
SRS Precode Update Delay Reduction Policy	NRDUCellPdschPrecode.SrsPrecodeUptDelayReduPol	None	Retain the default value.
SU Non-AS UE Precode Switch	NRDUCellPdschPrecode.SuNonAsPrecodeSw	DL_NAS_PRECODE_ENH_SW	Select this option if precoding enhancement for non-antenna-selection UEs is required.
Non-AS UE Precode Enhancement MCS Thld	NRDUCellPdschPrecode.NonAsPrecodeEnhMcsThld	None	Retain the default value.
SRS 1T Non-AS Precode Enhance Coefficient	NRDUCellPdschPrecode.Srs1TNonAsPrecodeEnhCoeff	None	Retain the default value.
SRS 2T Non-AS Precode Enhance Coefficient	NRDUCellPdschPrecode.Srs2TNonAsPrecodeEnhCoeff	None	Retain the default value.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling uplink and downlink SU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SuMimoMultipleLayerSw=DL_SU_MULTI_LAYER_SW-1&UL_SU_MULTI_LAYER_SW-1;
//Setting the maximum number of spatial multiplexing layers for uplink SU-MIMO
MOD NRDUCELLPUSCH: NrDuCellId=0, MaxMimoLayerCnt=LAYER_2;
//Setting the maximum number of spatial multiplexing layers for downlink SU-MIMO
MOD NRDUCELLPDSCH: NrDuCellId=0, MaxMimoLayerNum=LAYER_DEFAULT;
//Configuring downlink SU-MIMO enhancements
MOD NRDUCELLPDSCH: NrDuCellId=0, DLinkAdaptEnhancementSw=DL_RANK_ADAPT_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_DL_JT_SW-1,
SuMimoMultipleLayerSw=DL_SU_MULTI_LAYER_PWR_CTRL_SW-1&DL_SU_PWR_CTRL_ENH_SW-1&DL_SU_PREDICT_MOBILITY_ENH_SW-1&DL_SU_INTRA_COMP_MOB_ENH_SW-1;
MOD NRDUCELLDLAMC: NrDuCellId=0, SuMimoPwrCtrlProtectThld=100;
```

```
MOD NRDUCELLDMRS: NrDuCellId=0, DIDmrsSwitch=SU_DMRS_OH_ADAPT_DEDUCT_SW-1;  
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, DIPrecodeOptOnSrsDtxSw=SRS_PRECODE_OPT_SW-1,  
SrsPrecodeUptDelayReduPol=OFF,  
SuNonAsPrecodeSw=DL_NAS_PRECODE_ENH_SW-1,NonAsPrecodeEnhMcsThld=14,  
Srs1TNonAsPrecodeEnhCoeff=9,Srs2TNonAsPrecodeEnhCoeff=3;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink SU-MIMO enhancements  
MOD NRDUCELLPDSCH: NrDuCellId=0, DLinkAdaptEnhancementSw=DL_RANK_ADAPT_SW-0;  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CompSwitch=INTRA_GNB_DL_JT_SW-0,  
SuMimoMultipleLayerSw=DL_SU_MULTI_LAYER_PWR_CTRL_SW-0&DL_SU_PWR_CTRL_ENH_SW-0&DL_SU_PR  
EDICT_MOBILITY_ENH_SW-0&DL_SU_INTRA_COMP_MOB_ENH_SW-0;  
MOD NRDUCELLDMRS: NrDuCellId=0, DIDmrsSwitch=SU_DMRS_OH_ADAPT_DEDUCT_SW-0;  
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, DIPrecodeOptOnSrsDtxSw=SRS_PRECODE_OPT_SW-0,  
SuNonAsPrecodeSw=DL_NAS_PRECODE_ENH_SW-0;  
//Disabling uplink and downlink SU-MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
SuMimoMultipleLayerSw=DL_SU_MULTI_LAYER_SW-0&UL_SU_MULTI_LAYER_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Check whether SU-MIMO has taken effect. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > User Performance Monitoring > User Common Monitoring**, and check the number of layers corresponding to each codeword, as shown in [Table 5-7](#).

Table 5-7 SU-MIMO activation verification

Function	Monitoring Item	Description
Uplink SU-MIMO	Code0 UL Average Layer Num	This function has taken effect if the following condition is met: $1 < \text{Code0 UL Average Layer Num} \leq \min(\text{Number of gNodeB RX antennas, Number of UE TX antennas})$.
Downlink SU-MIMO	- Code0 DL Average Layer Num - Code1 DL Average Layer Num	This function has taken effect if the following condition is met: $1 < (\text{Code0 DL Average Layer Num} + \text{Code1 DL Average Layer Num}) \leq \min(\text{Number of gNodeB TX antennas, Number of UE RX antennas})$.

 NOTE

In the **User Common Monitoring** dialog box, you are advised to set **UEID Type** to **Random Value** or **STMSI** in NSA networking and set **UEID Type** to **5G-Random Value** or **5G-STMSI** in SA networking.

To further observe function effects, you can use the following indicators. These indicators apply to both SU-MIMO and MU-MIMO. They can be used to observe whether SU-MIMO has taken effect only when MU-MIMO is not enabled.

- Maximum number of downlink layers on a PRB in a cell = **N.ChMeas.MIMO.DL.Transmission.Layer.Max**
- Maximum number of uplink layers on a PRB in a cell = **N.ChMeas.MIMO.UL.Trans.Layer.Max**

5.4.3 Network Monitoring

Observing Average UE Throughput

Observe the average UE throughput increases of uplink SU-MIMO and downlink SU-MIMO using the following KPIs:

- **User Uplink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**

Observing Single-UE Throughput

Observe the single-UE throughput increases using real-time monitoring items on the MAE-Access.

- For uplink SU-MIMO, choose **User Performance Monitoring > User Common Monitoring > Uplink MAC Throughput**. Compare the uplink MAC throughput before and after the function is enabled.
- For downlink SU-MIMO, choose **User Performance Monitoring > User Common Monitoring > Downlink MAC Throughput**. Compare the downlink MAC throughput before and after the function is enabled.

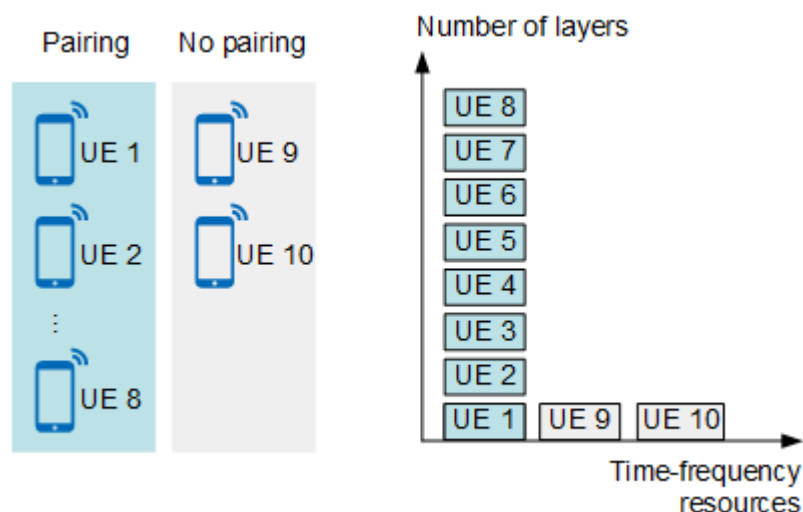
6 MU-MIMO (Low-Frequency TDD)

6.1 Principles

6.1.1 MU-MIMO Basics

Multi-user MIMO (MU-MIMO) implements spatial multiplexing of time-frequency resources for multiple UEs in both uplink and downlink. When multiple UEs share the same time-frequency resources, as shown in [Figure 6-1](#), the more orthogonal the channels between UEs, the smaller the interference and the higher the cell capacity and spectral efficiency in both uplink and downlink.

Figure 6-1 MU-MIMO



Spatial multiplexing for multiple UEs works on the PDSCH, PDCCH, and PUSCH. The corresponding MU-MIMO functions are as follows:

- PDSCH MU-MIMO

This function is controlled by the **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter.

- PDCCH MU-MIMO
This function is controlled by the **PDCCH_MU_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter.
- PUSCH MU-MIMO
This function is controlled by the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter.

When PDSCH, PDCCH, or PUSCH MU-MIMO is enabled, the actual maximum number of MU-MIMO layers for the corresponding channel is determined by both a parameter and the number of gNodeB antennas, as described in [Table 6-1](#). The parameters for these channels are as follows:

- PDSCH: **Downlink Maximum MIMO Layer Number Quota**
(**NRDUCellPdsch.MaxMimoLayerNum**)
- PDCCH: **Maximum Pairing Layer Number**
(**NRDUCellPdcch.MaxPairLayerNum**)
- PUSCH: **Uplink Maximum MIMO Layer Count Quota**
(**NRDUCellPusch.MaxMimoLayerCnt**)

Table 6-1 Actual maximum number of spatial multiplexing layers in MU-MIMO

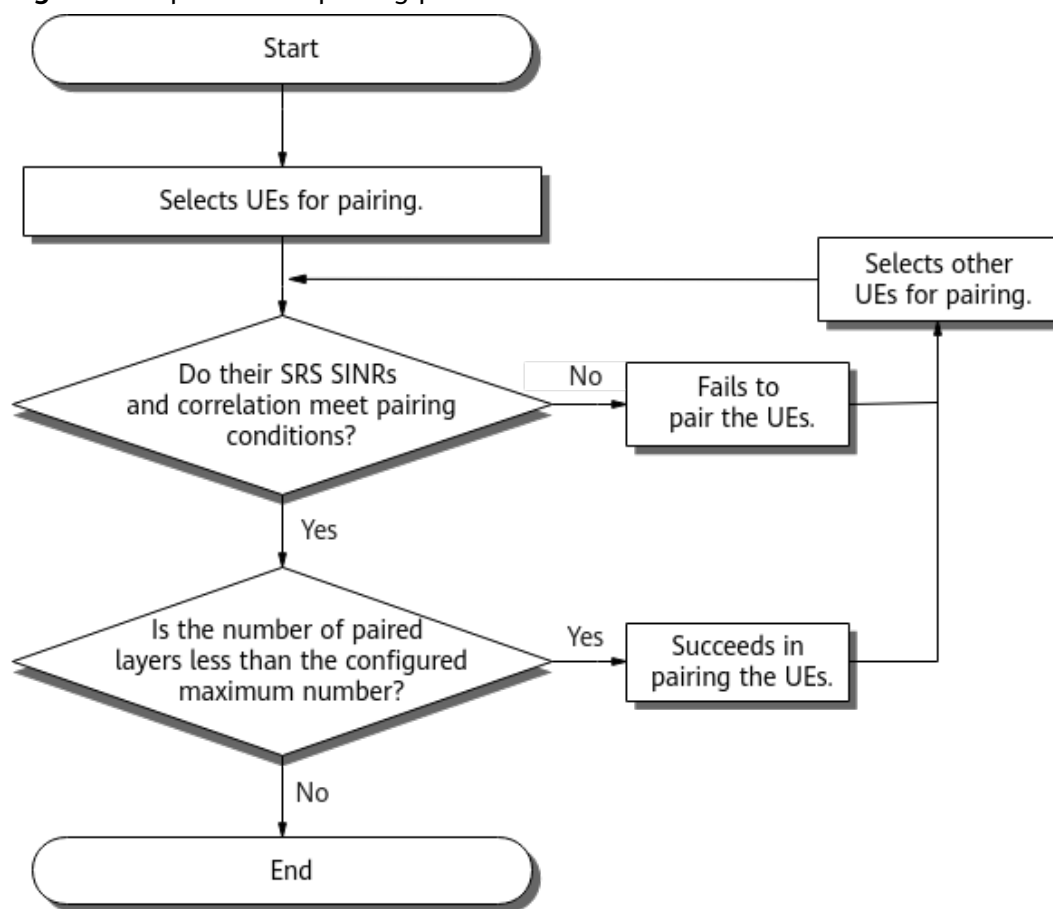
RAT	Number of gNodeB Antennas	Maximum Number of PDSCH MU-MIMO Layers	Maximum Number of PDCCH MU-MIMO Layers	Maximum Number of PUSCH MU-MIMO Layers
Low-frequency TDD	64T64R	Min(Downlink Maximum MIMO Layer Number Quota, 16)	Min(Maximum Pairing Layer Number, 4)	Min(Uplink Maximum MIMO Layer Count Quota, 8)
Low-frequency TDD	32T32R	Min(Downlink Maximum MIMO Layer Number Quota, 16)	Min(Maximum Pairing Layer Number, 4)	Min(Uplink Maximum MIMO Layer Count Quota, 8)
Low-frequency TDD	8T8R	4 (independent of Downlink Maximum MIMO Layer Number Quota)	Not supported	4 (independent of Uplink Maximum MIMO Layer Count Quota)

RAT	Number of gNodeB Antennas	Maximum Number of PDSCH MU-MIMO Layers	Maximum Number of PDCCH MU-MIMO Layers	Maximum Number of PUSCH MU-MIMO Layers
Note: - The number of gNodeB antennas is specified by the NRDUCellTrp.TxRxMode parameter. The RF module hardware capability needs to be considered during the configuration. MU-MIMO is not supported in scenarios where the number of gNodeB antennas is not one in the preceding table. - It is recommended that the Downlink Maximum MIMO Layer Number Quota parameter be set to a value less than or equal to 16 for a 64T64R gNodeB. If a larger value is used, more units of NR0S0DLEPU00 Massive MIMO DL 2-Layers Extended Processing Unit License (NR) will be consumed. For details about the consumption rules, see 6.3.1 Licenses.				

6.1.2 MU-MIMO Spatial Multiplexing Procedures

The spatial multiplexing procedures of PDSCH MU-MIMO and PUSCH MU-MIMO can be considered as UE pairing procedures. The gNodeB selects appropriate UEs for pairing according to pairing rules and schedules the UEs. The spatial multiplexing procedures of PDSCH MU-MIMO and PUSCH MU-MIMO are the same. [Figure 6-2](#) uses PDSCH MU-MIMO as an example.

Figure 6-2 Spatial multiplexing procedure of PDSCH MU-MIMO



The signal quality of SRS and the correlation between UEs need to be considered during UE pairing.

- If the signal quality of SRS is good (for example, the SINR is high and the signal fluctuation is slight) and the correlation between UEs is low, inter-UE interference can be effectively reduced and the UEs can be paired for MU-MIMO. In this situation, MU-MIMO can fully utilize good channel conditions to increase the system capacity.
- If the signal quality of SRS is poor (for example, the SINR is low or the signal fluctuation is large) or the correlation between UEs is high, inter-UE interference cannot be effectively reduced. In this situation, MU-MIMO may decrease the cell throughput. Therefore, such UEs will not be selected for pairing.

The correlation between UEs is inversely related to the distance between them; as the distance between UEs decreases, their correlation intensifies, and as the distance increases, their correlation diminishes. The correlation between UEs can be measured by spatial isolation or channel correlation. Spatial isolation is measured by the RSRP difference between the optimal beams of UEs. Channel correlation is calculated based on uplink feedback and can be considered as channel similarity.

The pairing rules of PDSCH MU-MIMO can be set using parameters.

- The rules described in [Table 6-2](#) apply when SRS SINRs of to-be-paired UEs are evaluated.

Table 6-2 Rules for evaluating whether UEs can be paired based on their SRS SINRs

Scenario	Rule	Parameter
A UE initially accesses the network.	If the SRS SINR of the UE is greater than or equal to the SRS PreSINR threshold for downlink MU-MIMO, the UE can participate in pairing; otherwise, it cannot.	The SRS PreSINR threshold for downlink MU-MIMO is specified by the NRDUCellDLMimo.DlMuMimoSrsPreSinrThld parameter.
A UE has accessed the network.	If the SRS SINR of the UE changes, the following rules apply: • If the SRS SINR is greater than the SRS PreSINR threshold for downlink MU-MIMO plus 3 dB, the UE can participate in pairing. • If the SRS SINR is less than the SRS PreSINR threshold for downlink MU-MIMO minus 3 dB, the UE cannot participate in pairing. • If the SRS SINR is in the range of [SRS PreSINR threshold for downlink MU-MIMO – 3 dB, This threshold + 3 dB], the previous pairing decision result is used.	

- The rules described in [Table 6-3](#) apply when isolation and correlation are evaluated if both are configured as the factors for downlink MU-MIMO grouping (by setting **NRDUCellPdsch.DlMuMimoGroupMode** to **ISOLATION_CORRELATION**).

Table 6-3 Rules for evaluating whether UEs can be paired based on their isolation and correlation

Scenario	Rule	Parameter
PMI-based weights are used for a UE.	The UE can be paired with another UE if the spatial isolation between them is greater than the spatial isolation threshold for downlink MU-MIMO using PMI-based weights.	The spatial isolation threshold for downlink MU-MIMO using PMI-based weights is specified by the NRDUCellPdsch.DlPmiMuMimoSpacelsThld parameter.

Scenario	Rule	Parameter
SRS-based weights are used for a UE.	• The UE can be paired with another UE if the spatial isolation between them is greater than the spatial isolation threshold for downlink MU-MIMO using SRS-based weights. • The UE can also be paired with another UE if the spatial isolation between them is less than or equal to the spatial isolation threshold for downlink MU-MIMO using SRS-based weights and the channel correlation between them is less than the channel isolation threshold for downlink MU-MIMO.	• The spatial isolation threshold for downlink MU-MIMO using SRS-based weights is specified by the NRDUCellPdsch.DlSrsMuMimoSpatialIsoThld parameter. • The channel isolation threshold for downlink MU-MIMO is specified by the NRDUCellPdsch.DlSrsMuMimoChannelIsoThld parameter.
Other scenarios	Pairing is not allowed.	None

6.1.3 PDSCH MU-MIMO Enhancements

PDSCH MU-MIMO can be enhanced from multiple dimensions, such as the number of paired layers and scheduling, to improve the post-pairing performance and cell throughput.

6.1.3.1 Optimization of the Number of Paired Layers

[Table 6-4](#) describes the functions for optimization of the number of paired layers.

Table 6-4 Functions for optimization of the number of paired layers

Function Name	Function Switch	Function Description
Downlink SRS-based MU-MIMO rank mode configuration	NRDUCellDLRank.DISrsMuMimoRankMode	For low-frequency TDD cells with at least 8T8R, the rank for a UE using SRS-based weights to take part in pairing can be determined using different methods, depending on the setting of this parameter: • If this parameter is set to RI_BASED_MU_RANK, the rank for a UE is adaptively selected based on the RI reported by the UE. • If this parameter is set to FIX_MU_RANK, the rank for a UE is the downlink MU-MIMO rank configured for UEs using SRS-based weights^a. The involved parameter is as follows: a: The downlink MU-MIMO rank for UEs using SRS-based weights is specified by the NRDUCellDLRank.DISrsMuMimoRank parameter.
Downlink PMI-based MU rank selection	NRDUCellDLRank.DIPmiMuMimoRank	For low-frequency TDD cells with at least 8T8R, the effective rank for a UE using PMI-based weights to take part in pairing is the minimum between the RI reported by the UE and the value of this parameter.
Downlink MU pairing optimization in UE gathering scenarios	NRDUCellPdsch.DIMuGatherOptSw	For low-frequency TDD cells with at least 8T8R, the UE pairing policy is optimized so that more large-packet UEs can be paired for MU-MIMO to increase the downlink cell throughput.

Function Name	Function Switch	Function Description
Downlink small-packet group merging	DL_SMALL_PACKET_MERGE_SW option of the NRDUCellDlMimo.DLMuMimoSchSupplementSw parameter	For low-frequency TDD cells with at least 4T4R, small-packet UE groups can be merged in scenarios where small-packet UEs and large-packet UEs need to be paired. This reduces the interference between paired UEs but may decrease the number of downlink paired layers.
Downlink small-packet pairing enhancement	DL_SMALL_PACKET_PAIR_ENH_SW option of the NRDUCellDlMimo.DLMuMimoSchSupplementSw parameter	For low-frequency TDD cells with at least 8T8R, more small-packet UEs can be paired based on their traffic volumes.
Downlink low time-domain correlation UE pairing	NRDUCellPdsch.DlLowTimeCorrMuSw	For low-frequency TDD massive MIMO cells, this function allows UEs with drastic channel quality fluctuation to be selected for pairing.

6.1.3.2 Scheduling Optimization

Table 6-5 describes the scheduling optimization functions.

Table 6-5 Scheduling optimization functions

Optimization Method	Function Name	Function Switch	Function Description
Spectral efficiency optimization	Downlink MIMO MCS optimization policy	NRDUCellPdsch.DlMimoMcsOptPolicy	For low-frequency TDD cells in downlink MU-MIMO and CoMP scenarios, the MCS index selection optimization policy can be selected based on the value of this parameter.
	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO	DL_SE_SU_MU_MIMO_ADAPT_SW option of the NRDUCellDlMimo.DLMuMimoSchSupplementSw parameter	For low-frequency TDD cells with at least 8T8R, if the spectral efficiency gain of MU-MIMO compared with SU-MIMO is less than the threshold indicated by the NRDUCellDlMimo.DlMuBackToSuSeThld parameter, SU-MIMO is selected.

Optimization Method	Function Name	Function Switch	Function Description
	Downlink MU-MIMO SINR optimization	NRDUCellDLAmc.DLMuMimoSirscaleFactor	For low-frequency TDD massive MIMO cells, the post-pairing SINRs of antenna selection UEs with large delay spread or large delay-spread change over time can be optimized based on information such as the channel conditions of UEs and the correlation between UEs so that MCS indexes can be selected more accurately for transmission. LampSite base stations do not support this function.
	Downlink MCS index table adaptation enhancement	DL_MCS_TABLE_A DAPT_ENH_SW option of the NRDUCellDIMimo.DLMuMimoSch SupplementSw parameter	For low-frequency TDD cells with at least 8T8R, the conditions for switching between MCS index tables are optimized to increase the downlink UE throughput.
	Downlink mobile MU-MIMO enhancement	DL_MOBILE_MU_ENH_SW option of the NRDUCellAlgoSwitch.UeFeatureBasedPairEnhSw parameter	For low-frequency TDD massive MIMO cells, the MCS indexes of paired UEs in movement can be adjusted based on their CQIs. If the signal quality of SRS is good, the MCS indexes can better match the actual channel quality, which increases the throughput of moving UEs. If the signal quality of SRS is poor, the cell throughput may decrease when channel estimation is inaccurate and channel quality fluctuates sharply. LampSite base stations do not support this function.

Optimization Method	Function Name	Function Switch	Function Description
Resource allocation optimization	Downlink MU-MIMO RB number estimation	NRDUCellDmrs.DIMuEstRbPolicy	For low-frequency TDD cells with at least 8T8R, the required DMRS resources are accurately determined during RB resource allocation in MU-MIMO pairing, thereby reducing RB resource consumption.
Priority optimization	Scheduling priority optimization in heavy-load scenarios	HEAVY_LOAD_SC H_PRI_OPT_SW option of the NRDUCellDISch.DISchAlgoSwitch parameter	For low-frequency TDD cells with at least 8T8R, the average downlink UE throughput can be increased by adjusting the scheduling priorities of UEs in heavy-load scenarios.

6.1.3.3 Power Optimization

MU-MIMO power allocation: This function is controlled by the **NRDUCellPdsch.MuMimoPwrAllocSwitch** parameter. For low-frequency TDD massive MIMO cells, the power of each layer can be adjusted for two UEs in rank-3 or rank-4 scheduling based on the signal quality of the respective layer. While ensuring the performance of the first two layers, this function improves the performance of the other layers and further increases the downlink cell throughput.

6.1.3.4 Other Optimizations

Table 6-6 describes the other optimization functions.

Table 6-6 Other optimization functions

Function Name	Function Switch	Function Description
Configuration of the downlink PRB usage threshold for downlink MU-MIMO	NRDUCellDIMimo.DIMuPrbUsageThld	For low-frequency TDD cells with at least 8T8R, downlink MU-MIMO can be triggered for a cell based on the measured downlink PRB usage of the cell.

Function Name	Function Switch	Function Description
Beam number threshold for PMI-based weight UEs to be selected for downlink MU-MIMO pairing	NRDUCellDlMimo.DlMuPmiBeamNumThld	For low-frequency TDD cells with at least 8T8R, a UE using PMI-based weights is not selected for pairing if the number of beams maintained for the UE exceeds the value of NRDUCellDlMimo.DlMuPmiBeamNumThld . This increases the downlink cell throughput.
Precoding-based interference suppression	NRDUCellPdschPrecodingInterfSupprValue	For low-frequency TDD cells, this function adjusts the interference suppression value for paired UEs in downlink MU-MIMO to increase the downlink cell throughput.
Pairing delay spread optimization	NRDUCellPdsch.DlSrsMuMimoFreqSelThld	For low-frequency TDD cells, this function controls the delay spread threshold for downlink MU-MIMO UEs to enter the pairing procedure so that UEs whose delay spread is less than or equal to this threshold are more likely to enter the pairing procedure, increasing the downlink UE throughput.

6.1.4 PUSCH MU-MIMO Enhancements

PUSCH MU-MIMO can be enhanced from multiple dimensions, such as the number of paired layers and scheduling, to improve the post-pairing UE throughput.

6.1.4.1 Optimization of the Number of Paired Layers

[Table 6-7](#) describes the functions for optimization of the number of paired layers.

Table 6-7 Functions for optimization of the number of paired layers

Function Name	Function Switch	Function Description
Pairing of UEs supporting different numbers of front-loaded DMRS symbols in the uplink	DIFF_FLDMRS_MAX_LENGTH_PAIR_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter	For low-frequency TDD cells with at least 8T8R, this function enables UEs supporting different numbers of front-loaded DMRS symbols to be paired for uplink MU-MIMO, increasing the uplink UE throughput. However, the uplink UE throughput may decrease if the proportion of UEs supporting two front-loaded DMRS symbols is large and the number of paired layers in the cell is close to or greater than the maximum number of layers supported by one front-loaded DMRS symbol.
Uplink MU-MIMO cross-slot pairing	MU_MULTI_SLOT_PAIR_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter	For low-frequency TDD massive MIMO cells, this function enables UEs in multiple slots to be paired, but frequency selective scheduling is not performed in uplink heavy-load scenarios. This increases the uplink UE throughput in uplink heavy-load scenarios but may decrease the downlink UE throughput when air interface resources are insufficient. LampSite base stations do not support this function.

6.1.4.2 Scheduling Optimization

Table 6-8 describes the scheduling optimization functions.

Table 6-8 Scheduling optimization functions

Function Name	Function Switch	Function Description
Frequency selective scheduling optimization for large-packet UEs	BIG_PKT_UE_FREQ_SEL_SCH_OPT_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter	For low-frequency TDD cells with at least 8T8R, this function introduces a condition for frequency selective scheduling to take effect. Specifically, frequency selective scheduling is not performed for a UE if the proportion of RBs required by the UE is greater than or equal to the threshold specified by NRDUCellUIMimo.MuFreqSelSchRatioThld . This prevents the RBs intended for large-packet UEs from being truncated by UEs with a higher priority.

Function Name	Function Switch	Function Description
Uplink scheduling enhancement	UL_SCHEDULE_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	For low-frequency TDD cells with at least 8T8R, uplink MU-MIMO pairing based on space domain optimization can take effect. The gNodeB selects a UE with the highest scheduling priority and determines whether the UE can take part in pairing. If the UE meets pairing conditions, the gNodeB searches for UEs that meet space-domain beam pairing requirements and pairing conditions from the remaining UEs to be scheduled. If the number of paired layers does not reach the maximum number after the searching is complete, the gNodeB selects UEs that meet correlation requirements and pairing conditions from the remaining UEs to be scheduled. In addition, the switch for optimal paired RB selection^a is used to specify whether to support correlation-based pairing determination when there are a large number of MU-MIMO UEs and whether to support frequency selective scheduling for MU-MIMO UEs. After completing UE pairing, the gNodeB allocates RBs to UEs. How RBs are allocated depends on the setting of the switch for enhanced uplink interference randomization-based scheduling^b. • If this switch is turned on, the start positions of scheduling for adjacent cells are fixed strictly based on their physical cell identifiers (PCIs), and the cells search for RBs from different start positions and allocate RBs to UEs. This reduces inter-cell interference. • If this switch is turned off, all cells search for RBs from the same start position and allocate RBs. The involved parameters are as follows: a: The switch for optimal paired RB selection is controlled by the OPT_PAIR_RB_SEL_SW option of the

Function Name	Function Switch	Function Description
		NRDUCellULMimo.UlMuMimoAlgoSwitch parameter. b: The switch for enhanced uplink interference randomization-based scheduling is controlled by the UL_INTRF_RANDOMIZED_SCH_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter.
Adaptive switching between MU-MIMO and SU-MIMO based on the frequency selective characteristics of channels	FREQ_SEL_CHARA_C_MU_SU_ADAPT_SW option of the NRDUCellULMimo.UlMuMimoAlgoSwitch parameter	For low-frequency TDD cells with at least 8T8R, this function identifies the frequency selective characteristics of channels for adaptive switching between MU-MIMO and SU-MIMO, which improves the accuracy of pairing decision.

Function Name	Function Switch	Function Description
Pairing optimization for small-packet UEs and pairing optimization for large-packet UEs	SML_PKT_ALIGN_P AIR_OPT_SW or BIG_PKT_ALIGN_P AIR_OPT_SW option of the NRDUCellUMimo. UIMuMimoAlgoSwitch parameter	Large-packet UEs and small-packet UEs can be distinguished based on the RB proportion threshold for identifying large- and small-packet UEs in alignment pairing^a. If the number of RBs required by a UE divided by the number of available PUSCH RBs is less than this threshold, the UE is identified as a small-packet UE. Otherwise, the UE is identified as a large-packet UE. For low-frequency TDD cells with at least 8T8R: • If the switch for alignment pairing optimization for small-packet UEs^b is turned on, a small-packet UE cannot participate in pairing after RB number reduction. In addition, there is a condition for the UE to participate in pairing after RB number increase. Specifically, the UE can participate in alignment pairing after RB number increase only if the number of its occupied RBs, when increased by a configured proportion^c, is greater than or equal to the number of RBs occupied by UEs that are already scheduled. • If the switch for alignment pairing optimization for large-packet UEs^d is turned on, there is a condition for a large-packet UE to participate in pairing after RB number reduction. Specifically, the UE can participate in alignment pairing after RB number reduction only if the number of its occupied RBs, when reduced by a configured proportion^e, is less than or equal to the number of RBs occupied by UEs that are already scheduled. The involved parameters are as follows: a: The RB proportion threshold for identifying large- and small-packet UEs in alignment pairing is specified by the NRDUCellUMimo.AlignPairBigSmlPktRbThld parameter.

Function Name	Function Switch	Function Description
		b: The switch for alignment pairing optimization for small-packet UEs is controlled by the SML_PKT_ALIGN_PAIR_OPT_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter. c: The proportion by which the number of a small-packet UE's RBs is increased for alignment pairing is specified by the NRDUCellUIMimo.AlignPairSmlPktRbExtRatio parameter. d: The switch for alignment pairing optimization for large-packet UEs is controlled by the BIG_PKT_ALIGN_PAIR_OPT_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter. e: The proportion by which the number of a large-packet UE's RBs is reduced for alignment pairing is specified by the NRDUCellUIMimo.AlignPairBigPktRbRedRatio parameter.
AMC optimization for uplink MU-MIMO	MU_AMC_OPT_SW option of the NRDUCellUIMimo.UlAmcAlgoSwitch parameter	For low-frequency TDD cells with at least 8T8R, this function selects MCS indexes for MU-MIMO UEs only in a common outer-loop manner, improving the MCS index selection accuracy.

6.1.4.3 Beam Domain Optimization

Table 6-9 describes the beam domain optimization functions.

Table 6-9 Beam domain optimization functions

Function Name	Function Switch	Function Description
Uplink MU-MIMO beam-domain pairing optimization	MU_BEAM_PAIR_OPT_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter	For low-frequency TDD cells with at least 8T8R, this function improves the accuracy of CEU selection for pairing in the beam domain.

Function Name	Function Switch	Function Description
Uplink MU-MIMO beam-domain pairing prohibition	UL_MU_BEAM_PAIR_FORBID_SW option of the NRDUCellUIMimo.UlMuMimoAlgoSwitch parameter	For low-frequency TDD 8T8R cells, this function prohibits beam-domain pairing and keeps only correlation-based pairing during pairing for PUSCH MU-MIMO. This reduces interference between paired UEs. This function requires uplink scheduling enhancement to be enabled.

6.1.4.4 Other Optimizations

Table 6-10 describes the other optimization functions.

Table 6-10 Other optimization functions

Function Name	Function Switch	Function Description
Correlation-based scheduling for uplink MU-MIMO	NRDUCellUIMimo.UlMuMimoCorrThld	For low-frequency TDD cells with at least 8T8R, UEs can be paired only when the correlation between them is less than or equal to the threshold indicated by this parameter. This reduces the probability that UEs with high correlation are paired incorrectly.
RB-usage-based scheduling for uplink MU-MIMO	NRDUCellUIMimo.MuRbRatioThld	For low-frequency TDD cells with at least 8T8R, PUSCH MU-MIMO can be attempted when the cell RB usage is greater than or equal to the threshold indicated by this parameter.

Function Name	Function Switch	Function Description
SINR-based scheduling for uplink MU-MIMO	NRDUCeLLULMimo.UlMuMimoSinrThld	For low-frequency TDD cells with at least 8T8R, UEs can be paired only when their SINRs are greater than or equal to the threshold indicated by this parameter. This reduces the probability that CEUs are paired incorrectly.
Uplink MU-MIMO in UE gathering scenarios	NRDUCeLLULMimo.UlMuGatherOptGainThld	This function estimates the spectral efficiency that would be achieved with and without UE pairing for uplink MU-MIMO. The base station then uses the estimation result to adaptively determine whether to perform such pairing. Specifically, UEs can be paired for MU-MIMO only if the ratio of the estimated spectral efficiency with MU-MIMO to that with SU-MIMO is higher than the threshold specified by NRDUCeLLULMimo.UlMuGatherOptGainThld .

6.2 Network Analysis

6.2.1 Benefits

In medium- and heavy-load scenarios where cell resources are insufficient, MU-MIMO increases the average UE and cell throughputs. It is recommended that MU-MIMO be enabled when the cell load is high. The cell load is high when the uplink or downlink PRB usage is high, for example, higher than or equal to 60%. In medium- and light-load scenarios where cell resources are sufficient, MU-MIMO does not increase the average UE and cell throughputs.

Under the same traffic volume, the increases in the average UE and cell throughputs are related to the number of paired RBs and the number of paired layers. The greater the numbers, the larger the increases.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

[Table 6-11](#) describes the network impacts of MU-MIMO.

Table 6-11 Network impacts of MU-MIMO

Function	Network Impact
PDSCH MU-MIMO	• After an increase in the downlink MU-MIMO pairing rate, there may be a decrease in the average downlink UE throughput if the data volume of tail packets accounts for a large proportion in medium-load scenarios or if the channel quality of UEs is good and UEs are distributed in a centralized manner. • After an increase in the downlink MU-MIMO pairing rate, there will be decreases in the average downlink MCS index and average rank of UEs, and a possible decrease in the downlink throughput of some UEs. • After an increase in the downlink MU-MIMO pairing rate, there will be an increase in the interference between paired UEs and accordingly an increase in the downlink BLER in the cell. • After an increase in the downlink MU-MIMO pairing rate, there will be increases in the number of scheduled UEs in the downlink and the consumption of CCE resources. If CCE resources for the uplink are limited, there will be an increase in the CCE allocation failure rate for uplink transmission and decreases in the average uplink cell throughput and average uplink UE throughput. • After an increase in the downlink MU-MIMO pairing rate, there will be an increase in the number of uplink status reports. If PUSCH PRB resources are limited, there will be decreases in the average uplink cell throughput and average uplink UE throughput. • The PRB utilization may decrease in medium- and heavy-load scenarios. It fluctuates normally in light-load scenarios. • When a UE accesses or is handed over to an MU-MIMO cell from the cell edge, the access or handover success rate of the UE slightly decreases.

Function	Network Impact
PUSCH MU-MIMO	- After an increase in the uplink MU-MIMO pairing rate, there will be a decrease in the average uplink MCS index of UEs and a possible decrease in the average uplink throughput of some UEs. - After an increase in the uplink MU-MIMO pairing rate, there will be an increase in the interference between paired UEs and accordingly an increase in the uplink BLER in the cell. - After an increase in the uplink MU-MIMO pairing rate, there will be an increase in the number of downlink status reports. If PDSCH PRB resources are limited, there will be decreases in the average downlink cell throughput and average downlink UE throughput. - The PRB utilization may decrease in medium- and heavy-load scenarios. It fluctuates normally in light-load scenarios. - When a UE accesses or is handed over to an MU-MIMO cell from the cell edge, the access or handover success rate of the UE slightly decreases.
PDCCH MU-MIMO	Interference between UEs increases, the CCE allocation success rate may decrease, and the service drop rate slightly increases.

6.3 Requirements

6.3.1 Licenses

The license for MU-MIMO Basic Pairing has been purchased.

Feature ID	Feature Name	Model	Sales Unit
FOFD-010010	MU-MIMO Basic Pairing	NR0S00MU MM00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

For TDD, if a gNodeB has 32T32R or more antennas, the license for the number of spatial multiplexing layers is also required. The maximum number cannot exceed the licensed number. One license unit corresponds to two layers. For details about licensing rules, see *License Control Item Lists*.

Model	Description	Sales Unit
NR0S0DLEPU00	Massive MIMO DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell
NR0S0ULEPU00	Massive MIMO UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell

 NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific limitation on the maximum number of available RBs	MAX_RB_LIMIT_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	Neither PDSCH MU-MIMO nor PUSCH MU-MIMO can be enabled together with the function of operator-specific limitation on the maximum number of available RBs.
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Neither PUSCH MU-MIMO nor PDSCH MU-MIMO can be enabled together with Cell Combination.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	PDCCH MU-MIMO is not supported in high-speed cells.
Low-frequency TDD	Uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellUISch.VoiceULSchSwitch parameter	None	Uplink slot aggregation cannot be enabled if the UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected and the UL_SCHEDULE_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter is deselected.
Low-frequency TDD	Slot aggregation	gNBDUSchParamGroup.SlotAggregationInd set to UL_SUPPORT	None	When PUSCH MU-MIMO is enabled, slot aggregation cannot be enabled (that is, the gNBDUSchParamGroup.SlotAggregationInd parameter cannot be set to UL_SUPPORT).
Low-frequency TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellUISch.VoiceULSchSwitch parameter	<i>VoNR</i>	3GPP R15-compliant uplink slot aggregation cannot be enabled if the UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected and the UL_SCHEDULE_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter is deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	3GPP R16-compliant uplink slot aggregation cannot be enabled if the UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected and the UL_SCHEDULE_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter is deselected.
Low-frequency TDD	Weight optimization in scenarios of aggregation into 2T	AGG_2T_WEIGHT_OPT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	PDSCH MU-MIMO or PUSCH MU-MIMO cannot be enabled together with "weight optimization in scenarios of aggregation into 2T".
Low-frequency TDD	Level-2 low jitter and high reliability	LOW_JIT_HIGH_RELIAB_LEVEL2_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	PDSCH MU-MIMO or PUSCH MU-MIMO cannot be enabled together with "level-2 low jitter and high reliability".
Low-frequency TDD	Uplink DMRS Type	NRDUCellPusch.UlDmrsType	None	If the NRDUCellPusch.MaxMimoLayerCnt parameter is set to LAYER_10 or LAYER_12 , the NRDUCellPusch.UlDmrsType parameter must be set to TYPE2 .

6.3.2.3 Function Impacts

- Functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	UEs performing low latency and high reliability services do not support MU-MIMO.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When MU-MIMO takes effect and only one type of UEs (UEs of Internet users or non-Internet users) are paired at a layer, there is no mutual impact between PRBs of two types of UEs. Therefore, if the PRB usage exceeds the average PRB usage threshold, lowering the scheduling priority is not required.
Low-frequency TDD	Downlink MU-MIMO QoS guarantee	DL_MU_MIMO_QOS_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	After downlink MU-MIMO QoS guarantee is enabled, the average downlink cell throughput and number of downlink MU-MIMO layers may decrease.
Low-frequency TDD	Downlink MU-MIMO optimization for FWA private line	PRIVATE_LINE_DL_MU_MIMO_OPT_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	After downlink MU-MIMO optimization for FWA private line is enabled, the downlink throughput of eMBB UEs in the cell and the number of downlink MU-MIMO layers in the cell may decrease.
Low-frequency TDD	Downlink MU-MIMO layer isolation	DL_MU_MIMO_ISOLATION_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	If downlink MU-MIMO layer isolation takes effect after PDSCH MU-MIMO is enabled, eMBB users cannot be paired with FWA users for MU-MIMO, and the benefits of PDSCH MU-MIMO decrease.
Low-frequency TDD	Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_LAYER_SW option of the NRDUCellUIRank.UlRankAlgoSw parameter	<i>FWA</i>	If uplink 4-layer transmission for a single FWA user has taken effect, PUSCH MU-MIMO does not take effect for the user.
Low-frequency TDD	Basic VoNR functions	VONR_SW option of the NRCCellAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	VoNR UEs cannot be involved in MU-MIMO pairing.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic uplink slot aggregation	REDCAP_UL_COVERAG E_ENH_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	RedCap UEs with dynamic uplink slot aggregation in effect do not participate in PUSCH MU-MIMO.
Low-frequency TDD	Downlink large-packet alignment pairing for RedCap UEs	DL_BIG_PKT_ALIGN_PAIR_SW option of the NRDUCellRedCap.RedCapSchAlgoSwitch parameter	<i>RedCap</i>	When downlink small-packet pairing enhancement is enabled for a cell, more small-packet UEs can be paired for MU-MIMO. If downlink large-packet alignment pairing for RedCap UEs is also enabled for the cell and both non-RedCap UEs and RedCap UEs are scheduled in the cell, large-packet RedCap UEs are less likely to be paired. As a result, there will be a smaller increase and even a decrease in the average downlink throughput of RedCap UEs. Therefore, it is not recommended that downlink small-packet pairing enhancement be enabled when downlink large-packet alignment pairing for RedCap UEs has been enabled.

- Functions related to carrier management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CarrierAlgoSwitch parameter	<i>Carrier Aggregation</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for PDSCH MU-MIMO in the SCells. As a result, the downlink cell capacity decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for PUSCH MU-MIMO in the SCells. As a result, the uplink cell capacity decreases.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for PDSCH MU-MIMO in the SCells. As a result, the downlink cell capacity decreases.
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If CA UEs do not transmit SRSs in their SCells, they cannot be paired for PUSCH MU-MIMO in the SCells. As a result, the uplink cell capacity decreases.
Low-frequency TDD	Multi-frequency convergent scheduling	MULTI_FREQ_CONVERGENT_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When multi-frequency convergent scheduling and PDSCH MU-MIMO are both enabled and the PDSCH MU-MIMO pairing rate is high, the benefits of multi-frequency convergent scheduling decrease.

- Functions related to coverage improvement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	VMIMO in a combined-RRU 4T4R cell	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	When the NRDUCellAlgoSwitch.VmimoSwitch parameter is set to ON for a 4T4R cell served by combined RRUs, the beams of two TRPs can use the same time-frequency resources of PUSCH or PDSCH for UEs in non-overlapping areas.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Probability-based uplink MCS selection	UL_PBMCSSEL_SW option of the NRDUCellUIMc.UlAmcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If both PUSCH MU-MIMO and probability-based uplink MCS selection are enabled, incorrect pairing of cell-edge UEs affects the uplink user-perceived rate of such UEs. To restrict the pairing of cell-edge UEs for uplink MU-MIMO, it is recommended that the NRDUCellUIMimo.UlMuMimoSinrThld parameter be set to -20.
Low-frequency TDD	RRC-reconfiguration-based dynamic waveform switching	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If both PUSCH MU-MIMO and RRC-reconfiguration-based dynamic waveform switching are enabled, UEs using the DFT-s-OFDM waveform and those using the CP-OFDM waveform cannot be paired for MU-MIMO. As a result, the MU-MIMO pairing rate decreases and the gains provided by PUSCH MU-MIMO may decrease.

- Functions related to coordination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	UEs involved in intra-base-station joint transmission do not support PDSCH MU-MIMO.
Low-frequency TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Coordinated beamforming (CBF) is not performed in slots with MU-MIMO pairing. If the MU-MIMO pairing rate is high, the CBF activation rate is low.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM UEs do not support PDSCH MU-MIMO.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink type 1 resource allocation for UEs performing small-packet services	SMALL_PKT_TYPE1_SC H_SW option of the NRDUCellPdsch.DlPdschAlgoSwitch parameter	<i>Downlink Scheduling</i>	When PDSCH MU-MIMO is enabled, downlink type 1 resource allocation for UEs performing small-packet services does not take effect for MU-MIMO UEs.
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch	<i>Downlink Scheduling</i>	When more than one UE requires scheduling in a cell and PDCCH MU-MIMO does not take effect, rate-matching-pattern-configuration-based PDCCH rate matching offers a low gain and may even have a negative impact. When only one UE requires scheduling in the cell, whether PDCCH MU-MIMO takes effect does not affect PDCCH rate matching.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	For UEs in non-overlapping areas of distributed massive MIMO cells, the beams of different TRPs can use the same time-frequency resources of PUSCH, PDSCH, or PDCCH for spatial multiplexing if the corresponding UL_MU_MIMO_SW , DL_MU_MIMO_SW , or PDCCH_MU_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	In high-speed railway scenarios, PDSCH MU-MIMO is supported only in common or hyper cells. PDSCH MU-MIMO in high-speed railway scenarios is a trial function and is supported only by 8T8R RRUs. In high-speed railway scenarios, PUSCH MU-MIMO is supported only in common or hyper cells. PUSCH MU-MIMO in high-speed railway scenarios is a trial function and is supported only by 8T8R RRUs and massive MIMO modules. Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free. Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully

RAT	Function Name	Function Switch	Reference	Description
				understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If MU-MIMO has been enabled and then the power saving BWP function is enabled, there will be: (1) increases in the number of downlink layers, downlink BLERs, and average number of RBs used for downlink pairing; (2) decreases in the number of uplink layers, uplink BLERs, and average number of RBs used for uplink pairing; (3) an increase in the number of scheduling times and decreases in the average cell throughput and UE throughput. If the power saving BWP function has been enabled and then downlink MU-MIMO is enabled, there will be a low probability that the average downlink cell throughput and UE throughput slightly decrease when there are a large proportion of small-packet services on the network.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	PDSCH MU-MIMO and PUSCH MU-MIMO are not supported for UEs working on BWP2 for power saving. Enabling time-domain power saving BWP after enabling PDSCH MU-MIMO or PUSCH MU-MIMO may cause the following impacts: • The average number of downlink MU-MIMO layers, average number of downlink layers, maximum number of downlink layers, downlink BLER, and average number of RBs for downlink MU pairing change. • The average number of uplink MU-MIMO layers, average number of uplink layers, maximum number of uplink layers, uplink BLER, and average number of RBs for uplink MU pairing change. • The numbers of uplink and downlink scheduling times decrease. • The average cell throughput and average UE throughput change. For the definitions of the preceding indicators, see <i>UE Power Saving</i>.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	PDSCH or PUSCH MU-MIMO works with NR DU cell resource management between network slices as follows: • If NRDUCellRes.ResStrategyType is set to TTI_ONLY for a network slice group, UEs in the group do not participate in pairing. • If NRDUCellRes.ResStrategyType is set to MEAN_ONLY for a network slice group, UEs in the group can participate in pairing. The following descriptions apply when pairing is performed: • The proportion of RBs used by UEs in a network slice group equals the proportion of layers paired for these UEs. For example, if there are a total of six paired layers and four of them are paired for UEs in a group, UEs in the group use 4/6 of all available RBs. • Assume that NRDUCellRes.DIRbAverageRatio and NRDUCellRes.UIRbAverageRatio are configured. The proportion of RBs used by UEs in a network slice group using the mean-type control policy can reach the configured proportion only when a sufficient number of UEs in the group perform full

RAT	Function Name	Function Switch	Reference	Description
				buffer services, in which case the pairing success rate will decrease for the UEs that are outside the network slice group and highly correlated with UEs in the group. This will negatively impact user experience because both the average downlink cell and UE throughputs will decrease.
Low-frequency TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW or INTRF_COORD_ALWAYS_WORKING_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When interference coordination between public and dedicated networks is enabled and the NRDUCellHighSpeed-Mob.DlIntrflLimitRbSchRatio parameter is set to a non-zero value, PDSCH MU-MIMO does not take effect.
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Dynamic RF channel shutdown is not performed in slots with PDSCH MU-MIMO pairing.
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TTI-level channel shutdown does not take effect in slots with PDSCH MU-MIMO pairing.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Percentage-based dynamic RB sharing among operators	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	When percentage-based dynamic RB sharing among operators takes effect together with PUSCH MU-MIMO or PDSCH MU-MIMO: The percentage of RBs occupied by each operator can be close to the configured RB percentage (specified by NRDUCellRes.DIRbAverageRatio and NRDUCellRes.UIRbAverageRatio) when the cell is fully loaded. In addition, when paired UEs that meet certain conditions are preferentially scheduled to guarantee pairing gains, the percentage of RBs occupied by an operator may exceed the configured RB percentage. As a result, the actual percentage of RBs used by an operator fluctuates around the configured percentage.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	PDSCH or PUSCH MU-MIMO works with QCI-based resource control as follows: - The proportion of RBs used by UEs in a QCI group equals the proportion of layers paired for these UEs. For example, if there are a total of six paired layers and four of them are paired for UEs in a group, UEs in the group use 4/6 of all available RBs. - Assume that NRDUCellRes.DIRbAverageRatio and NRDUCellRes.UIRbAverageRatio are configured. The proportion of RBs used by UEs in a QCI group can reach the configured proportion only when a sufficient number of UEs in the group perform full buffer services, in which case the pairing success rate will decrease for the UEs that are outside the QCI group and highly correlated with UEs in the group. This will negatively impact user experience because both the average downlink cell and UE throughputs will decrease.
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	If the PDSCH MU-MIMO pairing rate is high, inter-cell interference avoidance will less likely to take effect because it does not take effect in slots where UEs paired for PDSCH MU-MIMO are scheduled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink channel densified equalization for high-speed UEs	HS_UL_LOW_COST_DENSE_EQUAL_SW option of the NRDUCellHighSpeed-Mob.HighSpeedAlgorithmSw parameter	<i>High Speed Mobility</i>	"Uplink channel densified equalization for high-speed UEs" does not take effect for UEs involved in PUSCH MU-MIMO.
Low-frequency TDD	FSD transmission mode	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL and NRDUCellMultiTrp.DataTransMode set to FSD_MODE	<i>Hyper Cell</i>	Hyper cells In FSD transmission mode do not support PUSCH MU-MIMO, PDSCH MU-MIMO, or spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO.
Low-frequency TDD	Multi-beam reception enhancement	MULTI_BEAM_RX_ENH_SW option of the NRDUCellPusch.PuschAlgorithmExtSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Multi-beam reception enhancement does not take effect for UEs with PUSCH MU-MIMO in effect.
Low-frequency TDD	Enhanced uplink L4S congestion control based on rate adjustment	UL_L4SPLUS_FOR_FAIR_SW option of the NRDUCellUISch.UISchAlgorithmSwitch parameter	<i>Adaptive Managed Latency</i>	If both PUSCH MU-MIMO and "enhanced uplink L4S congestion control based on rate adjustment" are enabled, the maximum rate and fair rate calculated by the gNodeB for L4S UEs may be inaccurate.
Low-frequency TDD	Enhanced downlink L4S congestion control based on rate adjustment	DL_L4SPLUS_FOR_FAIR_SW option of the NRDUCellDISch.DISchAlgorithmEnhSwitch parameter	<i>Adaptive Managed Latency</i>	If both PDSCH MU-MIMO and "enhanced downlink L4S congestion control based on rate adjustment" are enabled, the maximum rate and fair rate calculated by the gNodeB for L4S UEs may be inaccurate.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules support this function. However, 2T2R and 4T4R RF modules support MU-MIMO only in virtual MIMO (VMIMO) scenarios. To learn which RF modules support NR TDD, see the technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

For details about VMIMO, see *Indoor Coverage*.

PDCCH MU-MIMO requires RF modules of 32T32R or higher.

6.3.4 Cells

- MU-MIMO is not supported in SDL cells (with **NRDUCell.DuplexMode** set to **CELL_SDL**).
- Neither PDSCH MU-MIMO nor PUSCH MU-MIMO is supported in combined cells (with **NRDUCell.NrDuCellNetworkingMode** set to **HYPER_CELL_COMBINE_MODE**).
- Neither PDSCH MU-MIMO nor PUSCH MU-MIMO is supported in a hyper cell (with **NRDUCell.NrDuCellNetworkingMode** set to **HYPER_CELL**) served by a low-frequency TDD macro base station if either of the following conditions is met for the cell: (1) **NRDUCellAlgoSwitch.VmimoSwitch** is set to **OFF**, and the values of **NRDUCellTrp.TxRxMode** do not include **8T8R**, **32T32R**, or **64T64R**. (2) **NRDUCellAlgoSwitch.VmimoSwitch** is set to **ON**, and the values of **NRDUCellTrp.TxRxMode** include a value other than **2T2R** and **4T4R**.
- Neither PDSCH MU-MIMO nor PUSCH MU-MIMO is supported in a cell with **NRDUCell.LampSiteCellFlag** set to **NO** and **NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL** if either of the following conditions is met for the cell: (1) The values of **NRDUCellTrp.TxRxMode** include **2R**, **4R**, **1T1R**, **1T2R**, or **2T4R**. (2) **NRDUCellTrp.TxRxMode** is set to **2T2R** or **4T4R**, **NRDUCellAlgoSwitch.VmimoSwitch** is set to **OFF**, and the **DM_MIMO_SERVICE_SWITCH** option of **NRDUCellAlgoSwitch.DmMimoSwitch** is deselected.
- PUSCH MU-MIMO is not supported in 2R or 4R SUL cells (with **NRDUCell.DuplexMode** set to **CELL_SUL** and **NRDUCellTrp.TxRxMode** set to **2R** or **4R**).

- If **NRDUCellPusch.MaxMimoLayerCnt** is set to **LAYER_10** or **LAYER_12**, **NRDUCell.LampSiteCellFlag** must be set to **YES** and the **DM_MIMO_SERVICE_SWITCH** option of **NRDUCellAlgoSwitch.DmMimoSwitch** must be selected.
- MU-MIMO is not supported when **NRDUCell.DeployPosition** is set to **DEPLOY_IAB**.

6.3.5 Others

None

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

[Table 6-12](#) and [Table 6-13](#) describe the parameters used for function activation and optimization, respectively.

Table 6-12 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Transmit and Receive Mode	NRDUCellTrp.TxRxMode	None	Set this parameter based on the actually used AAU/RRU model.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	UL_MU_MIMO_SW	It is recommended that this option be selected when cell load is high.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	DL_MU_MIMO_SW	It is recommended that this option be selected when cell load is high.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	PDCCH_MU_SW	Generally, retain the default value, that is, deselect this option. However, you are advised to select this option if the PDCCH CCE allocation failure rate is greater than 5%.

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Maximum MIMO Layer Number Quota	NRDUCellPdsch.MaxMimoLayerNum	None	Generally, retain the default value. You can adjust the setting based on network load, for example, you can configure a larger number of layers to increase capacity in heavy-load scenarios. It is recommended that this parameter be set to a value less than or equal to 16 for a 64T64R gNodeB. If a larger value is used, more units of NR0S0DLEPU00 Massive MIMO DL 2-Layers Extended Processing Unit License (NR) will be consumed. For details about the consumption rules, see 6.3.1 Licenses.
Uplink Maximum MIMO Layer Count Quota	NRDUCellPusch.MaxMimoLayerCnt	None	The value LAYER_4 is recommended. You can adjust the setting based on network load, for example, you can configure a larger number of layers to increase capacity in heavy-load scenarios.
Maximum Pairing Layer Number	NRDUCellPdsch.MaxPairLayerNum	None	Generally, retain the default value. You can adjust the setting based on network load, for example, you can configure a larger number of layers to increase capacity in heavy-load scenarios.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgorithmSwitch	UL_SCHEDULE_ENH_SW	This option is selected by default for new low-frequency TDD cells. For existing low-frequency TDD cells, select this option if uplink scheduling enhancement is required.

Table 6-13 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
DL PMI MU-MIMO Space Isolation Threshold	NRDUCellPdsch.DIPmiMuMimoSpacelsoThld	None	Retain the default value.
DL SRS MU-MIMO Space Isolation Threshold	NRDUCellPdsch.DLSrsMuMimoSpacelsoThld	None	Retain the default value.
DL MU-MIMO Channel Isolation Threshold	NRDUCellPdsch.DLSrsMuMimoChanIsoThld	None	Retain the default value.
DL MU-MIMO SRS PreSINR Threshold	NRDUCellDL Mimo.DLMuMimoSrsPreSinrThld	None	Use the recommended value.
DL MU-MIMO RB Estimation Policy	NRDUCellDmrs.DLMuEstRbPolicy	None	Retain the default value.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDL Mimo.DLMuMimoSchSupplementSw	DL_SMALL_PACKET_MERGE_SW	If downlink small-packet group merging is required, it is recommended that this option be selected to reduce the interference between paired UEs and increase the user-perceived rate.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDL Mimo.DLMuMimoSchSupplementSw	DL_SE_SU_MU_MIMO_ADAPT_SW	For low-frequency TDD massive MIMO cells, it is recommended that this option be selected if spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO is required; for low-frequency TDD cells with 8T8R, it is required that this option be deselected. The purpose is to improve the spectral efficiency after MU-MIMO is enabled.
Aligned Pair Big/Small Packet RB Threshold	NRDUCellUL Mimo.AlignPairBigSmlPktRbThld	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.UlMuMimoAlgoSwitch	MU_BEAM_PAIR_OPT_SW	Select this option if uplink MU-MIMO beam-domain pairing optimization is required.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.UlMuMimoAlgoSwitch	SML_PKT_ALIGN_PAIR_OPT_SW	This option is selected by default for new low-frequency TDD cells. For existing low-frequency TDD cells, select this option if pairing optimization is required for small-packet UEs.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.UlMuMimoAlgoSwitch	BIG_PKT_ALIGN_PAIR_OPT_SW	This option is selected by default for new low-frequency TDD cells. For existing low-frequency TDD cells, select this option if pairing optimization is required for large-packet UEs.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.UlMuMimoAlgoSwitch	BIG_PKT_UE_FREQ_SEL_SCH_H_OPT_SW	Select this option if frequency selective scheduling optimization for large-packet UEs is required.
Aligned Pair Small Packet RB Ext Ratio	NRDUCellULMimo.AlignPairSmlPktRbExtRatio	None	Retain the default value.
Aligned Pair Big Packet RB Reduce Ratio	NRDUCellULMimo.AlignPairBigPktRbRedRatio	None	Retain the default value.
MU Freq Sel Scheduling RB Ratio Threshold	NRDUCellULMimo.MuFreqSelSchRbRatioThld	None	It is recommended that this parameter be set to 40 if frequency selective scheduling optimization for large-packet UEs is required.
MU RB Ratio Threshold	NRDUCellULMimo.MuRbRatioThld	None	It is recommended that this parameter be set to 40 if PUSCH MU-MIMO is required.
DL Low Time Correlated MU-MIMO Switch	NRDUCellPdsch.DlLowTimeCorrMuSwitch	None	It is recommended that this switch be turned on for low-frequency TDD massive MIMO cells to improve the pairing efficiency.

Parameter Name	Parameter ID	Option	Setting Notes
DL MU-MIMO Group Mode	NRDUCellPdsch.DIMuMimoGroupMode	None	It is recommended that the value ISOLATION_CORRELATION be used to optimize the downlink pairing rules so that UEs with low correlation and low interference can be selected for pairing.
DL SRS MU-MIMO Rank	NRDUCellDLRank.DLSrsMuMimoRank	None	Retain the default value.
DL PMI MU-MIMO Rank	NRDUCellDLRank.DLPmiMuMimoRank	None	Retain the default value.
DL MU-MIMO PRB Usage Threshold	NRDUCellDLMimo.DIMuPrbUsageThld	None	Retain the default value.
DL MU-MIMO Back to SU-MIMO Gain Thld	NRDUCellDLMimo.DIMuBackToSuSeThld	None	Retain the default value.
DL PMI MU-MIMO UE Pairing Beam Num Thld	NRDUCellDLMimo.DIMuPmiBeamNumThld	None	Retain the default value.
DL Gathering MU-MIMO Optimization Switch	NRDUCellPdsch.DIMuGatherOptSw	None	Set this parameter based on the network plan.
DL MIMO MCS Optimization Policy	NRDUCellPdsch.DIMimoMcsOptPolicy	None	Retain the default value.
Uplink AMC Algorithm Switch	NRDUCellULAMc.UlAmcAlgoSw	MU_AMC_OPT_SW	Select this option if AMC optimization for uplink MU-MIMO is required.
UL MU-MIMO SINR Threshold	NRDUCellULMimo.UlMuMimoSinrThld	None	The value -20 is recommended if PUSCH MU-MIMO is required.

Parameter Name	Parameter ID	Option	Setting Notes
UL MU-MIMO Correlation Threshold	NRDUCellULMimo.UIMuMimoCorrThld	None	Retain the default value.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.UIMuMimoAlgoSwitch	DIFF_FLDMRS_MAX_LENGTH_PAIR_SW	It is recommended that this option be selected if UEs supporting different numbers of front-loaded DMRS symbols need to be paired in uplink MU-MIMO.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.UIMuMimoAlgoSwitch	MU_MULTI_SLOT_PAIR_SW	It is recommended that this option be selected in uplink heavy-load scenarios.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.UIMuMimoAlgoSwitch	OPT_PAIR_RB_SEL_SW	This option is selected by default for new low-frequency TDD cells. For existing low-frequency TDD cells, select this option if optimal paired RB selection is required.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UIPuschAlgoSwitch	UL_INTRF_RANDOMIZED_SCH_ENH_SW	This option is selected by default for new low-frequency TDD cells. For existing low-frequency TDD cells, select this option if uplink scheduling enhancement is required.
DL SRS MU-MIMO Rank Mode	NRDUCellDLRank.DLSrsMuMimoRankMode	None	It is recommended that this parameter be set to RI_BASED_MU_RANK if downlink SRS-based MU-MIMO rank mode configuration is required.
DL MU-MIMO SIR Scale Factor	NRDUCellDLIAmc.DLMuMimoSirScaleFactor	None	Use the recommended value. In addition, set the NRDUCellPdsch.DLMulInitialMcsAdjValue parameter to -2.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDLI Mimo.DLMuMimoSchSupplementSw	DL_MCS_TABLE_ADAPT_ENH_SW	Select this option if downlink MCS index table adaptation enhancement is required.

Parameter Name	Parameter ID	Option	Setting Notes
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDL MIMO.DLMu MIMO Sch SupplementSw	DL_SMALL_PACKET_PAIR_ENH_SW	Select this option if downlink small-packet pairing enhancement is required. This function is recommended when the cell is extremely lightly loaded. This function is not recommended in other scenarios because there may be negative impacts.
Precoding-based Intra Interference Suppression Value	NRDUCellPdschPrecode.PrecodingIntraInterferenceSupprValue	None	Use the recommended value.
DL SRS MU-MIMO Freq Selection Threshold	NRDUCellPdsch.DLSrsMuMIMO FreqSelThld	None	It is recommended that this parameter be set to 10 if pairing delay spread optimization is required.
Downlink Schedule Algorithm Switch	NRDUCellDLISch.DLSchAlgoSwitch	HEAVY_LOAD_SCHED_PRI_OPT_SW	Select this option if scheduling priority optimization in heavy-load scenarios is required.
UL MU-MIMO Algorithm Switch	NRDUCellUL MIMO.ULMu MIMO AlgoSwitch	UL_MU_BEAM_PAIR_FORBID_SW	Retain the default value.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	SR_SMART_SCH_SW	Select this option for uplink scheduling enhancement, so that UEs sending scheduling requests (SRs) can be paired with UEs running data services. This reduces the number of SR retransmissions and prevents the service drop rate from increasing. For details about this option, see <i>Uplink Scheduling</i> .

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.PuschPerformanceSwitch	UL_NONCONSEC_SCH_OPT_SW	Select this option for uplink scheduling enhancement, so that UEs involved in MU pairing can be allocated consecutive RBs. This reduces the padding rate and therefore prevents increases in the PRB usage and uplink interference as well as impacts on indicators. For details about this option, see <i>Uplink Scheduling</i> .
Uplink AMC Algorithm Switch	NRDUCellULAMc.ULAMcAlgorithmSwitch	UL_SCH_AVG_SINR_OPT_SW	Select this option for uplink scheduling enhancement, so that RB allocation to small-packet UEs with limited uplink power is more accurate. This shortens the data transmission delay for these UEs and increases the average uplink UE throughput. For details about this option, see <i>Uplink Scheduling</i> .
UL Gathering MU-MIMO Opt Gain Thld	NRDUCellULMimo.ULMuGatheringOptGainThld	None	Retain the default value.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3.2 Functions](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Setting Mandatory Parameters for MU-MIMO

Scenario 1: Massive MIMO

```
//Setting the TX/RX mode of a cell  
MOD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=32T32R;
```

```
//Activating MU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
MuMimoSwitch=UL_MU_MIMO_SW-1&PDCCH_MU_SW-1&DL_MU_MIMO_SW-1;
//Setting the maximum number of spatial multiplexing layers of PDSCH MU-MIMO
MOD NRDUCELLPDSCH: NrDuCellId=0, MaxMimoLayerNum=LAYER_DEFAULT;
//Setting the maximum number of spatial multiplexing layers of PDCCH MU-MIMO
MOD NRDUCELLPDCCH: NrDuCellId=0, MaxPairLayerNum=LAYER_2;
//Setting the maximum number of spatial multiplexing layers of PUSCH MU-MIMO
MOD NRDUCELLPUSCH: NrDuCellId=0, MaxMimoLayerCnt=LAYER_4;
```

Scenario 2: nTnR

```
//Setting the TX/RX mode of a cell
MOD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=8T8R;
//Activating MU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=UL_MU_MIMO_SW-1&DL_MU_MIMO_SW-1;
//Setting the maximum number of spatial multiplexing layers of PDSCH MU-MIMO
MOD NRDUCELLPDSCH: NrDuCellId=0, MaxMimoLayerNum=LAYER_DEFAULT;
//Setting the maximum number of spatial multiplexing layers of PUSCH MU-MIMO
MOD NRDUCELLPUSCH: NrDuCellId=0, MaxMimoLayerCnt=LAYER_4;
```

Setting Optional Parameters for MU-MIMO

The following provides only configuration optimization examples for MU-MIMO. You can determine the parameters to be configured based on network conditions.

Scenario 1: Massive MIMO

```
//(Optional) Configuring pairing thresholds for PDSCH MU-MIMO
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPmiMuMimoSpacelsoThld=140, DlSrsMuMimoSpacelsoThld=50,
DlSrsMuMimoChanlsoThld=30;
MOD NRDUCELLDLMIMO: NrDuCellId=0, DlMuMimoSrsPreSnrThld=-50;
//(Optional) Configuring PDSCH MU-MIMO enhancements
MOD NRDUCELLPDSCH: NrDuCellId=0, DlMuMimoGroupMode=ISOLATION_CORRELATION,
DlSrsMuMimoFreqSelThld=10, DlMuGatherOptSw=OFF, DlMimoMcsOptPolicy=OFF,
DlLowTimeCorrMuSw=ON, DlMuInitialMcsAdjValue=-2;
MOD NRDUCELLDLAMC: NrDuCellId=0, DlMuMimoSirScaleFactor=10;
MOD NRDUCELLDLMIMO: NrDuCellId=0, DlMuBackToSuSeThld=5, DlMuPmiBeamNumThld=3,
DlMuMimoSchSupplementSw=DL_SMALL_PACKET_MERGE_SW-1&DL_SE_SU_MU_MIMO_ADAPT_SW-1&DL_
MCS_TABLE_ADAPT_ENH_SW-1, DlMuPrbUsageThld=0;
MOD NRDUCELLDMRS: NrDuCellId=0, DlMuEstRbPolicy=PSEUDOORTHOG_DMRS_ADAPT_DEDUCT;
MOD NRDUCELLDLRANK: NrDuCellId=0, DlSrsMuMimoRank=RANK_2, DlPmiMuMimoRank=RANK_2,
DlSrsMuMimoRankMode=RI_BASED_MU_RANK;
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingIntrfSupprValue=DEFAULT;
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=HEAVY_LOAD_SCH_PRI_OPT_SW-1;
//(Optional) Enabling downlink small-packet pairing enhancement. This function is recommended when the
cell is extremely lightly loaded to avoid negative impacts.
MOD NRDUCELLDLMIMO: NrDuCellId=0,
DlMuMimoSchSupplementSw=DL_SMALL_PACKET_PAIR_ENH_SW-1;
//(Optional) Configuring PUSCH MU-MIMO enhancements (It is recommended that all functions of PUSCH
MU-MIMO enhancements be enabled. Enabling only some of the functions may cause negative impacts.)
MOD NRDUCELLULMIMO: NrDuCellId=0,
UlMuMimoAlgoSwitch=MU_BEAM_PAIR_OPT_SW-1&BIG_PKT_ALIGN_PAIR_OPT_SW-1&SML_PKT_ALIGN_PAI
R_OPT_SW-1&BIG_PKT_UE_FREQ_SEL_SCH_OPT_SW-1&MU_MULTI_SLOT_PAIR_SW-1&DIFF_FLDMRS_MAX_L
ENGTH_PAIR_SW-1&OPT_PAIR_RB_SEL_SW-1, AlignPairBigPktRbRedRatio=50,
AlignPairSmlPktRbExtRatio=200, AlignPairBigSmlPktRbThld=20, MuRbRatioThld=40,
MuFreqSelSchRbRatioThld=40, UlMuGatherOptGainThld=100;
MOD NRDUCELLULAMC: NrDuCellId=0,
UlAmcAlgoSw=MU_AMC_OPT_SW-1&UL_SCH_AVG_SINR_OPT_SW-1;
MOD NRDUCELLPUSCH: NrDuCellId=0,
UlPuschAlgoSwitch=UL_SCHEDULE_ENH_SW-1&SR_SMART_SCH_SW-1&UL_INTRF_RANDOMIZED_SCH_ENH
_SW-1, PuschPerformanceSwitch=UL_NONCONSEC_SCH_OPT_SW-1;
//(Optional) Configuring pairing thresholds for PUSCH MU-MIMO
MOD NRDUCELLULMIMO: NrDuCellId=0, UlMuMimoCorrThld=9, UlMuMimoSnrThld=-20;
```

Scenario 2: nTnR

```
//(Optional) Configuring pairing thresholds for PDSCH MU-MIMO
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPmiMuMimoSpacelsoThld=140, DlSrsMuMimoSpacelsoThld=50,
```

```

DLSrsMuMimoChanIsoThld=30;
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSrsPreSnrThld=-50;
//(Optional) Configuring PDSCH MU-MIMO enhancements
MOD NRDUCELLPDSCH: NrDuCellId=0, DLMuMimoGroupMode=ISOLATION_CORRELATION,
DLSrsMuMimoFreqSelThld=10, DLMuGatherOptSw=OFF, DLMimoMcsOptPolicy=OFF,
DLowTimeCorrMuSw=ON, DLMuInitialMcsAdjValue=-2;
MOD NRDUCELLDLMIMO: NrDuCellId=0,
DLMuMimoSchSupplementSw=DL_SMALL_PACKET_MERGE_SW-1&DL_SE_SU_MU_MIMO_ADAPT_SW-1&DL_
MCS_TABLE_ADAPT_ENH_SW-1, DLMuPrbUsageThld=0, DLMuBackToSuSeThld=5;
MOD NRDUCELLDMRS: NrDuCellId=0, DLMuEstRbPolicy=PSEUDOORTHOG_DMRS_ADAPT_DEDUCT;
MOD NRDUCELLDLRANK: NrDuCellId=0, DLSrsMuMimoRank=RANK_2, DLPmiMuMimoRank=RANK_2,
DLSrsMuMimoRankMode=RI_BASED_MU_RANK;
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingIntrfSupprValue=DEFAULT;
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=HEAVY_LOAD_SCH_PRI_OPT_SW-1;
//(Optional) Enabling downlink small-packet pairing enhancement. This function is recommended when the
cell is extremely lightly loaded to avoid negative impacts.
MOD NRDUCELLDLMIMO: NrDuCellId=0,
DLMuMimoSchSupplementSw=DL_SMALL_PACKET_PAIR_ENH_SW-1;
//(Optional) Configuring PUSCH MU-MIMO enhancements (It is recommended that all functions of PUSCH
MU-MIMO enhancements be enabled. Enabling only some of the functions may cause negative impacts.)
MOD NRDUCELLULMIMO: NrDuCellId=0,
ULMuMimoAlgoSwitch=MU_BEAM_PAIR_OPT_SW-1&BIG_PKT_ALIGN_PAIR_OPT_SW-1&SML_PKT_ALIGN_PAI
R_OPT_SW-1&BIG_PKT_UE_FREQ_SEL_SCH_OPT_SW-1&MU_MULTI_SLOT_PAIR_SW-1&DIFF_FLDMS_MAX_L
ENGTH_PAIR_SW-1&OPT_PAIR_RB_SEL_SW-1&UL_MU_BEAM_PAIR_FORBID_SW-1,
AlignPairBigPktRbRedRatio=50, AlignPairSmlPktRbExtRatio=200, AlignPairBigSmlPktRbThld=20,
MuRbRatioThld=40, MuFreqSelSchRbRatioThld=40, ULMuGatherOptGainThld=100;
MOD NRDUCELLULAMC: NrDuCellId=0,
ULAmcAlgoSw=MU_AMC_OPT_SW-1&UL_SCH_AVG_SINR_OPT_SW-1;
MOD NRDUCELLPUSCH: NrDuCellId=0,
ULPuschAlgoSwitch=UL_SCHEDULE_ENH_SW-1&SR_SMART_SCH_SW-1&UL_INTRF_RANDOMIZED_SCH_ENH
_SW-1, PuschPerformanceSwitch=UL_NONCONSEC_SCH_OPT_SW-1;
//(Optional) Configuring pairing thresholds for PUSCH MU-MIMO
MOD NRDUCELLULMIMO: NrDuCellId=0, ULMuMimoCorrThld=9, ULMuMimoSnrThld=-20;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Scenario 1: Massive MIMO

```
//Deactivating MU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
MuMimoSwitch=UL_MU_MIMO_SW-0&PDCCH_MU_SW-0&DL_MU_MIMO_SW-0;
```

Scenario 2: nTnR

```
//Deactivating MU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=UL_MU_MIMO_SW-0&DL_MU_MIMO_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Check whether MU-MIMO has taken effect. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > Cell Performance**

Monitoring. Select the corresponding base station and cell to start a monitoring task. If the value of a monitoring item listed in [Table 6-14](#) is greater than 1, the corresponding function has taken effect.

Table 6-14 MU-MIMO monitoring items

Function	Monitoring Item	Description
Downlink MU-MIMO monitoring	Max Number of DL Transmission Pair Layers	If the value is greater than 1, PDSCH MU-MIMO has taken effect.
Uplink MU-MIMO monitoring	Max Number of UL Transmission Pair Layers	If the value is greater than 1, PUSCH MU-MIMO has taken effect.
Air interface DCI monitoring	PdcchMaxMU LayerNum	If the value is greater than 1, PDCCH MU-MIMO has taken effect.

In addition, you can use the following indicators to estimate the effects of PDSCH MU-MIMO and PUSCH MU-MIMO.

- Total number of downlink MU-MIMO layers in a cell:
N.ChMeas.MIMO.DL.Pair.Layer
- Average number of downlink MU-MIMO layers in a cell:
N.ChMeas.MIMO.DL.Pair.Layer.Avg
- Total number of uplink MU-MIMO layers in a cell:
N.ChMeas.MIMO.UL.Pair.Layer
- Average number of uplink MU-MIMO layers in a cell:
N.ChMeas.MIMO.UL.Pair.Layer.Avg
- Indicators applicable to both SU-MIMO and MU-MIMO:
 - Maximum number of downlink layers on a PRB in a cell =
N.ChMeas.MIMO.DL.Transmission.Layer.Max
 - Maximum number of uplink layers on a PRB in a cell =
N.ChMeas.MIMO.UL.Trans.Layer.Max

6.4.3 Network Monitoring

After this function is enabled, you can observe the increase in cell capacity using the following KPIs:

- **User Uplink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**
- Average uplink cell throughput = **N.ThpVol.UL / N.RLC.ThpTime.UL.Cell**
- Average downlink cell throughput = **N.ThpVol.DL / N.RLC.ThpTime.DL.Cell**

7 Massive MIMO Multi-Layer Enhancement (Low-Frequency TDD)

7.1 Downlink Massive MIMO Multi-Layer Enhancement

7.1.1 Principles

Downlink massive MIMO multi-layer enhancement improves the downlink pairing performance and increases the downlink UE throughput in TDD massive MIMO cells with the maximum number of downlink paired layers (specified by the **NRDUCellPdsch.MaxMimoLayerNum** parameter) being at least eight.

This function is controlled by the **MMIMO_MULTILAYER_ENHANCE_SW** option of the **NRDUCellDLMimo.HighLayerMuMimoSw** parameter. After this option is selected and the maximum number of downlink paired layers in a cell is configured, the switches controlling the sub-functions need to be turned on for the corresponding sub-functions to take effect.

Downlink massive MIMO multi-layer enhancement optimizes interference suppression, scheduling, and weights. The following describes its functions.

Interference Suppression for a Large Number of Layers

Functions dedicated to interference suppression for a large number of layers mitigate inter-cell interference (SRS, PDSCH, and PUCCH interference) that increases in medium- or heavy-load scenarios. In this way, downlink pairing performance improves and the downlink UE throughput increases.

Table 7-1 describes the functions dedicated to interference suppression for a large number of layers.

Table 7-1 Functions dedicated to interference suppression for a large number of layers

Function Name	Function Switch	Function Description
Blind SRS interference suppression	SRS_BLIND_IS_SW option of the NRDUCellDIMimo.HighLayerMuMimoSw parameter	This function enables the gNodeB to implement SRS interference suppression based on the characteristic differences between interference signals and wanted signals, improving the weight accuracy in multi-layer scenarios and increasing the average downlink UE throughput. The proportion of UEs for which this function takes effect in a cell is controlled by the NRDUCellSrs.SrsBlindIsDegree parameter. This function is not supported by LampSite base stations.
Downlink anti-interference scheduling	DL_ANTI_INTRF_SCH_SW option of the NRDUCELLPDSCH.DLPdschAlgoSwitch parameter	This function optimizes resource allocation for multi-layer pairing, reducing inter-cell interference and increasing the downlink UE SINR. When both downlink anti-interference scheduling and PDSCH MU-MIMO are enabled, it is recommended that adaptive estimation mode for downlink MU-MIMO also be enabled to increase the accuracy of resource estimation for MU-MIMO. In this mode, the resource conditions both before and after pairing are considered during resource estimation for MU-MIMO, so that the estimation is more accurate. Adaptive estimation mode for downlink MU-MIMO is controlled by the DL_MU_ADAPT_ESTIMATE_MODE_SW option of the NRDUCellDIMimo.DLMuMimoSchSupplementSw parameter.
PUCCH interference coordination	PUCCH_INTRF_COORD_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	This function optimizes inter-cell PUCCH resource allocation to reduce inter-cell PUCCH interference, improve PUCCH demodulation performance, and improve the reliability of feedback on downlink scheduling.

Scheduling for a Large Number of Layers

In medium- and heavy-load scenarios where cell resources are insufficient, user experience is unsatisfactory. Given the same traffic volume, more paired RBs, more

appropriate power allocation to different UEs, and higher ranks lead to better pairing performance, more timely UE scheduling, and better user experience. In line with this principle, functions dedicated to scheduling for a large number of layers consider service, pairing, and channel conditions to adaptively adjust how the number of RBs, power, and ranks are determined for UEs paired at a large number of layers. These functions increase the downlink cell spectral efficiency and average downlink UE throughput.

Table 7-2 describes the functions dedicated to scheduling for a large number of layers.

Table 7-2 Functions dedicated to scheduling for a large number of layers

Function Name	Function Switch	Function Description
Downlink high-efficiency scheduling	DL_HIGH_EFF_SCH_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter	This function optimizes RB allocation when there are a large number of layers to improve the downlink spectral efficiency and average downlink UE throughput in a cell.
Downlink MU-MIMO pairing preference	MU_MIMO_PAIRING_PREFERRED_SW option of the NRDUCellDIMimo.HighLayerMuMimoSw parameter	This function preferentially pairs UEs in the space domain during resource allocation, increasing the spectral efficiency and number of paired RBs, and therefore increasing the average downlink UE throughput in a cell. The spectral efficiency threshold for switching from MU-MIMO to SU-MIMO is specified by the NRDUCellDIMimo.DIMuBackToSuSetHld parameter.
Downlink MU-MIMO power adjustment	DL_MU_MIMO_POWER_ADJ_SW option of the NRDUCellDIMimo.HighLayerMuMimoSw parameter	This function optimizes power allocation between paired UEs to improve demodulation performance after UE pairing and increase the average downlink UE throughput in a cell.
Frequency-domain high-resolution power control	FREQ_HIGH_RESOLUTION_PWR_CTRL_SW option of the NRDUCellPdsch.DLPdschAlgoSwitch parameter	This function adjusts power allocation in the frequency domain for multi-layer pairing to improve demodulation performance after UE pairing and increase the average downlink UE throughput in a cell.

Function Name	Function Switch	Function Description
MU rank boosting	MU_RANK_BOOSTING_SW option of the NRDUCellDlMimo.HighLayerMuMimoSw parameter	This function adaptively selects UEs with different ranks for pairing based on service, pairing, and channel conditions, increasing ranks for UEs, downlink cell spectral efficiency, and average downlink UE throughput in a cell. This function is not supported by LampSite base stations.

Function Name	Function Switch	Function Description
CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	This function optimizes the number of available PDCCH CCEs to increase the downlink spectral efficiency and average downlink UE throughput in a cell. It is recommended that the following functions be enabled together with CCE resource optimization: • PDSCH resource optimization during UE access (controlled by the UE_BWP0_PDSCH_RES_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter): This function ensures UE access when CCE resource optimization takes effect in a heavy-load cell (for example, a cell that serves more than 300 UEs). It enables faster scheduling of signaling on the PDSCH during UE access. This increases the access success rate in a cell but may decrease the average UE and cell throughputs in both the uplink and downlink. • PDCCH blind detection assignment optimization (controlled by the PDCCH_BLIND_DET_ASSIGN_OPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter): This function balances blind detection times across different PDCCH aggregation levels. This prevents uplink and downlink CCE allocation failures due to insufficient blind detection times, which may occur when CCE resource optimization is in effect and the assignment of blind detection times to each PDCCH aggregation level is uneven. Certain UEs are incompatible with CCE resource optimization. To address this compatibility issue, it is recommended that blacklist control for dynamic PDCCH symbol allocation be enabled by selecting the PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF option of the gNBUECompatSw.BlacklistCtrlExtS-

Function Name	Function Switch	Function Description
		witch parameter. This blacklist control function disables CCE resource optimization for such UEs.

Weights for a Large Number of Layers

Pairing at a large number of layers requires weights of higher accuracy than pairing at a small number of layers. To obtain more accurate beamforming weights, functions dedicated to weights for a large number of layers consider service characteristics of UEs to optimize SRS measurement resource allocation, accelerate SRS measurement update, and select optimal weight policies. In this way, MU-MIMO performance improves and the average downlink UE throughput in a cell increases.

Table 7-3 describes the functions dedicated to weights for a large number of layers.

Table 7-3 Functions dedicated to weights for a large number of layers

Function Name	Function Switch	Function Description
SRS measurement acceleration	SRS_MEAS_ACCELERATING_SW option of the NRDUCellIDMimo.HighLayerMuMimoSw parameter Note: SRS measurement acceleration for a cell served by a MetaAAU only requires the preceding option to be selected. The MMIMO_MULTILAYER_ENHANCE_SW option of the NRDUCellIDMimo.HighLayerMuMimoSw parameter does not need to be selected, and the maximum number of downlink paired layers in the cell does not need to be greater than or equal to 8.	This function enables the gNodeB to optimize SRS measurement resource allocation based on UE service characteristics, which accelerates SRS measurement update. In this way, MU-MIMO performance improves and the average downlink UE throughput in a cell increases.

Function Name	Function Switch	Function Description
Weight adaptation during BWP switching	BWP_SWITCHING_WEIGHT_ADAPT_SW option of the NRDUCellDLMimo.HighLayerMuMimoSw parameter	This function determines the weight policy to achieve the optimal user experience during BWP switching, increasing the average downlink UE throughput.
Downlink hybrid precoding	DL_HYBRID_PRECODING_SW option of the NRDUCellPdschPrecode.PrecodingAlgorithmSwitch parameter	This function selects precoding modes based on sub-band SRSSs, increasing the average downlink cell throughput and average downlink UE throughput.

7.1.2 Network Analysis

7.1.2.1 Benefits

It is recommended that all functions of downlink massive MIMO multi-layer enhancement be enabled when the downlink PRB usage of a cell is greater than 30% and the SRS interference (**N.SRS.NI.Avg**) is greater than -110 dBm. If all the functions are enabled, the average downlink UE throughput increases. Enabling only one of the functions cannot achieve the optimal gains.

Under the same conditions, the higher the load, the higher the gains. Under the same traffic volume, the higher the SRS interference, downlink rank 3/4 proportion, and downlink MU-MIMO pairing rate, the higher the gains.

In light-load scenarios (for example, the PRB usage is less than 30%) and heavy-load scenarios (for example, the PRB usage is greater than 70%), some functions take effect in a lower proportion of occasions and provide lower or even no gains.

7.1.2.2 Impacts

The impacts between this function and other functions are described in [7.1.3.2.3 Function Impacts](#).

Table 7-4 describes the network impacts of the functions of downlink massive MIMO multi-layer enhancement.

Table 7-4 Network impacts of the functions of downlink massive MIMO multi-layer enhancement

Function Name	Network Impact
Interference suppression for a large number of layers	• The gain in the average downlink UE throughput decreases due to the limitation of the proportion of SRS-based weights that take effect in a cell with SRS remote interference avoidance enabled. • When UEs in the overlapping area between cells are scattered, a larger value configured for the blind SRS interference suppression degree will result in a larger increase in the average downlink UE throughput in the cells. When UEs in the overlapping area between cells are concentrated, a larger value may result in a decrease in the average downlink UE throughput in the cells. • Enabling this function on a medium- or light-load network may have the following impacts: – The average downlink rank of UEs decreases, the average downlink PRB usage of cells increases, the average number of extended PRBs used by the PDSCH (indicated by N.PRB.DL.ExtendUsed.Avg) increases, and the downlink MU-MIMO pairing rate may fluctuate. – When the padding rate at the MAC layer decreases, the traffic at the MAC layer also decreases. As a result, the average downlink cell throughput decreases. • The number of available PUSCH RBs decreases and accordingly the average uplink cell throughput and average uplink UE throughput decrease when one of the following conditions is met: (1) the uplink load of the cell is heavy (the uplink PRB usage is greater than or equal to 70%); (2) the uplink load of the cell is medium (the uplink PRB usage is greater than 40% and less than 70%) and there are burst services in the uplink in the cell. • If the settings of the BWP2 power saving switch are inconsistent between cells, the downlink cell and UE throughputs may not increase. • The average downlink cell power (indicated by N.DL.Pwr.Avg) may decrease. • The downlink IBLER of the cell and the uplink and downlink MCS indexes of the cell may fluctuate. • After PUCCH interference coordination is enabled, the cell will be reestablished.

Function Name	Network Impact
Scheduling for a large number of layers	• Assume that the cell is under medium load and the proportion of traffic volume of tail packets is large, or that the channel quality of UEs is good and UEs are centrally distributed. After the downlink MU-MIMO pairing rate increases, the average downlink UE throughput and average downlink cell throughput increase. However, the downlink throughput of some UEs decreases, and accordingly the number of samples with low average downlink UE throughput in the cell increases. • After the downlink MU-MIMO pairing rate increases, the following impacts may occur: the number of paired layers and the pairing probability in the cell increase, the interference between paired UEs in the cell increases, and therefore the downlink IBLER of the cell increases; the average downlink MCS index of UEs decreases, and the downlink PRB usage and average downlink rank of UEs fluctuate. • When a UE accesses or is handed over to an MU-MIMO cell from the cell edge, the access or handover success rate of the UE fluctuates. • The average downlink cell power (indicated by N.DL.Pwr.Avg) may decrease. • CEUs for which this function takes effect may cause strong interference to neighboring cells. • The downlink CCE usage and equivalent CCE usage of a cell may fluctuate. The average downlink CQI of the cell may fluctuate. The proportion of downlink 256QAM in the cell may decrease, while the proportion of downlink 64QAM or downlink QPSK may increase. The delay at the MAC layer and the delay of the first packet at the RLC layer of the cell may fluctuate. • The probability of uplink CCE allocation failures in a cell increases when the cell is heavily loaded. As a result, the average uplink cell throughput and average uplink UE throughput decrease. • When the downlink load of a cell is medium (30% < downlink PRB usage < 50%), the number of PDCCH CCEs available for the downlink in the cell may decrease. • If PDSCH resource optimization during UE access is enabled when CCE resource optimization has taken effect, the access success rate may increase, but the average UE and cell throughputs in both the uplink and downlink may decrease. • If PDCCH blind detection assignment optimization is enabled when CCE resource optimization has taken effect, the uplink and downlink CCE allocation success rates and average uplink and downlink UE throughputs increase, except in

Function Name	Network Impact
	scenarios with a large number of UEs at the cell edge, where these indicators may decrease.
Weights for a large number of layers	The possible impacts are as follows: • The number of RRC connection reconfiguration messages over the air interface increases. The service drop rate increases; the stronger the interference, the larger the increase in the service drop rate. • The DRX sleep time of UEs (indicated by N.Cdrx.Sleep.Dur.Total) decreases. • The downlink MU-MIMO pairing rate in a cell fluctuates.

7.1.3 Requirements

7.1.3.1 Licenses

The maximum number of layers for massive MIMO is subject to the licensed number of layers. For details, see *License Control Item Lists*.

Model	Description	Sales Unit
NR0S0DLE PU00	Massive MIMO DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

- Interference suppression for a large number of layers

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	PDSCH MU-MIMO is required for downlink massive MIMO multi-layer enhancement to take effect.
Spectral efficiency estimation	DL_SE_SU_MU_MIMO_ADAPT_SW option of the NRDUCellDlMimo.DLMuMimoSchSupplementSw parameter	<i>MIMO (TDD)</i>	Downlink anti-interference scheduling takes effect only when spectral efficiency estimation is enabled.
UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	Enabling blind SRS interference suppression requires UE-characteristic-based SRS period adaptation.
PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	None	Enabling PUCCH interference coordination requires PUCCH RB adaptation.
Cell-level PUCCH long format	NRDUCellPucch.StructureType set to LONG_STRUCTURE .	<i>Channel Management</i>	PUCCH interference coordination can be enabled only when a cell-level PUCCH long format is configured.

- Scheduling for a large number of layers

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	PDSCH MU-MIMO is required for downlink massive MIMO multi-layer enhancement to take effect.
DL SRS MU-MIMO Rank	NRDUCellDlRank.DlSrsMuMimoRank	<i>MIMO (TDD)</i>	MU rank boosting takes effect only when the NRDUCellDlRank.DlSrsMuMimoRank parameter is set to RANK_4 .

Function Name	Function Switch	Reference	Description
PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	CCE resource optimization can be enabled only when PDCCH symbol number adaptation is enabled.
Downlink cell bandwidth	NRDUCell.DlBandwidth	<i>Scalable Bandwidth</i>	CCE resource optimization requires the NRDUCell.DlBandwidth parameter to be set to CELL_BW_80M , CELL_BW_90M , or CELL_BW_100M .
Maximum Aggregated PDCCH Power Offset	NRDUCellChnPower.MaxPdcchAggPwrOffset	None	CCE resource optimization takes effect if the NRDUCellChnPower.MaxPdcchAggPwrOffset parameter is set to a value greater than or equal to 30.
Common Control Resource RB Number	NRDUCellCommonControlResourceSet.CommonCtrlResRbNum	None	CCE resource optimization takes effect if the NRDUCellCommonControlResourceSet.CommonCtrlResRbNum parameter is set to RB48 .

- Weights for a large number of layers

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	PDSCH MU-MIMO is required for downlink massive MIMO multi-layer enhancement to take effect.
DRX	BASIC_DRX_SW option of the NRDUCellUEPowerSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	Enabling SRS measurement acceleration requires DRX.

Function Name	Function Switch	Reference	Description
SRS- and PMI-based weight adaptation	DL_PMI_SRS_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>MIMO (TDD)</i>	SRS- and PMI-based weight adaptation must be enabled for downlink hybrid precoding to take effect.
UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	Enabling SRS measurement acceleration requires UE-characteristic-based SRS period adaptation.

7.1.3.2.2 Mutually Exclusive Functions

- Interference suppression for a large number of layers

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Blind SRS interference suppression and High-speed Railway Superior Experience cannot be both enabled.
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Blind SRS interference suppression cannot be enabled in hyper cells.

Function Name	Function Switch	Reference	Description
Uplink narrowband interference avoidance	UL_NARROWBAND_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	PUCCH interference coordination cannot be enabled when all the following conditions are met: • The UL_NARROWBAND_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter is selected. • The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_MANUAL. PUCCH interference coordination cannot be enabled when all the following conditions are met: • The PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter is selected. • The UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter is selected. • The DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter is selected. • The NRDUCellIntrfIdent.IntrfIdentificationMethod parameter is set to UL_DL_MANUAL.
External interference avoidance	UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter and NRDUCellIntrfIdent.IntrfIdentificationMethod set to UL_MANUAL	None	
PUCCH interference avoidance	PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	None	
Uplink cell-level interference avoidance	UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	
Downlink UE-level interference avoidance	DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	None	
Interference Identification Method	NRDUCellIntrfIdent.IntrfIdentificationMethod set to UL_DL_MANUAL	None	

Function Name	Function Switch	Reference	Description
DL Cell Interference Avoidance Policy	NRDUCellIntrfIdent.DlCellIntrfAvoidPol	None	<i>ntrfIdentification-Method</i> parameter is set to UL_DL_MANUAL .
Interference Direction	NRDUCellIntrfB and .Interference Direction	None	• The BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is deselected. PUCCH interference coordination cannot be enabled when all the following conditions are met: • The PUCCH_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter is selected. • The UL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent./ntrfOptimizationMethod parameter is selected. • The NRDUCellIntrfIdent./ntrfIdentification-Method parameter is set to UL_MANUAL. • The BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is deselected. PUCCH interference coordination cannot be enabled when all the following conditions are met: • The NRDUCellIntrfIdent.

Function Name	Function Switch	Reference	Description
			<i>DLCellIntrfAvoidPol</i> parameter is set to DL_INTRF_AVOID. • Among the options of the <i>NRDUCellIntrfIdent./intrfOptimizationMethod</i> parameter, only the UL_CELL_LEVEL_INTRF_AVOID option is selected. • The <i>NRDUCellIntrfIdent./intrfIdentificationMethod</i> parameter is set to UL_MANUAL. • The <i>NRDUCellIntrfBand./interferenceDirection</i> parameter is set to UPLINK_DOWNLINK.
Exclusive common PUCCH resources for RedCap UEs and eRedCap UEs	INDEP_REDCAP_COMMON_PUCCH_SW option of the <i>NRDUCellRedCap.RedCapAlgoSwitch</i> parameter	None	PUCCH interference coordination cannot be enabled together with "exclusive common PUCCH resources for RedCap UEs and eRedCap UEs".
CSI RB resource extension	CSI_RES_EXT_SW option of the <i>gNodeBAlgo.PucchSwitch</i> parameter	None	PUCCH interference coordination cannot be enabled together with CSI RB resource extension.

- Scheduling for a large number of layers

Function Name	Function Switch	Reference	Description
Bandwidth reconfiguration of NR active BWP	gNBDULteNrSpectShrCg.NrReservedRbStartIndex and gNBDULteNrSpectShrCg.NrReservedRbEndIndex	None	CCE resource optimization cannot be enabled when both LTE and NR are deployed, the LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter is selected, and the difference between the values of the gNBDULteNrSpectShrCg.NrReservedRbEndIndex and gNBDULteNrSpectShrCg.NrReservedRbStartIndex parameters is less than 189.
Common Control Resource RB Number	NRDUCellCoreset.CommonCtrlResRbNum set to RB96 or RB24	<i>Channel Management</i>	CCE resource optimization cannot be enabled when the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB96 or RB24 .
Slot configuration	NRDUCell.SlotAssignment set to 2_3_DSUUU	None	CCE resource optimization cannot be enabled when the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU .
Super uplink	SUPER_UL_TDM_SCH_SW , FLEX_SUPER_UPLINK_SW , or INTER_GNB_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	CCE resource optimization cannot be enabled when the SUPER_UL_TDM_SCH_SW , FLEX_SUPER_UPLINK_SW , or INTER_GNB_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is selected.

Function Name	Function Switch	Reference	Description
Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	CCE resource optimization cannot be enabled when the BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected.
Smart downlink repetition	DL_SMART_REPETITION_SW option of the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter	None	When CCE resource optimization is enabled, the smart downlink repetition switch must be turned off.
PLLC primary cell	PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	When CCE resource optimization is enabled, the PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter must be deselected.
Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Single DCI does not work with CCE resource optimization. They cannot be both enabled.
TRP-based weight selection for distributed massive MIMO	DM_MIMO_TRP_SEL_WEIGHT_SW option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	TRP-based weight selection for distributed massive MIMO does not work with frequency-domain high-resolution power control.

- Weights for a large number of layers

Function Name	Function Switch	Reference	Description
Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None	Downlink hybrid precoding cannot be enabled in a Bi-Carrier cell.
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Downlink hybrid precoding cannot be enabled in a cell with Fusion Cell enabled.

7.1.3.2.3 Function Impacts

- Interference suppression for a large number of layers

Function Name	Function Switch	Reference	Description
TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The proportion of occasions where this function takes effect may be affected after the downlink anti-interference scheduling function takes effect.
Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The proportion of occasions where this function takes effect may be affected after the downlink anti-interference scheduling function takes effect.
Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	The proportion of occasions where this function takes effect may be affected after the downlink anti-interference scheduling function takes effect.
RF channel power adaptation	RF_CHN_PWR_ADAPT_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Only downlink anti-interference scheduling takes effect when it is enabled together with RF channel power adaptation.
PRACH frequency-domain start position	NRDUCell.PrachFreqStartPosition	<i>Random Access Control</i>	The PRACH frequency-domain start position may affect the effect of PUCCH interference coordination.

Function Name	Function Switch	Reference	Description
Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DIRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has been enabled, downlink anti-interference scheduling does not take effect when rank 5 or higher is used for the user.

- Scheduling for a large number of layers

Function Name	Function Switch	Reference	Description
DL SRS MU-MIMO Rank Mode	NRDUCellDLRank.DISrsMuMimoRankMode set to FIX_MU_RANK	None	If the NRDUCellDLRank.DISrsMuMimoRankMode parameter is set to FIX_MU_RANK , MU rank boosting does not take effect.
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Frequency-domain high-resolution power control can take effect after RF channel intelligent shutdown takes effect.
Power reduction for CCUs	NRDUCellChnPwr.MaxUeSatPwrOffset or NRDUCellChnPwr.MaxBsEvmpwrOffset	<i>MIMO (TDD)</i>	After power reduction for CCUs takes effect, power is no longer allocated to CCUs for which frequency-domain high-resolution power control also takes effect.
Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Enabling CCE resource optimization may reduce the reliability of the low latency and high reliability function.
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	If CCE resource optimization is enabled, the proportion of occasions where Co-MM takes effect may decrease. It is recommended that the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter be selected for optimization.

Function Name	Function Switch	Reference	Description
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	If CCE resource optimization is enabled, the proportion of occasions where intra-base-station DL CoMP takes effect may decrease. It is recommended that the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter be selected for optimization.
UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	If UE bandwidth adaptation and CCE resource optimization are both enabled, the latter may not take effect. In this case, it is recommended that the PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter be also selected.

Function Name	Function Switch	Reference	Description
CCE resource allocation optimization	NRDUCellPdcchAlgo.PdcchUlCceRsvPeriod not set to NOT_CONFIG	<i>Channel Management</i>	Assume that one of the following options is selected: • AVERAGE_3DEIRP_CONTROL_SW option of the NRDUCellSmartPwrLock.SmartPowerLockSw parameter • INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter • INTER_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter • INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter If the CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter is selected, CCE resource allocation optimization does not take effect in slots with aperiodic CSI.
Rank optimization for cell-edge UEs	FAR_UE_RANK_OPT_SW option of the NRDUCellDLMimo.DLMuMimoSchSupplementSw parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Rank optimization for cell-edge UEs does not apply to UEs for which MU rank boosting is in effect.

- Weights for a large number of layers

Function Name	Function Switch	Reference	Description
DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	When SRS measurement acceleration is in effect, the DRX sleep time of UEs decreases and the effect of UE power saving deteriorates.

7.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For blind SRS interference suppression and MU rank boosting: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

For other functions:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable low-frequency RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.1.3.4 Cells

- CCE resource optimization
 - The cell must be a low-frequency TDD massive MIMO cell with a bandwidth of 80 MHz, 90 MHz, or 100 MHz.
 - If the **NRDUCell.LampSiteCellFlag** parameter is set to **YES**, the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter must be selected.
- Blind SRS interference suppression requires that the **NRDUCell.NrDuCellNetworkingMode** parameter be set to **NORMAL_CELL** or **FUSION_CELL**.
- Blind SRS interference suppression requires that the **NRDUCell.HighSpeedFlag** parameter not be set to **HIGH_SPEED**.
- Downlink hybrid precoding
 - The **NRDUCell.NrDuCellNetworkingMode** parameter must be set to a value other than **FUSION_CELL**.
 - If the **NRDUCell.LampSiteCellFlag** parameter is set to **YES**, the **NRDUCellTrp.TrpType** parameter must be set to **MASTER_DM_MIMO** or **MASTER_DM_MIMO_LIGHT**.

7.1.3.5 Others

- If CCE resource optimization is enabled and BWP2 is configured for UEs, it is recommended that BWP2 be configured with 51 RBs (**NRDUCellUePwrSaving.Bwp2MaxRbNum** set to **RB51**).
- Blind SRS interference suppression requires that RF modules support the eCPRI protocol. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

Table 7-5 describes the parameters used for function activation.

Table 7-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
High-Layer MU-MIMO Switch	NRDUCellDlMimo.HighLayerMuMimoSw	MMIMO_MULTILAYER_ENHANCE_SW	Select this option if downlink massive MIMO multi-layer enhancement is required.
Downlink Maximum MIMO Layer Number Quota	NRDUCellPdsch.MaxMimoLayerNum	None	Set this parameter to LAYER_8 or a value indicating a larger number of layers if downlink massive MIMO multi-layer enhancement is required.
High-Layer MU-MIMO Switch	NRDUCellDlMimo.HighLayerMuMimoSw	MU_RANK_BOOSTING_SW	Select this option if MU rank boosting is required.
High-Layer MU-MIMO Switch	NRDUCellDlMimo.HighLayerMuMimoSw	DL_MU_MIMO_POWER_ADJ_SW	Select this option if downlink MU-MIMO power adjustment is required.

Parameter Name	Parameter ID	Option	Setting Notes
High-Layer MU-MIMO Switch	NRDUCellDlMimo.HighLayerMuMimoSw	MU_MIMO_PAIRING_PREFERRED_SW	Select this option if downlink MU-MIMO pairing preference is required.
High-Layer MU-MIMO Switch	NRDUCellDlMimo.HighLayerMuMimoSw	SRS_BLIND_IS_SW	Select this option if blind SRS interference suppression is required.
High-Layer MU-MIMO Switch	NRDUCellDlMimo.HighLayerMuMimoSw	SRS_MEAS_ACCELERATING_SW	Select this option if SRS measurement acceleration is required.
High-Layer MU-MIMO Switch	NRDUCellDlMimo.HighLayerMuMimoSw	BWP_SWITCHING_WEIGHT_ADAPT_SW	Select this option if weight adaptation during BWP switching is required.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgoSwitch	DL_ANTI_INTRF_SCH_SW	Select this option if downlink anti-interference scheduling is required.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgoSwitch	FREQ_HIGH_RESOLUTION_PWR_CTRL_SW	Select this option if frequency-domain high-resolution power control is required.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DlPdschAlgoSwitch	DL_HIGH_EFF_SCH_SW	Select this option if downlink high-efficiency scheduling is required.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Switch	NRDUCellPucch.<i>PucchAlgoSwitch</i>	PUCCH_INTRF_COORD_SW	You are advised to select this option for both the local cell and neighboring cells when inter-cell PUCCH interference is high. You are advised to deselect this option in other scenarios. Before selecting this option for cells, you are advised to plan their PCIs based on the values of PCIs mod 3. Otherwise, interference cannot be reduced. Note: This option is a high-risk option. Selecting it will cause the cell to be reestablished.
DL MU-MIMO Back to SU-MIMO Gain Thld	NRDUCellDIMimo.<i>DLMUBackToSuSeThld</i>	None	It is recommended that this parameter be set to 0 if downlink MU-MIMO pairing preference is required.
SRS Blind Intrf Suppression Degree	NRDUCellSrs.<i>SrsBlindISDegree</i>	None	It is recommended that this parameter be set to 4 if blind SRS interference suppression is required.

Parameter Name	Parameter ID	Option	Setting Notes
Fixed AMC Step Value	NRDUCellPdsch.FixedAmcStepValue	None	If the cell IBLER is less than 5%, the value 30 is recommended. In other scenarios, the default value is recommended.
Precoding Algorithm Switch	NRDUCellPdschPrecodingAlgorithmSwitch	DL_HYBRID_PRECODING_SW	Select this option if downlink hybrid precoding is required.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcchPdcchAlgorithmEnhancementSwitch	CCE_RESOURCE_OPT_SW	Select this option if CCE resource optimization is required.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcchPdcchAlgorithmEnhancementSwitch	UE_BWP0_PDSCH_RES_OPT_SW	You are advised to select this option if the access success rate needs to be increased for a heavy-load cell (for example, a cell that serves more than 300 UEs) with CCE resource optimization enabled. You are advised to deselect this option in other scenarios.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Switch	NRDUCellPdcch.PdcchAlgorithmSwitch	PDCCH_BLIND_DET_ASSIGN_OPT_SW	You are advised to select this option if CCE resource optimization is required. The purpose is to balance blind detection times across different PDCCH aggregation levels, so as to prevent uplink and downlink CCE allocation failures due to insufficient blind detection times.
Blacklist Control Extension Switch	gNBUECompatSw.BlacklistCtrlExtSwitch	PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF	Select this option if there are UEs incompatible with CCE resource optimization on the network.
Small Packet Type1 Robust Scheduling Policy	NRDUCellDISchRes.SmallPktType1RobustSchedulingPolicy	None	It is recommended that this parameter be set to LEVEL3 if downlink massive MIMO multi-layer enhancement is required.
Downlink Schedule Algorithm Switch	NRDUCellDISch.DLScheduleAlgorithmSwitch	TAIL_PKT_MCS_OPT_SW	It is recommended that this option be selected if downlink massive MIMO multi-layer enhancement is required.

Parameter Name	Parameter ID	Option	Setting Notes
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDLMimo.DLMuMimoSchSupplementSw	DL_MU_ADAPT_ESTIMATE_MODE_SW	When both downlink anti-interference scheduling and PDSCH MU-MIMO are enabled, it is recommended that this option be selected to increase the accuracy of resource estimation for MU-MIMO.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to [7.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the maximum number of downlink MIMO layers
MOD NRDUCELLPDSCH: NrDuCellId=0, MaxMimoLayerNum=LAYER_8;
//Turning on the downlink massive MIMO multi-layer enhancement switch
MOD NRDUCELLDLMIMO: NrDuCellId=0, HighLayerMuMimoSw=MMIMO_MULTILAYER_ENHANCE_SW-1;
//Enabling downlink anti-interference scheduling, frequency-domain high-resolution power control, and
downlink high-efficiency scheduling
MOD NRDUCELLPDSCH: NrDuCellId=0,
DiPdschAlgoSwitch=DL_ANTI_INTRF_SCH_SW-1&FREQ_HIGH_RESOLUT_PWR_CTRL_SW-1&DL_HIGH_EFF_SC
H_SW-1;
//Configuring the robust scheduling policy during type-1 resource allocation for small-packet services
MOD NRDUCELLDLSCHRES: NrDuCellId=0, SmallPktType1RobustSchPol=LEVEL3;
//Configuring MCS optimization for tail packets
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchAlgoSwitch=TAIL_PKT_MCS_OPT_SW-1;
//Enabling MU rank boosting
MOD NRDUCELLDLMIMO: NrDuCellId=0, HighLayerMuMimoSw=MU_RANK_BOOSTING_SW-1;
//Enabling PUCCH interference coordination
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_INTRF_COORD_SW-1;
//Enabling downlink MU-MIMO power adjustment, downlink MU-MIMO pairing preference, SRS
measurement acceleration, and weight adaptation during BWP switching
MOD NRDUCELLDLMIMO: NrDuCellId=0, DiMuBackToSuSeThId=0,
HighLayerMuMimoSw=MU_MIMO_PAIRING_PREFERRED_SW-1&SRS_MEAS_ACCELERATING_SW-1&BWP_SW
ITCHING_WEIGHT_ADAPT_SW-1&DL_MU_MIMO_PWR_ADJ_SW-1;
//Enabling blind SRS interference suppression
MOD NRDUCELLDLMIMO: NrDuCellId=0, HighLayerMuMimoSw=SRS_BLIND_IS_SW-1;
MOD NRDUCELLSRS: NrDuCellId=0, SrsBlindIsDegree=4;
```

```
//Enabling downlink hybrid precoding
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingAlgoSwitch=DL_HYBRID_PRECODING_SW-1;
//Enabling CCE resource optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=CCE_RESOURCE_OPT_SW-1;
//Enabling PDSCH resource optimization during UE access if the access success rate needs to be increased
for a heavy-load cell (for example, a cell that serves more than 300 UEs) with CCE resource optimization
enabled
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=UE_BWP0_PDSCH_RES_OPT_SW-1;
//((Recommended when CCE resource optimization is enabled) Enabling PDCCH blind detection assignment
optimization, which aims to balance blind detection times across different PDCCH aggregation levels, so as
to prevent uplink and downlink CCE allocation failures due to insufficient blind detection times
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-1;
//Enabling blacklist control for dynamic PDCCH symbol allocation if there are UEs incompatible with CCE
resource optimization on the network (using the UeFeatureValueContent parameter set to
ec7fee769ff7e0fff0 as an example)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff0",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF-1;
//((Optional; recommended when both downlink anti-interference scheduling and PDSCH MU-MIMO are
enabled) Enabling adaptive estimation mode for downlink MU-MIMO to increase the accuracy of resource
estimation for MU-MIMO
MOD NRDUCELLDLMIMO: NrDuCellId=0,
DLMuMimoSchSupplementSw=DL_MU_ADAPT_ESTIMATE_MODE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink anti-interference scheduling, frequency-domain high-resolution power control, and
downlink high-efficiency scheduling
MOD NRDUCELLPDSCH: NrDuCellId=0,
DlPdschAlgoSwitch=DL_ANTI_INTRF_SCH_SW-0&FREQ_HIGH_RESOLUT_PWR_CTRL_SW-0&DL_HIGH_EFF_SC
H_SW-0;
//Disabling MU rank boosting
MOD NRDUCELLDLMIMO: NrDuCellId=0, HighLayerMuMimoSw=MU_RANK_BOOSTING_SW-0;
//Disabling PUCCH interference coordination
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_INTRF_COORD_SW-0;
//Disabling downlink MU-MIMO power adjustment, MU-MIMO pairing preference, SRS measurement
acceleration, and weight adaptation during BWP switching
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuBackToSuSeThld=5,
HighLayerMuMimoSw=MU_MIMO_PAIRING_PREFERRED_SW-0&SRS_MEAS_ACCELERATING_SW-0&BWP_SW
ITCHING_WEIGHT_ADAPT_SW-0&DL_MU_MIMO_PWR_ADJ_SW-0;
//Disabling blind SRS interference suppression
MOD NRDUCELLDLMIMO: NrDuCellId=0, HighLayerMuMimoSw=SRS_BLIND_IS_SW-0;
//Disabling downlink hybrid precoding
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, PrecodingAlgoSwitch=DL_HYBRID_PRECODING_SW-0;
//Disabling blacklist control for dynamic PDCCH symbol allocation
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF-0;
//Disabling CCE resource optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=CCE_RESOURCE_OPT_SW-0;
//Turning off the downlink massive MIMO multi-layer enhancement switch
MOD NRDUCELLDLMIMO: NrDuCellId=0, HighLayerMuMimoSw=MMIMO_MULTILAYER_ENHANCE_SW-0;
//Disabling PDSCH resource optimization during UE access
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=UE_BWP0_PDSCH_RES_OPT_SW-0;
//Disabling PDCCH blind detection assignment optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-0;
//Disabling adaptive estimation mode for downlink MU-MIMO
MOD NRDUCELLDLMIMO: NrDuCellId=0,
DLMuMimoSchSupplementSw=DL_MU_ADAPT_ESTIMATE_MODE_SW-0;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

Functions related to interference suppression for a large number of layers have taken effect for a cell if its average downlink rank decreases and its average downlink MCS index increases.

Functions related to scheduling for a large number of layers have taken effect for a cell if its downlink IBLER increases and its average transmit power of RF modules decreases.

Functions related to weights for a large number of layers have taken effect for a cell if its average downlink MCS index and average downlink rank increase.

7.1.4.3 Network Monitoring

Monitor the following throughput indicators:

- **User Downlink Average Throughput (DU)**
- Average uplink cell throughput = $N.ThpVol.UL / N.RLC.ThpTime.UL.Cell$
- Average downlink cell throughput = $N.ThpVol.DL / N.RLC.ThpTime.DL.Cell$
- Average uplink user-perceived rate = $(N.ThpVol.UL - N.ThpVol.UE.UL.NonSpecSmallPkt) / N.ThpTime.UE.UL.RmvNonSpecSmallPkt$

In addition, monitor the following network monitoring indicators:

- Average downlink MCS index = $SUM(k \times N.ChMeas.PDSCH.MCS.k) / SUM(N.ChMeas.PDSCH.MCS.k)$
 $k = 0$ to 28
- Downlink PRB usage = $N.PR.B.DL.Used.Avg / N.PR.B.DL.Avail.Avg$
- Average number of layers paired for downlink MIMO = $N.ChMeas.MIMO.DL.Pair.Layer.Avg$
- Downlink IBLER = $(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler) / (N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)$
- Access success rate = $N.RRC.SetupReq.Succ / N.RRC.SetupReq.Att$
- SA UE context drop rate = $N.UECntx.AbnormRel / (N.UECntx.AbnormRel + N.UECntx.NormRel)$
- DRX sleep time = $N.Cdrx.Sleep.Dur.Total$

7.2 Uplink Massive MIMO Multi-Layer Enhancement

7.2.1 Principles

Uplink massive MIMO multi-layer enhancement improves the uplink pairing performance and increases the uplink UE throughput in TDD massive MIMO cells

with the maximum number of uplink paired layers (specified by the **NRDUCellPusch.MaxMimoLayerCnt** parameter) being at least four.

This function is controlled by the **UL_MMIMO_MULTILAYER_ENHANCE_SW** option of the **NRDUCellULMimo.UlHighLayerMuMimoSwitch** parameter. After this option is selected and the maximum number of uplink paired layers in a cell is configured, the switches controlling the sub-functions need to be turned on for the corresponding sub-functions to take effect.

Uplink massive MIMO multi-layer enhancement optimizes scheduling and pairing. The following describes its functions.

Scheduling for a Large Number of Layers

When a large number of layers are paired in a massive MIMO cell, interference significantly affects uplink measurement, demodulation performance, and eventually user experience. To increase the uplink user-perceived rates, functions dedicated to scheduling for a large number of layers are introduced. These functions consider UE channel conditions, scheduling characteristics, and differences in channel quality between beams during UE pairing to determine the optimal MCSs, beams, and number of RBs for paired UEs.

Table 7-6 describes the functions dedicated to scheduling for a large number of layers.

Table 7-6 Functions dedicated to scheduling for a large number of layers

Function Name	Function Switch	Function Description
Intelligent uplink multi-layer SINR prediction	UL_SU_SINR_INTEL_PREDICT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	This function enables the gNodeB to start training intelligent SINR prediction models for uplink SU-MIMO. The models are used to select the optimal MCS based on UE characteristics, increasing the average uplink UE throughput. The status of the models can be monitored using the prediction error. (The upper limit to the prediction error is specified by the NRDUCellAiAlgo.UISuPredictErrorUpperLimit parameter.) This function stops taking effect when the proportion of RBs paired for MU-MIMO is greater than the threshold specified by the NRDUCellAiAlgo.UISuMimoModelSuspendThld parameter. After intelligent uplink multi-layer SINR prediction is enabled, it is recommended that adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table (detailed in <i>Modulation Schemes</i>) and PUSCH time-domain enhancement (detailed in 4.1.3.1 Uplink Optimization) be enabled together to improve the performance of CEUs.
Multi-layer demodulation performance enhancement	MULTILAYER_DEMOD_ENH_SW option of the NRDUCellUIMimo.UlHighLayerMuMimoSwitch parameter	After this option is selected for a low-frequency TDD 64T64R cell, the function enhances reception and processing capabilities for paired UEs, selects the optimal beams for pairing, improves demodulation performance for these UEs, and increases the average uplink UE throughput.
Resource-based adaptive uplink scheduling	RES_BASED_ADAPT_ULSCH_SW option of the NRDUCellUISch.UlSchAlgoSwitch parameter	This function enables the gNodeB to allocate a more appropriate number of RBs to each UE during scheduling based on the amount of buffered data, increasing the average uplink UE throughput.

Function Name	Function Switch	Function Description
Latency-based adaptive uplink scheduling	LATENCY_BASED_ADAPT_ULSCH_SW option of the NRDUCellULSch.UlSchAlgoSwitch parameter	This function accelerates the scheduling of UE reports on buffered data, shortening the scheduling delay and increasing the average uplink UE throughput.

Pairing of a Large Number of Layers

When a large number of layers are paired in a massive MIMO cell, uplink pairing is subject to the RB lengths and ranks of UEs, affecting pairing performance in the cell. To improve pairing performance and increase the uplink user-perceived rate, functions dedicated to pairing of a large number of layers are introduced. These functions consider the RB lengths and ranks of UEs to adaptively adjust RB allocation and ranks for paired UEs.

Table 7-7 describes the functions dedicated to pairing of a large number of layers.

Table 7-7 Functions dedicated to pairing of a large number of layers

Function Name	Function Switch	Function Description
Flexible MU-MIMO pairing	MU_MIMO_FLEX_PAIR_SW option of the NRDUCellULMimo.UlHighLayerMuMimoSwitch parameter	This function allows UEs with different RB lengths to be paired when certain conditions are met, increasing the average uplink cell throughput.
Adaptive MU-MIMO rank selection	MU_MIMO_RANK_ADAPTIVE_SEL_SW option of the NRDUCellULMimo.UlHighLayerMuMimoSwitch parameter	This function checks whether rank 2 produces a higher spectral efficiency than rank 1 during scheduling of rank 2 UEs. If not, rank 1 is used for scheduling.

7.2.2 Network Analysis

7.2.2.1 Benefits

It is recommended that all functions of uplink massive MIMO multi-layer enhancement be enabled when the uplink PRB usage of a cell is greater than 30% and there is uplink interference. If all the functions are enabled, the average uplink UE throughput increases. Enabling only one of the functions cannot achieve the optimal gains.

Under the same conditions, the gains increase with the load, uplink interference, number of UEs, number of paired RBs, and uplink CCE allocation success rate.

In light-load scenarios (for example, the PRB usage is less than 30%) and heavy-load scenarios (for example, the PRB usage is greater than 70%), some functions take effect in a lower proportion of occasions and provide lower or even no gains.

7.2.2.2 Impacts

The impacts between this function and other functions are described in [7.2.3.2.3 Function Impacts](#).

Table 7-8 describes the network impacts of the functions of uplink massive MIMO multi-layer enhancement.

Table 7-8 Network impacts of the functions of uplink massive MIMO multi-layer enhancement

Function Name	Network Impact
Scheduling for a large number of layers	The possible impacts are as follows: • The uplink IBLER, RBLER, and retransmission proportion of a cell fluctuate. • The padding rate at the MAC layer of a cell decreases. As a result, the uplink traffic volume at the MAC layer of the cell and the average uplink throughput of the cell decrease. • The average uplink UE throughput fluctuates when air interface resources are insufficient (for example, when the uplink cell PRB usage is greater than 70% or uplink CCE congestion occurs). • The gain in the uplink user-perceived rate decreases when the proportion of UEs in the cell center increases. • After uplink cell performance is improved, the number of downlink status reports increases, which decreases the average downlink cell throughput and average downlink UE throughput.

Function Name	Network Impact
Pairing of a large number of layers	• The increase in the uplink MU-MIMO pairing rate in a cell may have the following impacts: – The number of paired layers and the pairing rate in the cell increase, and the interference between paired UEs in the cell increases. As a result, the uplink IBLER of the cell increases, the average uplink MCS index and average uplink rank of UEs decrease, the average uplink throughput decreases and the PUSCH RSRP increases for some UEs, the uplink PRB usage of the cell decreases, and the number of uplink paired RBs in the cell fluctuates. – The padding rate at the MAC layer of the cell decreases. As a result, the uplink traffic volume at the MAC layer of the cell and the average uplink throughput of the cell decrease. – The number of downlink cell status reports increases, which decreases the average downlink cell throughput and average downlink UE throughput. • When a UE accesses or is handed over to an MU-MIMO cell from the cell edge, the access or handover success rate of the UE slightly decreases.

7.2.3 Requirements

7.2.3.1 Licenses

The maximum number of layers for massive MIMO is subject to the licensed number of layers. For details, see *License Control Item Lists*.

Model	Description	Sales Unit
NR0S0ULE PU00	Massive MIMO UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch . MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Uplink massive MIMO multi-layer enhancement requires PUSCH MU-MIMO.
Uplink Maximum MIMO Layer Count Quota	NRDUCellPusch . MaxMimoLayerCnt	<i>MIMO (TDD)</i>	Uplink massive MIMO multi-layer enhancement requires the NRDUCellPusch . MaxMimoLayerCnt parameter to be set to LAYER_4 , LAYER_6 , LAYER_8 , LAYER_10 , or LAYER_12 .
Uplink scheduling enhancement	UL_SCHEDULE_ENH_SW option of the NRDUCellPusch . ULPuschAlgoSwitch parameter	None	Pairing of a large number of layers requires uplink scheduling enhancement.
Alignment pairing optimization for small-packet UEs	SML_PKT_ALIGN_PAIR_OPT_SW option of the NRDUCellULMimo . ULMuMimoAlgorithmSwitch parameter	<i>MIMO (TDD)</i>	Flexible MU-MIMO pairing requires alignment pairing optimization for small-packet UEs.
Alignment pairing optimization for large-packet UEs	BIG_PKT_ALIGN_PAIR_OPT_SW option of the NRDUCellULMimo . ULMuMimoAlgorithmSwitch parameter	<i>MIMO (TDD)</i>	Flexible MU-MIMO pairing requires alignment pairing optimization for large-packet UEs.

Function Name	Function Switch	Reference	Description
Frequency band	NRDUCell.FrequencyBand	<i>Cell Management</i>	Flexible MU-MIMO pairing can be enabled only when the NRDUCell.FrequencyBand parameter is set to N38 , N40 , N41 , N77 , N78 , or N79 .

7.2.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Slot configuration	NRDUCell.SlotAssignment set to 2_3_DSUUU	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , multi-layer modulation performance enhancement cannot be enabled.
Distributed massive MIMO (low-frequency TDD)	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	If the DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter is selected, intelligent uplink multi-layer SINR prediction cannot be enabled.
MU non-alignment pairing	MU_NONALIGNMENT_PAIR_SW option of the NRDUCellUIMimo.UlMuMimoAlgorithmSwitch parameter	None	If flexible MU-MIMO pairing is enabled, MU non-alignment pairing must be disabled.
MU DMRS non-orthogonal multiplexing	MU_DMRS_NONORTH_MULTIPLEX_SW option of the NRDUCellDmrs.UlDmrsSwitch parameter	None	If flexible MU-MIMO pairing is enabled, MU DMRS non-orthogonal multiplexing must be disabled.

Function Name	Function Switch	Reference	Description
Uplink DMRS type	NRDUCellPusch.UIDmrsType	None	If flexible MU-MIMO pairing is enabled, the NRDUCellPusch.UIDmrsType parameter must be set to TYPE1 .

7.2.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When RF channel intelligent shutdown is in effect, intelligent uplink multi-layer SINR prediction does not take effect.
PUSCH scheduling based on SRS interference avoidance	SRS_INTRF_AVOID_PUSCH_SCH_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	After PUSCH scheduling based on SRS interference avoidance, remote interference adaptive suppression, or remote interference resistance for PUSCH demodulation takes effect, the proportion of occasions where intelligent uplink multi-layer SINR prediction takes effect will be affected and accordingly the gains of intelligent uplink multi-layer SINR prediction may be lowered.
Remote interference adaptive suppression	RIM_ADAPT_SUPPRESSION_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	
Remote interference resistance for PUSCH demodulation	RIM_PUSCH_DEMOD_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	
Closed-loop PUSCH antenna selection	PUSCH_CLOSED_LOOP_ANT_SEL_SW option of the NRDUCellChnCovAlgo.UICoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Closed-loop PUSCH antenna selection and intelligent uplink multi-layer SINR prediction cannot take effect at the same time, and the former preferentially takes effect.

Function Name	Function Switch	Reference	Description
Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_SLOT_SW options of the NRDUCellSrs.SrsAlgorithmSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	After proactive avoidance of remote interference takes effect, the proportion of occasions where intelligent uplink multi-layer SINR prediction takes effect will be affected and accordingly the gains of intelligent uplink multi-layer SINR prediction may be lowered.
Uplink PAMC	UL_PAMC_SWITCH option of the NRDUCellPusch.UlPuschAlgorithmSwitch parameter	None	If uplink PAMC and intelligent uplink multi-layer SINR prediction are both enabled, the uplink IBLER and uplink throughput of the cell may fluctuate.
Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	The proportion of occasions where UL CoMP takes effect may decrease after multi-layer demodulation performance enhancement takes effect in heavy-load overlapping areas where the conditions for UL CoMP to take effect are met.
Uplink scheduling on a per RB basis	UL_1RB_SCH_SWITCH option of the NRDUCellPusch.UlPuschAlgorithmSwitch parameter	None	Uplink scheduling on a per RB basis takes effect preferentially when it is enabled together with multi-layer demodulation performance enhancement.

7.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

All functions of uplink massive MIMO multi-layer enhancement are supported by 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. Among these functions, intelligent uplink multi-layer SINR prediction requires that 5900 series base stations not be configured with BBU5900C or BookBBU5901.

In addition, resource-based adaptive uplink scheduling, latency-based adaptive uplink scheduling, and adaptive MU-MIMO rank selection are also supported by DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards and baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Intelligent uplink multi-layer SINR prediction is supported by the UMPTe only in NR-only scenarios.

The model specifications for intelligent uplink multi-layer SINR prediction vary according to the main control board in use and its working mode. [Table 7-9](#) lists the model specifications.

Table 7-9 Model specifications

Main Control Board	Working Mode	Model Specifications
UMPTg	NR-only	36
	Co-MPT multiple modes	18
UMPTga	NR-only	18
	Co-MPT multiple modes	9
UMPTe	NR-only	18
	Co-MPT multiple modes	0

RF Modules

Intelligent uplink multi-layer SINR prediction, resource-based adaptive uplink scheduling, latency-based adaptive uplink scheduling, and adaptive MU-MIMO rank selection are supported by all NR TDD low-frequency RF modules with at least 32T32R. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Flexible MU-MIMO pairing is supported by eCPRI-compliant NR TDD low-frequency RF modules with at least 32T32R. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Multi-layer demodulation performance enhancement is supported by eCPRI-compliant NR TDD low-frequency 64T64R RF modules. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.2.3.4 Cells

- When uplink massive MIMO multi-layer enhancement is enabled:
 - NRDUCell.NrDuCellNetworkingMode** can only be set to **NORMAL_CELL**.
 - NRDUCell.HighSpeedFlag** can only be set to **LOW_SPEED**.

- When resource-based or latency-based adaptive uplink scheduling is enabled and the **NRDUCell.SlotAssignment** parameter is set to **4_1_DDDSU**, uplink massive MIMO multi-layer enhancement may provide smaller or even no gains. Therefore, it is recommended that the slot configuration not be 4:1 when resource-based or latency-based adaptive uplink scheduling is used.

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

Table 7-10 describes the parameters used for function activation.

Table 7-10 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL High-Layer MU-MIMO Switch	NRDUCellULMimo.UlHighLayerMuMimoSwitch	UL_MMIMO_MULTILAYER_ENHANCE_SW	Select this option if uplink massive MIMO multi-layer enhancement is required.
Uplink Maximum MIMO Layer Count Quota	NRDUCellPusch.MaxMimoLayerCnt	None	Set this parameter to LAYER_4 or a value indicating a larger number of layers before uplink massive MIMO multi-layer enhancement is used.
AI AMC Algorithm Switch	NRDUCellAiAlgo.AiAmcAlgoSwitch	UL_SU_SINR_INTELL_PREDICT_SW	Select this option if intelligent uplink multi-layer SINR prediction is required.
UL SU-MIMO Predict Error Upper Limit	NRDUCellAiAlgo.UlSuPredictErrorUpperLimit	None	The value 23 is recommended if intelligent uplink multi-layer SINR prediction is required.

Parameter Name	Parameter ID	Option	Setting Notes
UL SU-MIMO Model Suspend Threshold	NRDUCellAiAlgo.UISuMimoModelSuspendThld	None	The value 100 is recommended if intelligent uplink multi-layer SINR prediction is required.
UL High-Layer MU-MIMO Switch	NRDUCellULMimo.UlHighLayerMuMimoSwitch	MULTILAYER_DEMOD_ENH_SW	Select this option if multi-layer demodulation performance enhancement is required.
UL High-Layer MU-MIMO Switch	NRDUCellULMimo.UlHighLayerMuMimoSwitch	MU_MIMO_FLEX_PAIR_SW	Select this option if flexible MU-MIMO pairing is required.
UL High-Layer MU-MIMO Switch	NRDUCellULMimo.UlHighLayerMuMimoSwitch	MU_MIMO_RANK_ADAPTIVE_SEL_SW	Select this option if adaptive MU-MIMO rank selection is required.
UL Scheduling Algorithm Switch	NRDUCellULSch.UlSchedulingSwitch	RES_BASED_ADAPT_ULSCH_SW	Select this option if resource-based adaptive uplink scheduling is required.
UL Scheduling Algorithm Switch	NRDUCellULSch.UlSchedulingSwitch	LATENCY_BASED_ADAPT_ULSCH_SW	Select this option if latency-based adaptive uplink scheduling is required.
Flexible Pairing RB Ratio Threshold	NRDUCellULMimo.FlexPairRbRatioThld	None	Use the recommended value.
Flexible Pairing Large Packet RB Ratio Thld	NRDUCellULMimo.FlexPairLrgPktRbRatioThld	None	Use the recommended value.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to [7.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTE

If intelligent uplink multi-layer SINR prediction is enabled, it is recommended that both PUSCH time domain enhancement and adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table be enabled to improve the performance of CEUs. For MML command examples of the two functions, see *Modulation Schemes* and [4.4.1.2 Using MML Commands](#), respectively.

Activation Command Examples

```
//Setting the maximum number of uplink MIMO layers
MOD NRDUCELLPUSCH: NrDuCellId=0, MaxMimoLayerCnt=LAYER_4;
//Turning on the uplink massive MIMO multi-layer enhancement switch
MOD NRDUCELLULMIMO: NrDuCellId=0,
UlHighLayerMuMimoSwitch=UL_MMIMO_MULTILAYER_ENHANCE_SW-1;
//Enabling intelligent uplink multi-layer SINR prediction
MOD NRDUCELLAIALGO: NrDuCellId=0, AiAmcAlgoSwitch=UL_SU_SINR_INTEL_PREDICT_SW-1,
UISuPredictErrorUpperLimit=23, UISuMimoModelSuspendThld=100;
//Enabling multi-layer demodulation performance enhancement
MOD NRDUCELLULMIMO: NrDuCellId=0, UlHighLayerMuMimoSwitch=MULTILAYER_DEMOD_ENH_SW-1;
//Enabling flexible MU-MIMO pairing
MOD NRDUCELLULMIMO: NrDuCellId=0, UlHighLayerMuMimoSwitch=MU_MIMO_FLEX_PAIR_SW-1,
FlexPairRbRatioThld=100, FlexPairLrgPktRbRatioThld=20;
//Enabling adaptive MU-MIMO rank selection
MOD NRDUCELLULMIMO: NrDuCellId=0,
UlHighLayerMuMimoSwitch=MU_MIMO_RANK_ADAPTIVE_SEL_SW-1;
//Enabling resource-based adaptive uplink scheduling
MOD NRDUCELLULSCH: NrDuCellId=0, UISchAlgoSwitch=RES_BASED_ADAPT_ULSCH_SW-1;
//Enabling latency-based adaptive uplink scheduling
MOD NRDUCELLULSCH: NrDuCellId=0, UISchAlgoSwitch=LATENCY_BASED_ADAPT_ULSCH_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling intelligent uplink multi-layer SINR prediction
MOD NRDUCELLAIALGO: NrDuCellId=0, AiAmcAlgoSwitch=UL_SU_SINR_INTEL_PREDICT_SW-0;
//Disabling multi-layer demodulation performance enhancement
MOD NRDUCELLULMIMO: NrDuCellId=0, UlHighLayerMuMimoSwitch=MULTILAYER_DEMOD_ENH_SW-0;
//Disabling flexible MU-MIMO pairing
MOD NRDUCELLULMIMO: NrDuCellId=0, UlHighLayerMuMimoSwitch=MU_MIMO_FLEX_PAIR_SW-0;
//Disabling adaptive MU-MIMO rank selection
MOD NRDUCELLULMIMO: NrDuCellId=0,
UlHighLayerMuMimoSwitch=MU_MIMO_RANK_ADAPTIVE_SEL_SW-0;
//Disabling resource-based adaptive uplink scheduling
MOD NRDUCELLULSCH: NrDuCellId=0, UISchAlgoSwitch=RES_BASED_ADAPT_ULSCH_SW-0;
//Disabling latency-based adaptive uplink scheduling
MOD NRDUCELLULSCH: NrDuCellId=0, UISchAlgoSwitch=LATENCY_BASED_ADAPT_ULSCH_SW-0;
//Turning off the uplink massive MIMO multi-layer enhancement switch
MOD NRDUCELLULMIMO: NrDuCellId=0,
UlHighLayerMuMimoSwitch=UL_MMIMO_MULTILAYER_ENHANCE_SW-0;
```


7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

Scheduling for a large number of layers has taken effect in a cell if the uplink IBLER of the cell changes or the UE scheduling delay (indicated by the **N.ThpTime.UE.UL.RmvSmallPkt** counter) decreases. Among its functions, intelligent uplink multi-layer SINR prediction can also be observed using an MML command. Specifically, this function has taken effect if the **DSP NRDUCELLAISCHMODEL** command output shows that the **AI Model Status** is **Unavailable, Building, In Use, or Updating**.

Pairing of a large number of layers has taken effect in a cell if the number of uplink paired layers in the cell increases.

7.2.4.3 Network Monitoring

Monitor the following throughput indicators:

- **User Downlink Average Throughput (DU)**
- **User Uplink Average Throughput (DU)**
- Average uplink cell throughput = $N.ThpVol.UL / N.RLC.ThpTime.UL.Cell$
- Average downlink cell throughput = $N.ThpVol.DL / N.RLC.ThpTime.DL.Cell$

In addition, monitor the indicators listed in [Table 7-11](#).

Table 7-11 Indicators to be monitored

Indicator	Definition
Uplink PRB usage	$N.PR.B.UL.Used.Avg / N.PR.B.UL.Avail.Avg$
Average uplink MCS index	$SUM(k \times N.ChMeas.PUSCH.MCS.k) / SUM(N.ChMeas.PUSCH.MCS.k)$ $k = 0 \text{ to } 28$
Average uplink rank	$SUM(k \times N.PUSCH.InitTbUL.Rank.k) / SUM(N.PUSCH.InitTbUL.Rank.k)$ $k = 1 \text{ to } 2$
Average number of layers paired for uplink MIMO	$N.ChMeas.MIMO.UL.Pair.Layer.Avg$

Indicator	Definition
Uplink IBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)}$
Uplink RBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)}$
Average uplink CCE aggregation level	$\frac{(N.CCE.UL.AggLvl16Num \times 16 + N.CCE.UL.AggLvl8Num \times 8 + N.CCE.UL.AggLvl4Num \times 4 + N.CCE.UL.AggLvl2Num \times 2 + N.CCE.UL.AggLvl1Num)}{(N.CCE.UL.AggLvl16Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl1Num)}$
Uplink padding rate	$1 - (N.ThpVol.UL / N.ThpVol.UL.Cell)$
Uplink data transmission duration	N.ThpTime.UE.UL.RmvSmallPkt

8

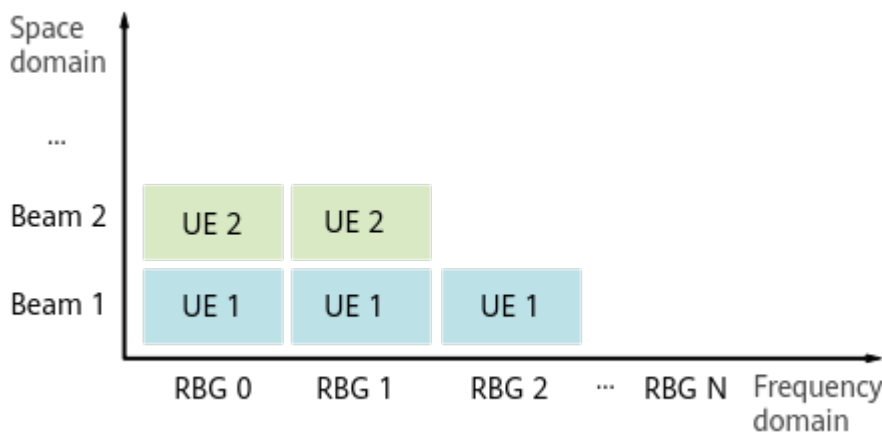
mmWave MU-MIMO (High-Frequency TDD)

8.1 Principles

8.1.1 Basic Principles of mmWave MU-MIMO

mmWave MU-MIMO implements spatial multiplexing of time-frequency resources for multiple UEs in both uplink and downlink. When multiple UEs share the same time-frequency resources, the more orthogonal the channels between UEs, the smaller the interference and the higher the capacity and spectral efficiency in both uplink and downlink. mmWave MU-MIMO requires TRXs to be split to transmit data for UEs covered by different analog beams. For UEs in the coverage of the same beam, frequency diversity scheduling can be performed but MU-MIMO cannot. [Figure 8-1](#) shows how this function works.

Figure 8-1 mmWave MU-MIMO



mmWave PUSCH MU-MIMO supports spatial multiplexing for multiple UEs only in 4T4R or 8T8R FWA scenarios. This function is controlled by the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.*MuMimoSwitch*

parameter. It is recommended that this function be enabled when the total bandwidth of active cells served by a single RF module is less than or equal to 200 MHz. In other scenarios, enabling this function cannot ensure performance.

The maximum number of spatial multiplexing layers for multiple UEs is four. This number can be specified by the **NRDUCellPusch.MaxMimoLayerCnt** parameter in the uplink.

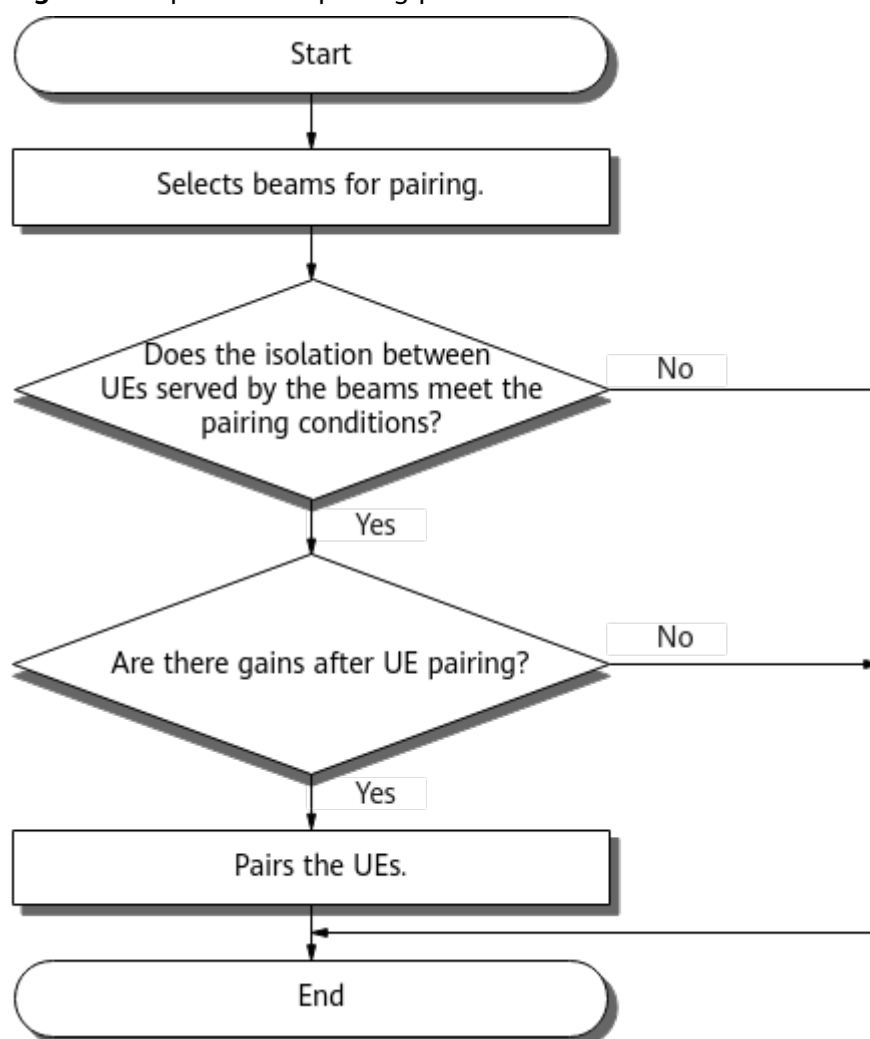
 NOTE

mmWave PUSCH MU-MIMO cannot take effect in a slot where there are both PRACH and PUSCH resources. This is because beams are insufficient in such a slot.

8.1.2 Spatial Multiplexing Procedure of mmWave MU-MIMO

The spatial multiplexing procedure of mmWave MU-MIMO can be considered as a UE pairing procedure. The gNodeB selects appropriate UEs for pairing according to pairing rules and schedules the UEs. [Figure 8-2](#) shows the spatial multiplexing procedure of mmWave MU-MIMO.

Figure 8-2 Spatial multiplexing procedure of mmWave MU-MIMO



The SRS quality of UEs needs to be considered during UE pairing.

- If the SRS quality of UEs is good (for example, the SINR is high and the signal fluctuation is small) and the measured inter-UE isolation is good, inter-UE interference can be ignored and MU-MIMO can increase the cell throughput.
- If the SRS quality of UEs is poor (for example, the SINR is low or the signal fluctuation is large) or the measured inter-UE isolation is poor, inter-UE interference cannot be effectively eliminated. In this situation, MU-MIMO may decrease the cell throughput and therefore such UEs will not be selected by the gNodeB for pairing.

Inter-UE isolation is indicated by the RSRP difference between their beams.

The pairing rules of mmWave PUSCH MU-MIMO can be set using parameters and are as follows:

- When a UE initially accesses a cell, measurement of the spatial isolation between beams is started if the SRS SINR before equalization of the UE is greater than the threshold specified by **NRDUCellUIMimo.UlMuSrsPreSnrThld** and the transmitted traffic volume for the UE is greater than the threshold specified by **NRDUCellUIMimo.UlMuIsolationMeasThld**; otherwise, the measurement is not performed and the UE is not selected for pairing.
- This UE can be successfully paired with another UE in a TTI only if the following conditions are met: (1) the spatial isolation between their optimal beams is greater than the threshold specified by **NRDUCellUIMimo.UlMuSpacelsoThld**; (2) the gain in the post-pairing throughput relative to the pre-pairing throughput is greater than the threshold specified by **NRDUCellUIMimo.UlMuBackToSuSeGainThld**.

8.2 Network Analysis

8.2.1 Benefits

mmWave PUSCH MU-MIMO:

- In light- or medium-load cells where resources are sufficient, mmWave PUSCH MU-MIMO may cause fluctuations in the average uplink UE and cell throughputs.
- In heavy-load cells where resources are insufficient, mmWave PUSCH MU-MIMO increases the average uplink UE and cell throughputs.
- It is recommended that mmWave PUSCH MU-MIMO be enabled when the uplink load of a cell is heavy. The uplink load of a cell is heavy when the uplink PRB usage is greater than or equal to 50%.
- Under the same traffic volume, the increases in the average uplink UE and cell throughputs are related to the number of paired RBs and the number of paired layers. The greater the numbers, the larger the increases.
- In CA scenarios, if there are only a few UEs on each component carrier (CC), the MU-MIMO pairing rate will be low and the benefits of mmWave PUSCH MU-MIMO will decrease. One example of such a scenario is when UEs are distributed on different CCs due to their limited uplink CA capabilities and consequently only a few UEs are scheduled on each CC.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

mmWave PUSCH MU-MIMO:

- After an increase in the uplink MU-MIMO pairing rate, there may be a decrease in the average uplink UE throughput in either of the following situations: (1) the data volume of tail packets accounts for a large proportion in light- or medium-load scenarios; (2) the channel quality of UEs is good and UEs are concentrated.
- After an increase in the uplink MU-MIMO pairing rate, there will be increases in the number of scheduled UEs in the uplink and the consumption of CCE resources. If CCE resources for the uplink are limited, there will be an increase in the CCE allocation failure rate for uplink transmission.
- After an increase in the uplink MU-MIMO pairing rate, there will be more downlink response packets. If PDSCH PRB resources are insufficient, there will be changes in the average downlink cell and UE throughputs, average downlink MCS index, average rank, and values of scheduling-related indicators such as the downlink IBLER.
- After an increase in the uplink MU-MIMO pairing rate, there will be an increase in the interference between paired UEs and accordingly an increase in the uplink BLER in the cell.
- After an increase in the uplink MU-MIMO pairing rate, the number of times each UE is scheduled will change, causing changes in the average uplink MCS index, average rank, and values of scheduling-related indicators such as the uplink IBLER.
- After an increase in the uplink MU-MIMO pairing rate, the downlink transmit power may decrease.

8.3 Requirements

8.3.1 Licenses

The following are license requirements for mmWave MU-MIMO.

Feature ID	Feature Name	Model	Sales Unit
FOFD-010010	MU-MIMO Basic Pairing	NR0S00M UMM00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

In addition, the license for the number of spatial multiplexing layers is also required. The maximum number cannot exceed the licensed number. One license

unit corresponds to two layers. For details about licensing rules, see *License Control Item Lists*.

Model	Description	Sales Unit
NR0SMMWULE00	mmWave UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per 100MHz per Cell

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
mmWave 3D coverage pattern	NRDUCellTrpMmwav-Beam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	mmWave PUSCH MU-MIMO cannot be enabled if the NRDUCellTrpMmwav-Beam.CoverageScenario parameter is set to SCENARIO_103 or CUSTOMIZED_BEAM_SCENARIO_106 .
Adaptive HARQ-ACK feedback on downlink transmission	ACK_NACK_REPORTING_PUSCH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	<i>Uplink Scheduling</i>	If adaptive HARQ-ACK feedback on downlink transmission is enabled, mmWave PUSCH MU-MIMO cannot be enabled.

Function Name	Function Switch	Reference	Description
DMRS spatial multiplexing for uplink MU-MIMO	MU_DMRS_SPATIAL_MULTIPLEX_SW option of the NRDUCellDmrs.UIDmrsSwitch parameter	None	If DMRS spatial multiplexing for uplink MU-MIMO is enabled, mmWave PUSCH MU-MIMO cannot be enabled.
Uplink MU-MIMO for large-packet UEs in mmWave cells	MMW_LRG_PKT_UL_MU_MIMO_SW option of the NRDUCellFeatureSw.MimoFeatureSwitch parameter	None	If uplink MU-MIMO for large-packet UEs in mmWave cells is enabled, mmWave PUSCH MU-MIMO cannot be enabled.

8.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	After Flexible Dense Beam is enabled together with mmWave PUSCH MU-MIMO, the cell throughput changes because the serving beams of UEs and the measured analog beam interference between UEs change.
Operator-specific RB management	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	In RAN sharing with common carrier, mmWave PUSCH MU-MIMO will not take effect if operator-specific RB management is enabled.
mmWave 3D coverage pattern	NRDUCellTrpMmwav-Beam.CoverageScenario set to SCENARIO_104	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the wide-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwav-Beam.CoverageScenario parameter set to SCENARIO_104), the gains of mmWave PUSCH MU-MIMO may decrease because coverage areas of SSB and CSI-RS beams may change.

Function Name	Function Switch	Reference	Description
NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch . NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	When "NR DU cell resource management between network slices" is enabled together with mmWave PUSCH MU-MIMO, UEs in a network slice group cannot participate in pairing.
Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch . CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	mmWave PUSCH MU-MIMO does not take effect for inter-FR CA UEs.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards as well as the UBBPg and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

RF modules must be the HAAU5323/HAAU5235w or later HAAU series modules. For details about HAAU series modules, see the AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

8.3.4 Cells

mmWave PUSCH MU-MIMO requires the **NRDUCell.SlotAssignment** parameter to be set to **4_1_DDSU**.

8.3.5 Others

mmWave PUSCH MU-MIMO applies to FWA-only scenarios and does not work for eMBB UEs. For details about FWA, see *FWA*.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

The parameters used for function activation are described in [Table 8-1](#). No parameters are involved in function optimization.

Table 8-1 Parameters used for activating mmWave PUSCH MU-MIMO

Parameter Name	Parameter ID	Option	Setting Notes
Transmit and Receive Mode	NRDUCellTrp .TxRxMode	None	Set this parameter based on the actually used AAU/RRU model. In FWA scenarios, mmWave MU-MIMO can be enabled only when this parameter is set to 4T4R or 8T8R .
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	UL_MU_MIMO_SW	It is recommended that this option be selected in FWA scenarios where the total bandwidth of active cells served by a single RF module is less than or equal to 200 MHz and the uplink cell load is heavy.
Uplink Maximum MIMO Layer Count Quota	NRDUCellPusch.MaxMimoLayerCnt	None	Set this parameter to LAYER_4 . mmWave PUSCH MU-MIMO cannot be enabled under other configurations.
UL MU-MIMO SRS PreSINR Threshold	NRDUCellULMimo.UlMuSrsPreSnrThld	None	Use the recommended value.
UL MU-MIMO Isolation Measure Threshold	NRDUCellULMimo.UlMuIsolationMeasureThld	None	Use the recommended value.
UL MU-MIMO Space Isolation Threshold	NRDUCellULMimo.UlMuSpaceIsoThld	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
UL MU-MIMO Back to SU- MIMO Gain Thld	NRDUCellUL Mimo.UIMuB ackToSuSeGa inThld	None	Use the recommended value.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling mmWave PUSCH MU-MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=UL_MU_MIMO_SW-1;  
//Setting pairing thresholds for mmWave PUSCH MU-MIMO  
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuSrsPreSinrThld=0, UIMuIsolationMeasThld=2,  
UIMuSpacelsoThld=50, UIMuBackToSuSeGainThld=5;  
//Setting the maximum number of uplink MIMO layers  
MOD NRDUCELLPUSCH: NrDuCellId=0, MaxMimoLayerCnt=LAYER_4;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating mmWave PUSCH MU-MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=UL_MU_MIMO_SW-0;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

Use a monitoring item to check whether mmWave PUSCH MU-MIMO has taken effect. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > Cell Performance Monitoring**. Select the corresponding base station and cell to start a monitoring task. If the value of the monitoring

item listed in [Table 8-2](#) is greater than 1, the corresponding function has taken effect.

Table 8-2 mmWave PUSCH MU-MIMO

Function	Monitoring Item	Description
Uplink MU-MIMO monitoring	Max Number of UL Transmission Pair Layers	If the value is greater than 1, mmWave PUSCH MU-MIMO has taken effect.

In addition, you can use the following indicator to estimate the effects of mmWave PUSCH MU-MIMO.

Number of uplink MU-MIMO layers in a cell: **N.ChMeas.MIMO.UL.Pair.Layer**

8.4.3 Network Monitoring

You can observe the increase in cell capacity using the following indicators:

- **User Uplink Average Throughput (DU)**
- **Cell Uplink Average Throughput (DU)**

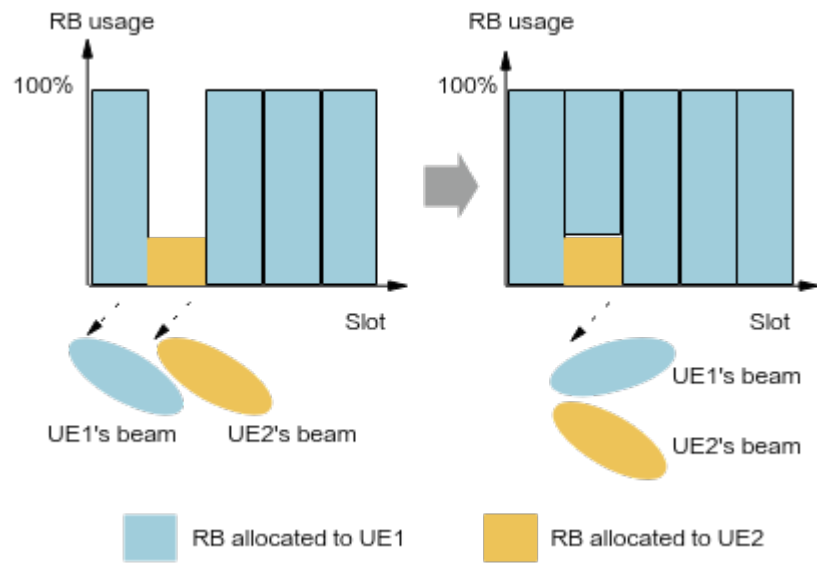
9 Multi-Beam FDM (High-Frequency TDD)

9.1 Principles

Hybrid beamforming, including analog beamforming, is used in high frequency bands. Generally, all TX/RX channels form one analog beam to enhance coverage. Therefore, only one analog beam is used in a high-frequency cell at a time. ("beam" hereinafter refers to "analog beam" unless otherwise stated.) In addition, only one optimal beam can be selected for a UE in the cell. For details about how to select an optimal beam, see *mmWave Beam Management (High-Frequency TDD)*. As multiple beams cannot be scheduled at the same time, only UEs in one beam can be scheduled in one slot and spectrum resources cannot be fully utilized. To improve spectral efficiency, multi-beam frequency division multiplexing (FDM) is introduced.

Multi-beam FDM is enabled when the **VOL_BASED_BEAM_MULTIPLEX_SW** option of the **NRDUCellAlgoSwitch.BeamMultiplexSwitch** parameter is selected. Different TX/RX channels form different beams. An RF module with four TX/RX channels can form a maximum of two beams while an RF module with eight TX/RX channels can form a maximum of four beams. If a UE served by a beam does not occupy all RB resources and its coverage is not limited, this UE can be scheduled together with another UE in another beam in the same slot, as shown in [Figure 9-1](#).

Figure 9-1 Multi-beam FDM



For burst services, multi-beam FDM may not take effect when there are restrictions on PDCCH beams. To avoid this, it is recommended that the **PUSCH_DTX_AGG_LVL_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter be selected.

9.2 Network Analysis

9.2.1 Benefits

When network load is heavy, it is recommended that this function be enabled to increase the downlink PRB utilization, average downlink UE throughput, and downlink cell traffic volume.

Network load is heavy if the proportion of scheduled downlink slots is greater than 80% during busy hours.

Proportion of scheduled downlink slots = $\frac{N.ThpTime.DL.Cell}{125/60/}$
Measurement period (minutes)/Number of downlink slots within 1s

The following table describes the numbers of downlink slots within 1s with the 4:1 slot configuration used in high frequency bands.

Networking	Number of SSB Beams	Number of Downlink Slots Within 1s
SA	6	5950
	8	5800
	12	5500
	16	5200

Networking	Number of SSB Beams	Number of Downlink Slots Within 1s
NSA	6	6250
	8	6200
	12	6100
	16	6000

9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

The average transmit power of RF modules changes after multi-beam FDM takes effect.

The single-beam gains of UEs in the cell center will decrease and the downlink IBLER of the cell will increase after multi-beam FDM takes effect.

As an antenna array is divided into multiple sub-arrays, the MCS indexes of non-cell-edge UEs decrease. When traffic is suppressed, however, the number of scheduling times for large-packet UEs increases, and the overall MCS index of the cell increases as the MCS indexes of large-packet UEs are higher than those of other UEs. Therefore, the average MCS index of the cell will decrease or increase, depending on the proportion of non-cell-edge large-packet UEs.

After UEs under multiple beams are scheduled at the same time, the values of the indicators related to uplink and downlink scheduling may change. These indicators include:

- Average uplink and downlink ranks of the cell
- Average uplink MCS index of the cell
- PDCCH CCE usage of the cell
- Average downlink throughput of UEs
- Average CCE aggregation level of UEs
- First packet transmission delay at the RLC layer

The downlink cell throughput may change after multi-beam FDM takes effect. Specifically, if there are a large number of small-packet UEs in a cell, the numbers of scheduled UEs and PRBs increase and the average downlink cell throughput and average downlink cell PRB usage increase after multi-beam FDM takes effect. If there are a large number of large-packet UEs in a cell and only some component carriers (CCs) of these UEs are activated, the number of scheduled slots in each involved cell increases after multi-beam FDM takes effect; however, the average downlink cell throughput decreases because the average MCS index decreases.

9.3 Requirements

9.3.1 Licenses

There are no license requirements for basic functions.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

None

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav-Beam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	After the Flexible Dense Beam function is enabled, the number of data transmission beams may increase and the beam multiplexing degree may decrease. Therefore, it is recommended that multi-beam FDM be enabled together with this function to improve the beam multiplexing degree. Otherwise, the Cell Downlink Average Throughput (DU) and User Downlink Average Throughput (DU) will decrease due to a reduction in the beam multiplexing degree.
Load-based adaptive downlink scheduling	NRDUCellDISch.DISchPolicy set to ADAPTIVE	None	If multi-beam FDM and load-based adaptive downlink scheduling are both enabled (in any sequence), the average downlink UE throughput may fluctuate, depending on the UE distribution and traffic model.

Function Name	Function Switch	Reference	Description
Downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm	NRDUCellDlsch.DlBeamPriorityBasedSch set to HEPF	<i>Downlink Scheduling</i>	If multi-beam FDM and downlink scheduling based on beam priorities defined by the Hybrid EPF algorithm are enabled at the same time, there will be a lower proportion of occasions where multi-beam FDM takes effect.
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Multi-beam FDM does not take effect when RF channel intelligent shutdown takes effect.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

RF modules must be HAAU5213, HAAU5222, HAAU5323, or HAAU5235w.

9.3.4 Others

None

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

[Table 9-1](#) describes the parameters used for multi-beam FDM activation.

Table 9-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Beam Multiplex Switch	NRDUCellAlgoSwitch.BeamMultiplexSwitch	VOL_BASED_BEAM_MULTIPLEX_SW	It is recommended that this option be selected to improve spectral efficiency.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcchPdcchAlgoEnhSwitch	PUSCH_DTX_AGG_LVL_ADAPT_SW	It is recommended that this option be selected to increase the PDCCH beam utilization.
DL Coverage Algorithm Switch	NRDUCellChnCoverageAlgoSwitch	CSIRS_2PORT_COV_ENH_SW	It is recommended that this option be selected if there are UEs supporting 2-port CSI-RS measurement on the network.

9.4.1.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling multi-beam FDM
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, BeamMultiplexSwitch=VOL_BASED_BEAM_MULTIPLEX_SW-1;
//Turning on the PUSCH_DTX_AGG_LVL_ADAPT_SW
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-1;
//((Optional) Turning on the 2-port CSI-RS coverage enhancement switch if there are 2-port CSI-RS UEs on the network
MOD NRDUCCELLCHNCOVALGO: NrDuCellId=0, DlCoverageAlgoSwitch=CSIRS_2PORT_COV_ENH_SW-1;
```

Deactivation Command Examples

```
//Turning off the PUSCH_DTX_AGG_LVL_ADAPT_SW
MOD NRDUCCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-0;
//Disabling multi-beam FDM
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, BeamMultiplexSwitch=VOL_BASED_BEAM_MULTIPLEX_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

None

9.4.3 Network Monitoring

Observe the increases in the downlink PRB utilization, average downlink UE throughput, and downlink cell traffic volume.

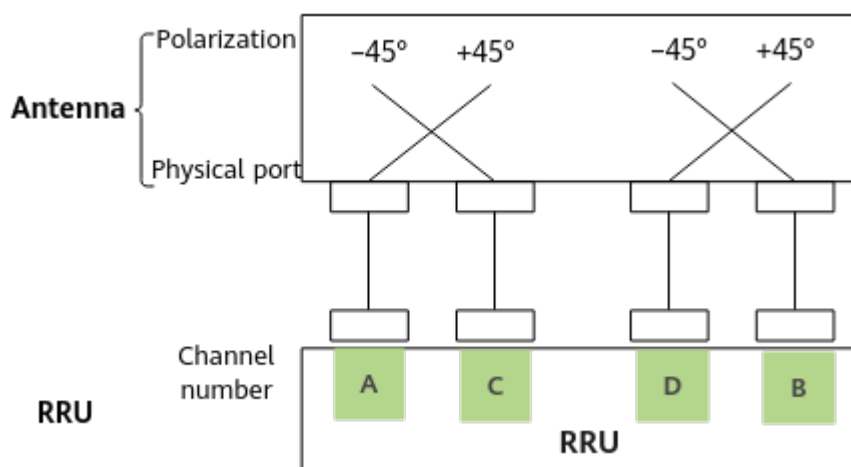
- Downlink PRB utilization = $(\text{N.PR.B.DL.Used.Avg} / \text{N.PR.B.DL.Avail.Avg}) \times 100\%$
- Average downlink UE throughput = **User Downlink Average Throughput (DU)**
- Downlink cell traffic volume = **Downlink Traffic Volume (DU)**

10 Inter-Cell Cable Sequence Detection

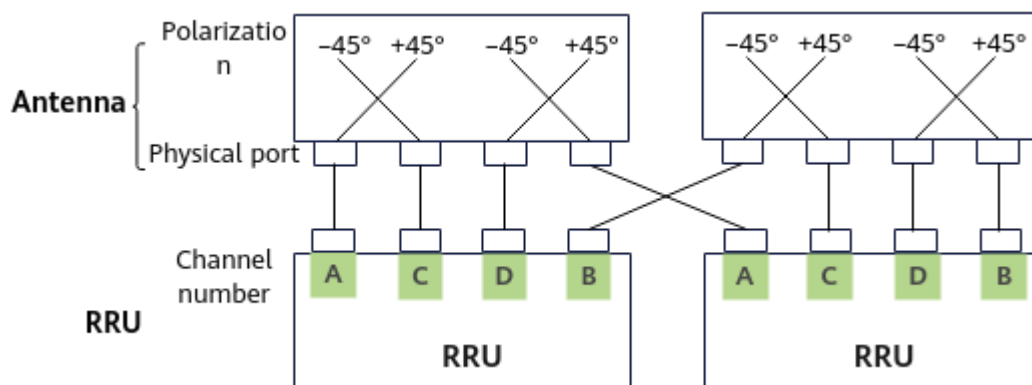
10.1 Principles

Generally, the four channels of a 4R RRU are connected to those of a four-channel antenna, as shown in [Figure 10-1](#).

Figure 10-1 Correct RRU connections



However, due to construction problems on the live network, an RRU channel of a cell may be connected to a polarized antenna channel of another cell, as shown in [Figure 10-2](#). Such partial inter-cell cross-connections are incorrect and will severely affect the downlink multi-antenna performance. If this problem is not solved during site deployment, troubleshooting needs to be performed later, increasing the number of site visits and delivery costs. To address this issue, the inter-cell cable sequence detection function is introduced.

Figure 10-2 Incorrect RRU connections**NOTE**

Partial inter-cell cross-connections refer to cases where only some RRU channels of one cell are connected to a polarized antenna of another cell. For 4R RRUs, partial inter-cell cross-connections can involve one to three incorrect connections. Inter-cell cable sequence detection applies in all these scenarios. [Figure 10-2](#) shows the scenario where one connection is incorrect.

Full inter-cell cross-connections refer to cases where all RRU channels of one cell are connected to a polarized antenna of another cell. Inter-cell cable sequence detection is not supported in full inter-cell cross-connection scenarios.

The inter-cell cable sequence detection function is used to determine whether an RRU channel of a cell is connected to an antenna channel of another cell based on the correlation between the signals received by RRU channels in these intra-base-station cells.

The inter-cell cable sequence detection function is started by running the **STR ANTENNAPORTOPTDET** command with **AntPortOptDetPolicy** set to **INTER_CELL_DETECT** and takes effect only when both of the following conditions are met:

- At least two cells are properly activated on the same logical site.
- There are UEs performing uplink services in the cells involved in the detection within the detection time. Otherwise, the detection results are unreliable.

NOTE

- The maximum detection time is 30 minutes.
- To improve the effective rate of detection results during the initial phase of site deployment, it is recommended that the detection be performed three times, 15 minutes after cell activation, for each site.

After this function is enabled, cells are selected for detection, and the signal correlation between the RRU channels of a cell is calculated to determine whether the RRU is also connected to the antenna of another cell.

1. Selecting cells for detection

The gNodeB searches intra-base-station cells for properly activated cells that meet detection requirements. It starts a detection only after qualified cells are found. The gNodeB selects to-be-checked cells based on the following principles:

- Number of cells:
 - A maximum of five frequency bands are supported on a site.
 - A maximum of 36 cells are supported in a frequency band.
- Cell frequency: The frequencies of to-be-checked cells working on the same frequency band must be the same. If the cells served by a base station work on different frequencies in the same frequency band, cells with the lowest frequency are preferentially selected for detection. (After qualified cells are selected, cells working on other frequencies will not be checked.)
- Cell type:
 - The cell TX/RX mode is 4T4R.
 - The **NRDUCell.NrDuCellNetworkingMode** parameter is set to **NORMAL_CELL**.
- Cell status:
 - At least two of the cells supported in a frequency band are activated, and channel shutdown or channel derating does not take effect. (Channel derating is triggered by channel faults.)
 - The average value of interference plus noise measured three times within 15 minutes is less than or equal to -90 dBm in to-be-checked cells.

The detection results may be inaccurate in the following scenarios:

- The checked cells are covered by an indoor distributed antenna system (DAS).
- Cell services are abnormal or voltage standing wave ratio (VSWR) alarms are reported in the checked cells.

If a cell does not meet the conditions, the subsequent inter-cell cable sequence detection will not be performed in this cell and the corresponding error is provided in the detection result. For details, see [10.4.2 Activation Verification](#).

2. Calculating the signal correlation between RRU channels of the to-be-checked cell
 - If the RRU of the cell is also connected to the antenna of another cell, there is an inter-cell cross-connection.
 - Otherwise, inter-cell cross-connections do not exist.

3. Querying the detection results

Run the **DSP ANTENNAPORTOPTDET** command with **AntPortOptDetPolicy** set to **INTER_CELL_DETECT**.

- If the operation is successful, **Operation succeeded** is displayed. In addition, the detection progress is periodically updated. When the progress reaches 100%, the detection results are automatically displayed on the MML command execution page.
- If the operation fails, **No cell is available for the operation** or **The cable connection check is in progress** is displayed, depending on the cause of the detection failure. **No cell is available for the operation** indicates that no cell

meets the conditions for inter-cell cable sequence detection. **The cable connection check is in progress** indicates that inter-cell cable sequence detection or antenna port detection is being performed in some cells.

Inter-cell cable sequence detection can be stopped by running the **STP ANTENNAPORTOPTDET** command with **AntPortOptDetPolicy** set to **INTER_CELL_DETECT**.

10.2 Network Analysis

10.2.1 Benefits

The inter-cell cable sequence detection function enables customers to learn the connections between RRUs and antennas without visiting sites. Customers need to visit sites for troubleshooting only if inter-cell cross-connections are detected.

The signal correlation between RX channels is affected by many factors, such as UE distribution, external interference, and cell service status.

Inter-cell cross-connections can be corrected onsite to improve downlink multi-antenna performance. For UEs accessing the network, the access success rate increases. For UEs performing data transmission, the downlink throughput increases.

It is not recommended that this function be enabled in the current version because this function needs to be optimized.

10.2.2 Impacts

The impacts between this function and other functions are described in [10.3.2.3 Function Impacts](#).

This function has no impact on the network.

10.3 Requirements

10.3.1 Licenses

None

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

None

10.3.2.2 Mutually Exclusive Functions

None

10.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Intelligent carrier shutdown	INTRA_GNB_MULT I_CARR_SD_SW or INTER_GNB_MULT I_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If intelligent carrier shutdown takes effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.
Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If timing carrier shutdown takes effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If RF channel intelligent shutdown takes effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.

Function Name	Function Switch	Reference	Description
Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	UL CoMP is not performed for UEs involved in cable sequence detection. If a UE meets both UL CoMP and cable sequence detection conditions, only UL CoMP is performed for the UE.

Function Name	Function Switch	Reference	Description
Fast carrier shutdown	- Capacity-layer cells: - In intra-base-station multi-carrier scenarios: INTRA_GNB_MULTI_CARR_SD_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter and FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter - In inter-base-station multi-carrier scenarios: INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter and FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter - Basic-layer cells:	<i>Energy Conservation and Emission Reduction</i>	If fast carrier shutdown takes effect during inter-cell cable sequence detection, the detection results will show that the detection fails in the corresponding cells.

Function Name	Function Switch	Reference	Description
	FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		

10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

NR TDD-capable RRUs that have 4T4R or have 2T2R but can be combined for 4T4R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Antennas

Combined RRUs are supported but combined antennas are not supported.

Cells

Inter-cell cable sequence detection has the following requirements on cells:

- Number of cells:
 - A maximum of five frequency bands are supported on a base station.
 - A maximum of 36 cells are supported in a frequency band.
- Cell frequency: The frequencies of to-be-checked cells working in the same frequency band must be the same. If the cells served by a base station work on different frequencies in the same frequency band, cells with the lowest frequency are preferentially selected for detection. (After qualified cells are selected, cells working on other frequencies will not be checked.)

- Cell type:
 - The cell TX/RX mode is 4T4R.
 - The **NRDUCell.NrDuCellNetworkingMode** parameter is set to **NORMAL_CELL**.
- Cell status:
 - At least two of the cells supported in a frequency band are activated, and channel shutdown or channel derating does not take effect. (Channel derating is triggered by channel faults.)
 - The average value of interference plus noise measured three times within 15 minutes is less than or equal to –90 dBm in to-be-checked cells.

The detection results may be inaccurate in the following scenarios:

- The checked cells are covered by an indoor distributed antenna system (DAS).
- Cell services are abnormal or voltage standing wave ratio (VSWR) alarms are reported in the checked cells.

10.3.4 Others

None

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

Table 10-1 describes the parameters used for function activation.

Table 10-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Antenna Port Optimize Detection Policy	AntPortOptDetPolicy parameter in the STR ANTENNAPORTOPTDET command	Set this parameter to INTER_CELL_DETECT .

10.4.1.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Performing detection on the entire base station  
STR ANTENNAPORTOPTDET: AntPortOptDetPolicy=INTER_CELL_DETECT;
```

Deactivation Command Examples

```
//Stopping detection  
STP ANTENNAPORTOPTDET: AntPortOptDetPolicy=INTER_CELL_DETECT;
```

10.4.2 Activation Verification

Method 1: Using MML Commands

Run the **DSP ANTENNAPORTOPTDET** command to query the intra-base-station inter-cell cable sequence detection results, IDs of the NR DU cells with cross-connections, and latest detection completion time. [Table 10-2](#) describes the inter-cell cable sequence detection results.

Table 10-2 Parameter values and descriptions of the inter-cell cable sequence detection results

Parameter Value	Meaning	Description
CROSSED	Crossed	There is a cross-connection between this cell and another cell.
CORRECT	Correct	There is no cross-connection between this cell and another cell.
ERROR_SERVICE_INTERRUPT	Error_Service Interruption	Services of this cell are interrupted during the detection.
ERROR_CHANNEL_SHUTDOWN	Error_Channel Shutdown	Channels of this cell are shut down during the detection.
ERROR_CHANNEL_DERATING	Error_Channel Derating	Channel derating occurs in this cell during the detection.
ERROR_INSUFFICIENT_DATA	Error_Insufficient Data	No sufficient detection data is collected in this cell during the detection.
ERROR_LOW_CORRELATION	Error_Low Correlation	The signal correlation between channels of this cell is too low during the detection.
ERROR_RRSI_NO_CHANGE	Error_No RSSI Change	The channel RSSIs of this cell remain unchanged during the detection and therefore no detection results can be obtained.

Parameter Value	Meaning	Description
ERROR_NO_DETECT_RESULT	Error_No Detect Result	Among intra-base-station intra-frequency cells, only this cell is activated during the detection and therefore the detection cannot continue.
ERROR_EXCESSIVE_INTERFERENCE	Error_Excessive Interference	The interference in this cell is greater than -90 dBm. The detection accuracy cannot be ensured, and the cable sequence detection cannot be performed.
ERROR_INVALID_RSSI_DATA	Error_Invalid RSSI Data	During the detection, the RSSIs of some channels of this cell indicate that signals are close to interference. The detection accuracy cannot be ensured, and the cable sequence detection cannot be performed.
DETECTING	Detecting	Cable sequence detection is being performed in this cell.
NOT_DETECTED	Not Detected	This cell is not selected for detection.

If a local cell has a cross-connection with another cell, the detection result of the local cell is **CROSSED** and the **Crossed NR DU Cell ID** indicates the ID of the other cell. If a cell has no cross-connection, the detection result of this cell is **CORRECT**. If the detection fails due to another reason, a corresponding detection result listed in [Table 10-2](#) is displayed.

If an inter-cell cable sequence detection succeeds, the detection results of the checked cells are displayed. If the next inter-cell cable sequence detection fails in some cells, the **Inter-Cell Line Sequence Detection Result** field shows the corresponding errors while the **Crossed NR DU Cell ID** and **Latest Detection Complete Time** fields show the previous successful detection results. After a cell is reset, deactivated, or reestablished, the previous detection result is retained. After the **RST APP** command is executed in a cell, the previous detection result is no longer retained.

Method 2: Using the MAE

On the MAE, perform **Inter-Cell Cable Sequence Detection**, set **RAT Type** to **gNodeB**, and manually select a site or cells to be checked. After the detection is complete, click **Export** to download the detection result file. If the **Inter-Cell Line Sequence Detection Result** field shows the detection results, see [Table 10-2](#) for the meaning and explanation.

10.4.3 Network Monitoring

None

11 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

12 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

13 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

14 Reference Documents

- 3GPP TS 38.104 V17.8.0: "General"
- 3GPP TS 38.214 V17.2.0: "UE sounding procedure for DL CSI acquisition"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Energy Conservation and Emission Reduction*
- *UE Power Saving*
- *URLLC*
- *License Control Item Lists*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *CoMP*
- *High Speed Mobility*
- *Multi-Operator Sharing*
- *Hyper Cell*
- *Cell Combination*
- *Cell Management*
- *Carrier Aggregation*
- *CA Experience Improvement*
- *Super Uplink*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *VoNR*
- *DRX*
- *Channel Management*
- *Random Access Control*
- *mmWave Beam Management (High-Frequency TDD)*
- *FWA*
- *Co-MM (Low-Frequency TDD)*
- *Positioning*

- *Multi-Frequency Smart Aggregation*
- *mmWave Beam Management (High-Frequency TDD)*
- *Turbo Uplink (Low-Frequency TDD)*
- *Scalable Bandwidth*
- *Remote Interference Management (Low-Frequency TDD)*
- *Modulation Schemes*
- *Green Site*
- *Uplink Scheduling*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *QCI-based Resource Control*
- *Fusion Cell (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Network Slicing*
- *Indoor Coverage*
- *RedCap*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Adaptive Managed Latency*